# Class 8

# Chemistry

## (Solutions)

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# 1. Atoms and Molecules

## Solutions

### LEVEL-I

**1. Correct option: (B)**

J.J.Bezelius was the scientist who suggested a method of representing elements using the English letters(capital as well as small)

**2. Correct option: (D)**

The symbol C stands for the element Carbon. It is derived from the English name of the element and the symbol is first letter of the name.

**3. Correct option: (C)**

Latin name of Hg is Hydrargyrum, in Greek Hydro means water and argyros means silver.Hg is a heavy silvery d-block element.

**4. Correct option: (A)**

Mendelevium is the element that named after the name of the scientist Mendeleev (some elements are named after the names of the scientists.)

**5. Correct option: (D)**

Neptunium is the element that named after the planet name in the following elements.

**6. Correct option: (A)**

Germanium is the element of the following elements named after the country's name.

**7. Correct option: (C)**

Atomic weight of a substance can be expressed in both gram atomic weight and gram atom.

**8. Correct option: (A)**

We know relative atomic mass of O atom =Mass of one O atom/one twelve th of mass of C-12 isotope.

=16amu/1amu

=16(amu s got cancelled)

**9. Correct option: (D)**

Chlorine molecule exist as $Cl_2$,

Gram molecular weight of chlorine is =2*atomic mass of Cl

=2*35.5g

=71.0g

**10. Correct option: (B)**

The molar mass of $H=H_2$=2*atomic mass of H

=2*1.008g

=2.016g

**11. Correct option: (A)**

One mole of molecules of Oxygen means, it contains $6.023*10^{23}$ molecules of Oxygen.

**12. Correct option: (B)**

The weight of 1 mole of $Na_2CO_3$=106g

The weight of 0.1 mole of $Na_2CO_3$=106g/0.1

=10.6g

**13. Correct option: (A)**

One mole of any substance contains $6.023*10^{23}$number of elementary particles.

**14. Correct option: (B)**

1 mole of hydrogen atom contains $6.023*10^{23}$ atoms of Hydrogen.

**15. Correct option: (D)**

Applying the concept of No. of molecules=W/M.wt*$N_A$

No. of molecules of 23g of $NO_2$=23/46*$N_A$=0.5$N_A$

A)8gm of oxygen($O_2$) no. of molecules=8/32$N_A$=0.25$N_A$

B)28gm of carbon,(CO)No. of Molecules=28/28$N_A$=$N_A$

c)16gmof $SO_2$,No.of molecules=16/64$N_A$=0.25$N_A$

d)22gm of $CO_2$,No.of Molecules=22/44=0.5$N_A$

**16. Correct option: (A)**

Given,
Mass of CO2=44gm
To find moles of oxygen,
Here,
Mole=mass/molar mass
=44/44=1 mole
So 1 mole of $CO_2$ contains
1 mole of Carbon and
2 moles of oxygen atoms.

**17. Correct option: (B)**

The molar volume of any gas at STP is 22.414 litre.

**18. Correct option: (D)**

The volume of two gram moles of oxygen at STP is 22.4*2
=44.8L

**19. Correct option: (A)**

The combining capacity of an element is called its valency.

**20. Correct option: (A)**

The electrons that present in outermost energy level are known as valence electrons.

**21. Correct option: (A)**

The number of electrons that present in outermost shell is one.so valency is one.

**22. Correct option: (B)**

Silver shows more than one valency that is known as variable valency. It shows both 1 and 2 valances and 1 is the lowest valency.

**23. Correct option: (A)**

Mercury also shows variable valency(1,2).the higher valency of Hg is 2

**24. Correct option: (D)**

Cu(Copper) also shows variable valancies(1,2).the valency in cupric chloride is 2

**25. Correct option: (B)**

Valency of Nitrogen is three as nitrogen has five electrons in its valence shell, it can therefore, accept three electrons to complete its octate. Hence the valency of nitrogen in $NH_3$ is 3.

**26. Correct option: (C)**

In $PCl_5$ P combine with 5 chlorine atoms, Hence the valence of phosphorus is 5 .

**27. Correct option: (C)**

In $CO_2$n,carbon is combined with two oxygen atoms. So,the valence of Carbon is 4(one oxygen atom has 2 valency)

## LEVEL-II

**1. Correct option: (D)**
Given all the statements are correct.
**2. Correct option: (A)**
The name of the element having atomic number 16 is Sulphur and the symbol is 'S'.
**3. Correct option: (C)**
The element Curium named after the scientist Madam Curie.
**4. Correct option: (B)**
The symbol 'Po' stands for Polonium.
**5. Correct option: (D)**
All are correct regarding A mole.
**6. Correct option: (B)**
No. of Moles=weight/M.Wt
M,Wt of $CuSO_4$ is= 64+32+64=160
So number of moles=1.60/160
=0.01 mole
**7. Correct option: (B)**
A) Mass of one gram atom of Iron=1gm
B)5 moles of N2=weight/M.wt
5=Wt/28
Mass=140g
C) $10^{23}$ carbon atoms
$6.023*10^{23}$ atoms are present in 12g of carbon
So $10^{23}$atoms are present in =$(10^{23}/6.023*10^{23})*12$
=12/6.023
=1.992g
D) Invalid MCQ
So 5 moles of $N^2$ has the highest mass.
**8. Correct option: (A)**
Atomic weight of H is 1u,therefore 1g hydrogen has 1 mole of hydrogen and 1 mole of hydrogen contains $6*10^{23}$ atoms.
Moles of He present in 4g of He is=mass/M.Wt
=4/4=1mole
Hence,4g of Helium also contains $6*10^{23}$ atoms.
**9. Correct option: (C)**
M.Wt of CaCO3 =100,Given weight=50g
No.of moles of $CaCO_3$=50/100=0.5
1 mole of $CaCO_3$ has 3 moles of Oxygen
0.5 moles of $CaCO_3$ has 1.5 moles of $O_2$
1 mole=$6.023*10^{23}$ particles
Therefore 50g of $CaCO_3$=$1.5*6.023*10^{23}$
=$9.035*10^{23}$
**10. Correct option: (B)**
A) weight of 0.1g silver=0.1g
B) 5 moles of N2=weight/M.wt
5=Wt/28
Mass=140g

C) Weight of copper=1gm

D) $10^{23}$ carbon atoms

$6.023*10^{23}$ atoms are present in 12g of carbon

So $10^{23}$atoms are present in $=(10^{23}/6.023*10^{23})*12$

$=12/6.023$

$=1.992g$

5 moles of $N_2$ has more weight than the remaining.

**11. Correct option: (B)**

The valence of suplhur is 6,as sulphur bonded to three oxygen atoms and we know that oxygen has the valency 2

3*2=6

**12. Correct option: (B)**

The atomic number 20 will combine most likely with atom having atomic number 16 as both will gain stable electronic configuration.

**13. Correct option: (B)**

Electronic configuration of F in a/tomic state is $1s^22s^22p5$. Seven electrons were present in outermost shell. If stability attained with 6 electrons F will lose one electron. Then F becomes $F^{+.}$

$F^+=1s^22s^22p4$

**14. Correct option: (A)**

The element X has 2 electrons in outermost shell and Y has 1 electron in outermost shell and valency of X is 2 and y is 1 .To get stability it forms the compound by the combination of X and Y as $XY_2$.

**15. Correct option: (B)**

Electronic configuration of $A=1s^22s^22p^63s^23p^1$

The outermost shell has 3 electrons (Valency is 3)

Electronic configuration of $B=1s^22s^22p^4$

The outermost shell has 6 electrons; hence valency is 8-6=2

| Element | A | B |
|---|---|---|
| valency | 3 | 2 |

Now the compound formed by cirss-cross method is $A_2B_3$.

**16) C 17) B 18) A 19) B 20) A 21) C**

**22) D 23) D 24) B 25) B 26) A 27) C**

**28) B 29) A 30) B 31) A,D 32) A,B,D**

**33) A,B 34) B,C 35) A,B 36) A 37) B 38) A**

**39) (a→P, R) (b→R) (c→ P, S) (d→Q, R)**

**40) (a→ S) (b→ P, R) (c → Q) (d→ R, P)**

**41) 9 42) 5**

# 2. Atomic Structure

## Solutions

### LEVEL – I

**1. Correct option: (B)**

John Dalton (father of atoms and atomic theory) proposed the atomic theory in the early 19th century, which explains that matter is composed of extremely small particles called atoms.

He also suggested that atoms are indivisible and indestructible particles that combine to form molecules and compounds. Dalton's atomic theory laid the foundation for our understanding of the composition and behavior of matter. But John Dalton elaborately explained about atoms. He also explained that atoms can be rearranged, combined, or separated in chemical reactions.

**2. Correct option: (A)**

The term "atom" was first introduced by the ancient Greek philosopher Democritus around the 5th century BCE. Democritus proposed that all matter is composed of indivisible and indestructible particles called atoms.

He believed that atoms are the primary building blocks of the matter in the universe and they are different in size, shape, and arrangement, giving rise to the variety of substances observed in the world.

**3. Correct option: (A)**

Elements in periodic table made up of very small particles of matter called atoms. These are the basic building blocks of elements. Each element consists of a specific type of atom that has unique number of protons, neutrons, electrons in the nucleus of atom. Elements can be distinguished from one another based on the number of protons or neutrons in their atoms, which is known as the atomic number represented by Z.

**4. Correct option: (D)**

Maharishi Kanada, he is an ancient Indian philosopher and he is known for his contributions to the atomic theory in the ancient Vedic period.

He proposed that matter is composed of extremely small particles called "Anu" (meaning atom) and "Paramanu" (meaning sub-atomic particle). According to Kanada, these particles are indivisible and can’t be described.

**5. Correct option: (A)**

According to Dalton's atomic theory, this was proposed by John Dalton, states that atoms are indivisible and indestructible particles with which all the matter is made. According to this theory, elements are composed of atoms, and each element is characterized by a unique type of atoms. Atoms of different elements differ in their properties, including mass, size, and chemical behavior.

**6. Correct option: (A)**

Democritus, he is an ancient Greek philosopher around the 5th century BCE. Who proposed the concept of atoms as the fundamental building blocks of matter. He suggested that matter is composed of tiny, indivisible particles called atoms. Democritus theorized that atoms are constantly moving and combine in different ways to form different substances. His ideas were only based on philosophical reasoning.

**7. Correct option: (B)**

The electron is a negatively charged particle, and its charge is approximately $-1.6022\times10^{-19}$ coulombs. This value is often rounded to $-1.6\times10^{(-19)}$ C for convenience.

**8. Correct option: (A)**

Anode rays were discovered by the German physicist Eugen Goldstein in the late 19th century. Eugen observed while studying these rays cathode rays in gas discharge tubes. He noticed that when a perforated cathode was used, a beam of positive ions traveled in the opposite direction, towards the cathode. These positive ions became known as anode rays.

**9. Correct option: (B)**

Proton is a subatomic particle that is found in the nucleus of an atom. It carries a positive charge. It is commonly associated with hydrogen atoms because the nucleus of a hydrogen atom consists of a single proton.

Option A) it is a nucleus of deuterium is incorrect. Deuterium is a hydrogen isotope that consists of a proton and a neutron in its nucleus.

Option C) it is an ionized hydrogen molecule is incorrect. A hydrogen molecule ($H_2$) consists of two hydrogen atoms bonded together, and a proton is not a hydrogen molecule.

Option D) it is a α-particle is incorrect. A α-particle is a helium nucleus consisting of two protons and two neutrons.

**10. Correct option: (B)**

The e/m value describes the ratio of the electron's charge (e) to its mass (m). The value of the e/m ratio for an electron is approximately $1.76\times10^{11} C/Kg$ .

**11. Correct option: (A)**

The correct option less than e/m value of an electron.

We know that magnitude of charge is the same for a proton and an electron but the mass of an electron is less than that of the mass of a proton.

**12. Correct option: (C)**

Specific charge is defined as the ratio of charge of a particle by its mass. Since the charge of an electron remains constant and its mass increases with speed, the specific charge remains the same.

**13. Correct option: (B)**

The specific charge is defined as the ration between charges to mass of a particle, for a cathode ray is constant.

In the case of a cathode ray, which consists of electrons, the charge and mass of an electron is constant. Regardless of the gas in the discharge tube or the atomic number of the elements involved, the charge-to-mass ratio of an electron remains the same.

**14. Correct options: (A and D)**

The e/m value of a proton is approximately $9.58\times10^{4} C/kg$ it represents the ratio of the charge (e) to the mass (m) of a particle. For a proton, the charge is approximately $1.6\times10^{-19} C$ and the mass is approximately $1.67\times10^{-27} Kg$ . Dividing the charge by the mass we get e/m value i.e. $5.8\times10^{4} C/Kg$ .

**15. Correct option: (B)**

The specific charge of a particle is defined as the ratio of the charge (e) to its mass (m). It is a fundamental characteristic of particles. Used to differentiate between different types of particles based on their charge-to-mass ratios.

**16. Correct option: (D)**

The charge of a neutron is 0.Neutrons are electrically neutral particles they reside in nucleus along with proton, meaning they have no net charge. They are composed of quarks (specifically, two down quarks and one up quark), and the combination of these quarks results in a neutral overall charge.

**17. Correct option: (A)**

Anode rays, also known as canal rays; they are produced when a high voltage is applied across a discharge tube containing a gas at low pressure. The nature of the gas present in the discharge tube determines the characteristics of the anode rays.

Different gases exhibit different behavior when influenced to electric fields. The gas molecules or atoms in the discharge tube can get ionized and form positive ions. These positive ions, known as anode rays, are accelerated towards the cathode due to the electric field. All the properties of anode rays depend on nature of gas used in discharge tube.

**18. Correct option: (B)**

James Chadwick, he a British physicist, discovered the neutron 1932. He conducted experiments that involved bombarding beryllium with alpha particles, which led to the observation of a neutral particle named as neutron with a mass similar to that of a proton.

**19. Correct option: (C)**

The mass of a neutron is slightly more than that of a proton. Both protons and neutrons are subatomic particles found in the nucleus of an atom. The mass of a neutron is approximately $1.675 \times 10^{-27} Kg$ kilograms, which is slightly greater than the mass of a proton, which is approximately $1.673 \times 10^{-27} Kg$ kilograms.

**20. Correct option: (D)**

To determine the highest specific charge among the given options, we need to calculate the specific charge for each element. Specific charge is defined as the charge-to-mass ratio.

A) $Na^{+}(A = 23)$ The atomic number of sodium is 11, and given charge is +1. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+1)}{(23)} \approx 0.0435$$

B) $Mg^{+2}(A = 24)$ The atomic number of magnesium is 12, given charge is +2. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+2)}{(24)} \approx 0.0833$$

C) $Al^{+2}(A = 27)$ The atomic number of aluminum is 13, given charge is +3. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+3)}{(27)} \approx 0.1111$$

D) $Si^{+4}(A = 28)$ The atomic number of silicon is 14, given charge is +4. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+4)}{(28)} \approx 0.1429$$

Comparing the calculated specific charges, we find that option D) $Si^{+4}(A = 28)$ has the highest specific charge.

**21. Correct option: (A)**

According to Thomson's model also known as “plum pudding “model of the atom, here the atom is represented as a sphere in shape. Where the atom is considered as a positively charged sphere with electrons embedded throughout it, much like the plums in a

pudding. This model was proposed by J.J. Thomson in the late 19th century before the discovery of the atomic nucleus and the subsequent development of the atomic structure.

**22. Correct option: (A)**

Thomson's model proposed that the atom consists of a positively charged sphere with electrons embedded in it. The distribution of positive and negative charges in the atom balanced out, resulting in an overall electrically neutral atom. Drawback of this model is it didn't explained how atoms do not collapse due to the repulsion between like charges.

However, Thomson's model does not explain the existence of orbital's (option C) or the concept of electron orbital's as understood in modern atomic theory.

**23. Correct option: (C)**

Thomson's atomic model is also known as the Plum pudding model (option B) and watermelon model (option C). This name was given to the model because it compared the structure of the atom to a positively charged "pudding" with negatively charged electrons embedded in it, resembling plums and watermelon fruit as positive charged matter and black seeds are negatively charged electrons. The model was proposed by J.J. Thomson in the late 19th century.

**24. Correct option: (A)**

Ernest Rutherford conducted the famous α-particle scattering experiment, also known as the gold foil experiment, in 1909. In this experiment, he projected a beam of α-particles (positively charged particles) on a thin gold foil and observed their behavior. According to Thomson's plum pudding model, the positively charged "pudding" should cause minimal deflection of the α-particles.

However, in Rutherford's observations showed that some α-particles were deflected at large angles and a few even bounced back. So Rutherford to propose a new atomic model. Based on the experimental results, Rutherford suggested that atoms have a small, dense, and positively charged nucleus at the center, which contains most of the atom's mass. He proposed that electrons orbit around the nucleus at a distance. This became known as the Rutherford atomic model or the nuclear model of the atom.

**25. Correct option: (C)**

A α-particle consists of two protons and two neutrons, resembling to the nucleus of a helium atom. It carries a positive charge of +2e. Due to their composition, α-particles have a relatively large mass compared to electrons and other subatomic particles. So an $\alpha-$ particle is doubly charged nuclei of Helium atom.

In certain radioactive decays, such as alpha decay, atomic nuclei emit α-particles as they undergo a transformation.

**26. Correct option: (D)**

In Rutherford's gold foil experiment, it was observed that the majority of the α-particles passed straight through the foil without any significant deflection. This observation led Rutherford to conclude that most of the atom is empty space.

He proposed that the positively charged α-particles were mostly encountering empty space within the atom, resulting in their relatively straight path.

Most part of the atom is empty, which allows the α-particles to pass through the foil with little deflection.

**27. Correct option: (A)**

The Rutherford atomic model was proposed by Ernest Rutherford in 1911 explained that the atom having a small, dense, and positively charged nucleus at the center. Electrons were thought to orbit around the nucleus in a manner similar to planets orbiting the Sun, hence the name "Planetary model."

This model brought a significant advancement from Thomson's Plum pudding model and provided a more accurate representation of the atom's structure. The Planetary model laid the foundation for further developments in atomic theory, leading to the Bohr model and eventually the quantum mechanical model.

**28. Correct option: (B)**

When an electron jumps from the N shell to the K shell (closer to the nucleus), energy is released. Electrons in an atom occupy specific energy levels or shells. The K shell is the innermost shell, while the N shell is further away from the nucleus. When an electron transitions from a higher energy level to a lower energy level, it releases energy in the form of electromagnetic radiation.

**29. Correct option: (A)**

The angular momentum of the electron (mvr) is quantized and can be represented as nh/2π, where n is a positive integer and h is Planck's constant.

Therefore, the correct answer is option nh/2π.

**30. Correct option: (A)**

In the Bohr model of the atom, electrons occupy discrete energy levels or shells. These energy levels are labeled with letters such as K, L, M, N, and so on, innermost shell is K.

According to the Bohr model, the energy of an electron in a specific shell is inversely proportional to the square of its principal quantum number (n). Therefore, as the value of n increases, the energy of the electron in that shell increases.

Since adjacent orbits refer to energy levels that are closest to each other, the energy difference between two adjacent orbits is minimum when considering the transition between the K and L shells.

**31. Correct option: (C)**

The angular momentum of the electron in the M-shell of the hydrogen atom is (option D) 4h/2π.

In the Bohr model of the atom, the angular momentum of an electron in a particular orbit is given by the product of its mass (m), velocity (v), and radius (r). Mathematically, it can be expressed as mvr.

For the hydrogen atom, the angular momentum is quantized, and it follows the relationship L = nh/2π, where n is the principal quantum number and h is Planck's constant.

In the case of the M-shell, the principal quantum number (n) would be 3. Hence, substituting n = 3 in the formula L = nh/2π, we get:

L = 3h/2π

**32. Correct option: (A)**

We have equation

$$\frac{m_e v^2}{r} = \frac{Zk_e e^2}{r^2} \rightarrow I \quad r = \frac{n^2 h^2}{4\prod^2 mZe^2}$$

We also have another equation $$mvr = \frac{nh}{2\prod} \rightarrow II$$

From solving equations I and II we get $r = \frac{n^2 h^2}{4\prod^2 mZe^2}$

**33. Correct option: (D)**

Bohr's atomic theory, also known as the Bohr model, according to that electrons in an atom occupy specific energy levels also called as stationary states. These stationary states are characterized by fixed energy values and specific distances from the nucleus.

According to Bohr's theory, electrons can transition between these stationary states by absorbing or emitting some amounts of energy in the form of photons. This concept of stationary states laid the foundation for understanding atomic structure and the quantization of energy levels in atoms.

**34. Correct option: (A)**

Formula for energy difference between two orbits is $|-13.6Z^2[\frac{1}{(n+1)^2}-\frac{1}{n^2}]|$

Greater the difference, smaller the "n" value. So $E_2 - E_1 = E$ is the correct answer.

**35. Correct option: (A)**

According to Bohr's atomic model $r\alpha\frac{n^2}{Z}$

So, radius of nth orbit is r= $r_1 n^2$ , as Z=1 for H atom.

**36. Correct option: (A)**

In Bohr's atomic model, the energy of an electron at an infinite distance from the nucleus is taken as Zero. According to the model, electrons in the hydrogen atom have zero energy when they are infinitely far away from the nucleus.

**37. Correct option: (B)**

The ratio of the radius of the $Li^{+2}$ ion in the 3rd energy level to that of the $He^+$ ion in the 2nd energy level can be calculated using the formula for the radius of an orbit in the Bohr model.

$r \propto \frac{n^2}{Z}$

Where:
r = radius of the orbit,
n = principal quantum number,
Z = atomic number.

For the $Li^{+2}$ ion in the 3rd energy level (n = 3), we have:

$\frac{r_1}{r_2} = \frac{3}{2} = 3:2 \quad r_1 = (\frac{3^2}{3}) = 3$

For the $He^+$ ion in the 2nd energy level (n = 2), we have $r_1 = (\frac{2^2}{2}) = 2$

Therefore, the ratio of the radii is: $\frac{r_1}{r_2} = \frac{3}{2} = 3:2$

**38. Correct option: (B)**

In the Bohr model of the atom, the difference between adjacent energy levels decreases as the number of orbits increases from the nucleus.

As the electron moves to higher energy levels, the energy spacing between these levels decreases. This is due to the inverse square relationship between the radius of the orbit and the energy level, it can be written as

$E \propto \frac{1}{n^2}$

Here E represents the energy level and n represents the principal quantum number. As n Increases, the energy levels get closer together, and the difference between adjacent energy levels decreases.

**39. Correct option: (B)**

For an object moving in the circular orbit, the centripetal force is given by $\frac{-mv^2}{r}$ And is equal to force of attraction between electron and proton.

**40. Correct option: (A)**

According to Bohr's postulates, the energy of an electron in a given orbit is inversely proportional to the square of the principal quantum number (n). Therefore, the energy of an electron in the first orbit (n = 1) is higher than the energy of an electron in the second orbit (n = 2).

When an electron jumps from a higher energy level to a lower energy level, it emits energy in the form of a photon. This energy emission causes the electron to lose energy and decrease its velocity.

Therefore, the change in velocity when an electron jumps from the first orbit to the second orbit is less than its original velocity. The change in velocity would be a reduction by some fraction.

**41. Correct option: (D)**

Bohr's model of the atom explained about the emission spectrum of hydrogen and provided a framework for understanding the stability of electron orbits. However, it also had limitations when applied to atoms with more than one electron, leading to the drawbacks as mentioned in the options.

A) **Spectrum of atoms having more than one electron**: Bohr's model was not able to predict the emission or absorption spectra of atoms with more than one electron. It failed to explain the complex spectral lines observed in such atoms.

B) **Zeeman effect**: This effect refers to the splitting of spectral lines in the presence of a magnetic field. Bohr's model was unable to account for this effect, which demonstrated the need for a more advanced model that considered electron spin and magnetic interactions.

C) **Stark effect**: The Stark effect refers to the splitting and shifting of spectral lines in the presence of an electric field. Similarly, Bohr's model did not explain this effect, highlighting its limitations.

**42. Correct option: (B)**

Bohr's model was specifically developed on the hydrogen atom model, which has only one electron. It successfully explained the spectral behavior of hydrogen. However, when it comes to atoms or ions with multiple electrons, Bohr's model fails to predict their properties and spectral behavior. As a result, it is not applicable to species like Be , which has two electrons.

**43. Correct option: (A)**

In physics, elliptical orbits are characterized by the eccentricity (e) of the ellipse, which is defined as the ratio of the distance between the foci of the ellipse to the major axis of the ellipse. The eccentricity is given by the equation: $e = c/a$

Where c represents the distance between the foci and represents the semi-major axis (half the major axis) of the ellipse.

In the context of the given options, option C) k/n matches this definition, with k representing the distance between the foci and n representing the semi-major axis of the ellipse.

**44. Correct option: (D)**

Let's break down the variables in this expression:

m = mass of the electron.

h= Planck's constant.

L is the orbital angular momentum of the electron.

This expression relates the mass of the electron, its orbital angular momentum, and Planck's constant. It states that the product of the mass and the square of the orbital angular momentum is equal to π (pi) times Planck's constant.

**45. Correct option: (A)**

According to Aufbau's principle, electrons in an atom can have a maximum of three quantum numbers.

Aufbau's principle states that electrons fill the available atomic orbitals in order of increasing energy levels. Each electron in an atom is described by a set of four quantum numbers they are principal quantum number (n), the azimuthal quantum number (l), the magnetic quantum number ($m_l$), and the spin quantum number ($m_s$).

However, Aufbau's principle focuses on the filling of orbital's, and it explains that each orbital is filled with a maximum of two electrons, with opposite spins, before moving to the next orbital.

Therefore, while each electron has four quantum numbers, Aufbau's principle indicates that only one quantum number is needed to describe the order of filling orbital's.

**46. Correct option: (B)**

According to the rules for the filling of electron orbital's in an atom, each shell can hold a maximum number of electrons equal to $2n^2$, where n represents the principal quantum number.

The principal quantum number (n) designates the energy level of an electron. For example, the first energy level (n=1) can hold a maximum of $2(1)^2 = 2$ electrons, the second energy level (n=2) can hold a maximum of $2(2)^2 = 8$ electrons and so on.

**47. Correct option: (C)**

Each orbital can accommodate a maximum of two electrons.

According to the Pauli Exclusion Principle, no two electrons in an atom can have the same set of four quantum numbers. Since an orbital is described by three quantum numbers (n, l, and $m_l$), each orbital can hold a maximum of two electrons with opposite spins (spin quantum number, $m_s$).

This means that in a given orbital, you can have one electron with a spin-up (+1/2) and another electron with a spin-down (-1/2). Therefore, the maximum number of electrons that can be accommodated in an orbital is two.

**48. Correct option: (C)**

The statement "Electrons will fill a set of degenerate orbital's by keeping their spins parallel" is based on Hund's rule.

Hund's rule is a principle in quantum mechanics that governs the filling of degenerate orbitals, which are orbital's with the same energy level. According to Hund's rule, when multiple degenerate orbital's are available, before pairing up electrons will first occupy separate orbital's within the set, each with the same spin (parallel spins) after pairing up with opposite spins. This arrangement minimizes electron-electron repulsion and stabilizes the atom.

This rule explains why, each orbital is occupied by one electron before any pairing occurs for example, when filling the p orbital's in a sub shell (such as the 2p sub shell). The electrons distribute themselves with parallel spins to maximize the stability of the atom.

**49. Correct option: (D)**

Aufbau's principle states that electrons fill the atomic orbitals in order of increasing energy levels. In this electronic configuration, the numbers represent the maximum number of electrons hat can be accommodated in each sub shell:

The first sub shell is the 1s sub shell, which can hold a maximum of 2 electrons.
The second sub shell is the 2s sub shell, which can also hold a maximum of 2 electrons.
The third sub shell is the 2p sub shell, which can accommodate a maximum of 6 electrons.
So, the given option satisfies the correct order of electronic configuration follows the principle of filling orbitals in order of increasing energy levels, which is the essence of Aufbau's principle.

**50. Correct option: (A)**

Pauli's exclusion principle states that "it is impossible for any two electrons to have the same set of all four quantum numbers."
This principle was proposed by Wolfgang Pauli, is a fundamental concept in quantum mechanics. It states that no two electrons in an atom can have identical quantum numbers. The four quantum numbers that describe an electron are the principal quantum number (n), azimuthal quantum number (l), magnetic quantum number ($m_l$), and spin quantum number ($m_s$).
According to Pauli's exclusion principle, any given set of quantum numbers must be unique for each electron in an atom.

**51. Correct option: (B)**

Half-filled or completely filled sub-shells impart greater stability to the atoms.
When a sub-shell is half-filled or completely filled, it results in a more stable electronic configuration. This stability arises from the balance of electron-electron repulsion and the arrangement of electrons in such a way that minimizes energy.
The stability of an atom is crucial in determining its chemical properties and reactivity. Atoms with stable electron configurations are less likely to be inert. This stability is especially significant for elements in the transition metals and noble gas groups, as they tend to have half-filled or completely filled sub-shells, contributing to their unique chemical behavior and inertness.

**52. Correct option: (A)**

The Nitrogen atom has total 7 electrons and the configuration corresponds to the ground state electronic configuration of a nitrogen atom. Let's break it down:
The first two electrons occupy the 1s orbital.
The next two electrons occupy the 2s orbital.
The remaining three electrons occupy the 2p orbitals, with one electron each in the 2px, 2py, and 2pz orbitals.
This arrangement accounts for the atomic number 7, which corresponds to nitrogen. Therefore, the given configuration represents a nitrogen atom in its ground state.

**53. Correct option: (D)**

Hund's rule deals with the distribution of electrons in degenerate orbitals.
Degenerate orbitals refer to orbitals that have the same energy level. According to Hund's rule, when electrons occupy degenerate orbitals, they will first fill each orbital with one electron before pairing up.
This rule is based on the principle that electrons repel each other due to their negative charges. By occupying separate orbitals, they minimize their repulsion and increase the overall stability of the atom or molecule.

**54. Correct option: (A)**

The systematic arrangement of electrons in orbitals in the increasing order of their energies is known as electronic configuration.

The electronic configuration of an atom or molecule explains how the electrons are distributed among the various orbitals and energy levels. It specifies the number of electrons in each shell, sub shell, and orbital. The electronic configuration provides important information about the chemical behavior and properties of an element.

**55. Correct option: (C)**

The atomic number of chromium is 24, which means it has 24 electrons. The electron configuration of an atom describes how these electrons are distributed among the energy levels and sub shells.

The noble gas that precedes chromium in the periodic table is argon (Ar), which has the electronic configuration $1s^2\ 2s^2\ 2p^6 3s^2\ 3p^6$.

To determine the electronic configuration of chromium, we need to add the remaining electrons after argon.

The 3d sub shell can hold up to 10 electrons, and the 4s sub shell can hold up to 2 electrons. Therefore, the electronic configuration of chromium is (Ar) $3d^5\ 4s^1$.

**56. Correct option: (B)**

The electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$ represents the element phosphorus. Phosphorus has an atomic number of 15, meaning it has 15 electrons. The electronic configuration describes the arrangement of these electrons in different energy levels and sub shells.

The first shell (n=1) has 2 electrons in the 1s sub shell ($1s^2$).

The second shell (n=2) has 2 electrons in the 2s sub shell ($2s^2$) and 6 electrons in the 2p sub shell ($2p^6$).

The third shell (n=3) has 2 electrons in the 3s sub shell ($3s^2$) and 3 electrons in the 3p sub shell ($3p^3$).

**57. Correct option: (B)**

For vanadium (V), the electronic configuration is:

$1s^2 2s^2 2p^6$

The valence electrons are those in the outermost energy level, which is the 4th energy level (n=4).

The sub shells in the 4th energy level are 4s, 3d, and 4p.

**58. Correct option: (C)**

Given electronic configuration is $1s^2 2s^2 2p^5 3s^1$ with atomic number 10 which represents Neon atom. But original electronic configuration of Neon atom is $1s^2 2s^2 2p^6$

So here one electron is transferred from 2p to 3s orbital which is a higher energy state. So the given electronic configuration is represents exited state of **NEON** atom.

**59. Correct option: (A)**

The electronic configuration of different atoms can be represented in the $n^{1^x}$ method.

In the $n^{1^x}$ method, the electronic configuration is written as a combination of the principal quantum number (n) and the azimuthal quantum number (l). The principal quantum number describes the energy level or shell of an electron, while the azimuthal quantum number describes the sub shell or orbital.

**60. Correct option: (A)**

The sequence in which electrons occupy various orbitals can be easily remembered with the help of Moeller's diagram.

Moeller's diagram is a graphical representation that shows the order in which electrons fill the orbitals of an atom. It is based on the Aufbau's principle, which states that electrons fill the lowest energy orbitals first before filling higher energy orbitals.
The diagram organizes the orbitals into different energy levels or shells and indicates the order in which they are filled. Each orbital is represented by a box, and arrows are used to represent electrons. The arrows point up or down to indicate the electron's spin.
By following the sequence in Moeller's diagram, one can easily determine the order in which electrons occupy various orbitals as you move across the periodic table. This helps in understanding the electron configuration of different elements and their placement in the periodic table.

**61. Correct option: (A)**

The electronic configuration of nitrogen is $1s^2 2s^2 2p^3$ .Atomic number of Nitrogen is 7.

In this electronic configuration, we can break it down as follows:

$1s^2$: This represents the first energy level (n = 1) and the 1s orbital, which can hold a maximum of 2 electrons. In the case of nitrogen, it has 2 electrons in the 1s orbital.

$2s^2$: This represents the second energy level (n = 2) and the 2s orbital, which can also hold a maximum of 2 electrons. Nitrogen has 2 electrons in the 2s orbital.

$2p^3$: This represents the second energy level (n = 2) and the 2p orbital, which can hold a maximum of 6 electrons. Nitrogen has 3 electrons in the 2p orbital.

Therefore, the electronic configuration of nitrogen is $1s1s^2 2s^2 2p^3$ .

**62. Correct option: (B)**

After the 3d sub-level is completely filled, the differentiating electron enters into the 4p sublevel. Therefore, option B) 4p is the correct answer.
In the electron configuration of atoms, the sublevels are filled in a specific order based on their increasing energy.
The order of filling is generally given by the Aufbau's principle, which states that lower energy levels are filled before higher energy levels.
The filling order for the sublevels is as follows:
1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, and so on.

**63. Correct option: (B)**

To determine the number of unpaired electrons in the Ni+2 ion, we need to know the electronic configuration of the neutral nickel atom (Ni) and then consider the loss of two electrons to form the $Ni^{+2}$ ion.

The electronic configuration of neutral nickel (Ni) is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$ .

When nickel loses two electrons to form the $Ni^{+2}$ ion, the electron configuration become $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$

In this configuration, we can see that there are 2 unpaired electrons in the 3d sublevel ( $3d^8$).

Therefore, the number of unpaired electrons in the $Ni^{+2}$ ion is 2.

**64. Correct option: (A)**

The maximum number of electrons that can be present in the M and N-shells, respectively, can be determined using the formula 2n^2, where n represents the principal quantum number.
For the M-shell:
The principal quantum number (n) for the M-shell is 3.

Using the formula $2n^2$, we have $2(3)^2 = 2(9) = 18$.

Therefore, the maximum number of electrons that can be present in the M-shell is 18.

For the N-shell:

The principal quantum number (n) for the N-shell is 4.

Using the formula $2n^2$, we have $2(4)^2 = 2(16) = 32$.

Therefore, the maximum number of electrons that can be present in the N-shell is 32.

## LEVEL-II

**1. Correct option: (B)** $IE = 13.6\frac{Z^2}{1^2}eV$

The $Li^{+2}$ ion is formed when two electrons are removed from the lithium (Li) atom. The electron configuration of lithium is $1s^2 2s^1$.

The energy required to remove an electron from its outermost orbital is called as ionization energy. The ionization energy of hydrogen (H) is given as 13.6 eV.

Since the electron configuration of the $Li^{+2}$ ion is the same as that of helium (He), which has a filled $1s^2$ shell, it requires a significantly higher energy to remove an electron from $Li^{+2}$ compared to removing an electron from hydrogen (H).

According to Bohr's model $IE = 13.6\frac{Z^2}{n^2}eV$

So, in this the E is equal to energy of electron in nth shell of atom whose atomic number is Z.

Same amount of energy with opposite sign must be given to electron in order to remove it.

$IE = 13.6\frac{Z^2}{1^2}eV$ (n=1)

Therefore, the correct answer is B) 122.4 eV.

**2. Correct option: (C)**

The ionization energy of a helium atom is the energy required to remove one electron from a neutral helium atom. This value is known to be approximately 24.6 eV. Once one electron is removed; the helium atom becomes a helium ion (He+), with a single remaining electron. To remove the second electron, we need to consider the energy required for this process.

The energy required to remove the second electron from a helium ion ($He^+$) is higher than the ionization energy of the helium atom because the remaining electron experiences a stronger attractive force from the positively charged nucleus.

The approximate energy required to remove the second electron from a helium ion is 54.4 eV.

So, to find the total energy required to ionize a helium atom and remove both electrons, we sum up the ionization energy of the helium atom (24.6 eV) and the energy required to remove the second electron (54.4 eV):

24.6 eV + 54.4 eV = 79 eV

**3. Correct option: (A)**

To find the radius of the first orbit of a helium atom, we can use the same formula as before, but with Z = 2 for helium:

$r_{helium} = 0.529 \times \frac{1^2}{2} = 0.2645$ Angstorm, here Z=2,n=1

Now, we need to convert the radius from angstroms to centimeters. We know that 1 angstrom (Å) is equal to $1\times10^{(-8)}$ centimeters (cm). Therefore:

$r_{helium}cm = 0.2645\times10^{(-8)}cm$ Calculating this value, we find:

$$r_{helium}cm = 0.2645\times10^{(-8)}cm$$

**4. Correct option: (C)**

Ratio of time period of moving electrons in circular orbit is $T\alpha\frac{n^3}{Z^2}$

As we know Z= constant. $T\alpha n^3$ And $R\alpha n^2$ (R=Radius)

We can relate time and radius by $T\alpha R^{\frac{3}{2}} \Rightarrow \frac{T_1}{T_2}$

$$\frac{T_1}{T_2} = (\frac{R_1}{R_2})^{\frac{3}{2}} = (\frac{R}{4R})^{\frac{3}{2}} = \frac{1}{8} = 1:8$$

**5. Correct option: (C)**

We know Potential energy of Hydrogen atom is -27.2eV.

If we consider, potential energy is zero means we increase Hydrogen energy level by 27.2ev.

Energy of an electron is given by the equation, is

$$E = \frac{-13.6}{n^2}$$

Total energy in the first excited state when n=1 is given by:

$$E_1 = \frac{-13.6}{n^2} = \frac{-13.6}{1^2} = -13.6eV$$

$$PE_1 = 2\times E_1$$

$$PE_1 = 2\times(-13.6) = -27.2eV$$

Total energy in the first excited state when n=2 are given by:

$$E_2 = \frac{-13.6}{n^2} = \frac{-13.6}{4^2} = -3.4eV$$

So, first excited state total energy is -3.4+27.2=23.8eV.

**6. Correct option: (A)**

To determine the excited state in which the electron is present, we can use the formula for the energy levels of the hydrogen atom:

$$E_n = -\frac{13.6}{n^2}eV \rightarrow I$$

Given potential energy of hydrogen atom is -6.8eV.

$$E_n = \frac{PE}{2} = \frac{-6.8}{2} = -3.4 \rightarrow II$$

By equating I and II equations we get $Cu^{+}$ and $Zn^{+2}$ $n^2 = 4$

$$n = 2$$

Where $E_n$ the energy of the nth energy level and n is is the principal quantum number. Since n=2 it present in 1st exited state.

**7. Correct option: (C)**

The potential energy of the electron present in $He^{+}$ can be expressed in terms of e (elementary charge) and r (distance between the electron and the nucleus).

In the case of $He^+$ (helium ion with a single proton), the potential energy is given by:

$E = -\frac{(2e^2)}{(4\pi\varepsilon_0 r)}$ Where e is the elementary charge, ε₀ is the vacuum permittivity (approximately $8.854\times10^{-12}\frac{C^2}{Nm^2}$), and r is the distance between the electron and the nucleus.

Therefore, the potential energy of the electron in $He^+$ is $E = -\frac{(2e^2)}{(4\pi\varepsilon_0 r)}$.

**8. Correct option: (D)**

I) $He^+$: $He^+$ is a helium ion resembles with configuration if Hydrogen. Bohr's theory is applicable here. Therefore, $He^+$ is applicable.

II) $Li^{+2}$: $Li^{+2}$ is a lithium ion with two protons and only one electron. Bohr's theory can be applied here because there is a single electron orbiting the nucleus. Therefore, $Li^{+2}$ is applicable.

III) Tritium: Tritium is a radioactive isotope of hydrogen with one proton and two neutrons. Bohr's theory is applicable to hydrogen-like species, so it can be applied to tritium. Therefore, Tritium is applicable.

IV) Be+2: Be+2 is a beryllium ion with four protons and two electrons. Bohr's theory cannot be applied here.

Based on the above analysis, the correct answer is:

D) I, II, III is correct option.

**9. Correct option: (B)**

Cathode rays are made up of negatively charged particles.

The discovery of cathode rays is attributed to J.J. Thomson, who conducted experiments using cathode ray tubes. These cathode rays were found to be streams of negatively charged particles called electrons.

**10. Correct option: (B)**

The constancy of the e/m (charge-to-mass) ratio for electrons indicates that electrons are universal particles of all matter.

It was determined through various experiments, including J.J. Thomson's experiments with cathode rays. These experiments showed that the e/m ratio for electrons remains constant regardless of the source or type of material being used.

This constancy of the e/m ratio led to the conclusion that electrons are fundamental particles present in all matter, not just limited to specific substances or produced only in discharged tubes.

**11. Correct option: (C)**

The mass of an electron is nearly equal to 1/1837 the mass of one atom of hydrogen.

Therefore, the correct answer is: C) $\frac{1}{1837}$

**12. Correct option: (C)**

When cathode rays were passed through an electric or magnetic field, they were observed to be deflected, which suggests that cathode rays consist of charged particles. This behavior is consistent with the particle nature of cathode rays, indicating that they are composed of charged particles (electrons) that respond to electric and magnetic fields.

Therefore, the correct answer is:

C) They get deflected by an electric and magnetic field.

**13. Correct option: (B)**

George Johnstone Stoney, an Irish physicist, proposed the term "electron" in 1891 to represent the fundamental unit of electric charge. He introduced this term while studying the properties of electricity and electrical phenomena. J. J. Thomson later confirmed the existence of these particles and their properties through his experiments with cathode rays.

**14. Correct option: (A)**

The charge-to-mass ratio of an electron is much higher than that of a proton, alpha particle (α- particle), or a neutron. Electrons have a relatively small mass compared to their charge, resulting in a higher charge-to-mass ratio.

The charge of an electron is -1 elementary charge (approximately$-1.6 \times 10^{-19}$C), and its mass is approximately$9.1\times10^{-31}kg$ . The charge-to-mass ratio for an electron is therefore:

$$\frac{-1.6 \times 10^{-19}\text{C}}{9.1 \times 10^{-31}\text{kg}}$$

**15. Correct option: (D)**

Here's the explanation for the order:

Neutron (n): The neutron has no charge (zero charge) and a non-zero mass, so its e/m value is the lowest.

Alpha particle (α): The alpha particle has a positive charge (twice the elementary charge) and a relatively large mass, so its e/m value is higher than that of a neutron.

Proton (p): The proton has a positive charge (equal to the elementary charge) and a smaller mass compared to the alpha particle, resulting in a higher e/m value than that of an alpha particle.

Electron (e): The electron has a negative charge (equal to the elementary charge) and a much smaller mass compared to all the other particles, giving it the highest e/m value.

Therefore, the correct answer is C) n, α, p, e

**16. Correct option: (B)**

The mass of one mole of electrons can be calculated using the molar mass of an electron.

The molar mass of an electron is the same as its atomic mass, which is approximately $9.10938356 \times 10^{(-31)}$ kilograms or $9.10938356 \times 10^{(-28)}$grams.

To convert this value to other units:

B) 0.55 mg (milligrams): To convert from grams to milligrams, we multiply by 1000. Therefore, $9.10938356 \times 10^{(-28)} grams$ is approximately 0.55 mg.

C) 1.005 g (grams): This value is not correct. The molar mass of an electron is much smaller than 1 gram.

D) 0.184 mg (milligrams): This value is not correct. The correct answer is B) 0.55 mg.

Therefore, the mass of one mole of an electron is approximately 0.55 milligrams (0.55 mg).

**17. Correct option: (C)**

The charge of one mole of electrons can be calculated using Avogadro's number and the charge of a single electron. Avogadro's number (NA) is approximately $6.022 \times 10^{23}$, representing the number of entities (such as atoms, molecules, or ions) in one mole of a substance.

The charge of a single electron (e) is approximately $1.602 \times 10^{(-19)}$ coulombs.

To calculate the charge of one mole of electrons:

Charge = Avogadro's number × Charge of a single electron

Charge = $6.022 \times 10^{23} \times 1.602 \times 10^{(-19)}$coulombs

Simplifying this expression: Charge ≈ $9.65 \times 10^4$ coulombs

Therefore, the charge of one mole of electrons is approximately $9.65 \times 10^4$ coulomb that is also equal to 1Faraday.

**18. Correct option: (C)**

Among the given options, the heaviest particle is Neutron

Neutrons are subatomic particles that have a mass of approximately $1.675\times10^{(-27)}$ kilograms. They are found in the nucleus of atoms and are neutral in charge.

Comparatively, the mass of an electron is much smaller, approximately $9.10938356\times10^{(-31)}$ kilograms. Protons have a mass similar to that of neutrons, approximately$1.6726219\times10^{-27}$) kilograms, but they are positively charged. Positrons, which are the antiparticles of electrons, have the same mass as electrons but have a positive charge.

**19. Correct option: (A)**

The maximum magnetic moment among the given options is:

A) $Fe^{+2}$ (Iron ion with a +2 charge)

The magnetic moment of an atom or ion is determined by the number of unpaired electrons and their spins. Iron (Fe) has a maximum of 4 unpaired electrons in its 3d orbital when it forms the $Fe^{+2}$ ions. This configuration results in a higher magnetic moment compared to the other options. In contrast, $Ni^{+2}$ (Nickel ion with a +2 charge) has 2 unpaired electrons, $Ti^{+3}$ (Titanium ion with a +3 charge) has 1 unpaired electron, and $Sc^{+3}$ (Scandium ion with a +3 charge) has no unpaired electrons.

Therefore, the correct answer is $Fe^{+2}$.

**20. Correct option: (B)**

The maximum number of electrons that can be accommodated in an atom can be determined by using the formula: Maximum number of electrons = $2n^2$

Where n is the highest principal quantum number.

In this case, the highest principal quantum number value is 4. Plugging this value into the formula: Maximum number of electrons = $2\times(4)^2=2\times16=32$

Therefore, the correct answer is 32.

**21. Correct option: (A)**

The electrons identified by the quantum numbers 'n' and are as follows:

I) n = 4, = 1: This refers to an electron in the fourth principal energy level (n = 4) with an angular momentum quantum number (= 1).

II) n = 4, = 0: This refers to an electron in the fourth principal energy level (n = 4) with an angular momentum quantum number (= 0).

III) n = 3, = 2: This refers to an electron in the third principal energy level (n = 3) with an angular momentum quantum number (= 2).

IV) n = 3, = 1: This refers to an electron in the third principal energy level (n = 3) with an angular momentum quantum number (= 1).

The correct order, from smallest to largest value of is $IV < II < III < I$

**22. Correct option: (B)**

1. Electrons of Scandium and Titanium ions $Sc^{+3}=18, Ti^{+4}=18$

2. Electrons of Manganese and Ferrous ions $Mn^{+2}=23, Fe^{+2}=24$

3. Electrons of Ferrous and Cobalt $Fe^{+2}=24, Co^{+3}=24$

4. Electrons of Copper and Zinc $Cu^{+}=28, Zn^{+2}=28$

**23. Correct Answer is: (D)**

# 3.STOICHIOMETRY

## Solutions

## Mole Concept

## LEVEL-I

**1. Correct option: (C)**

One atomic mass unit (amu) is defined as $^{1}/_{12}{}^{th}$ the mass of a carbon-12 atom. It is approximately equal to $1.66 \times 10^{-24}$ grams.

**2. Correct option: (B)**

A) General Atomic Weight (GAW) = Weight of 1 gm atom of an element

B) Relative Atomic Mass (RAM) = $\frac{\text{Mass of 1 atom of that element}}{^{1}/_{12}{}^{th}\text{ mass of C}-12\text{ atom}}$

C) Molecular weight refers to the sum of the atomic weights of all the atoms in a molecule.

D) Equivalent weight = $\frac{\text{Molecular weight}}{n}$

where, n = number of equivalents of the substance involved in the reaction.

**3. Correct option: (B)**

A) 1 Dalton = 1 amu

B) Pascal is the SI unit of pressure, not related to the atomic mass unit.

C) 1 Avogram = $1.66 \times 10^{-24}$ g = 1 amu

D) One Aston = $^{1}/_{16}{}^{th}$ the mass of a oxygen-16 atom

**4. Correct option: (A)**

Since a hydrogen atom contains only one proton and no neutrons, its atomic mass is approximately equal to the mass of a proton.

The mass of a proton is approximately $1.66\times10^{-24}$ grams.

According to the definition of an atomic mass unit (amu), 1 amu is equal to $1.66\times10^{-24}$ grams.

Therefore, the mass of one hydrogen atom is approximately equal to 1 amu.

**5. Correct option: (D)**

Given: 1. 1 mole of sodium (Na)

Calculation:

- Moles of Na = $\frac{\text{Mass of Na}}{\text{Molecular weight of Na}}$

Mass of Na = 23 g/mol x 1 mol = 23 g.

The calculated mass of sodium as 23 g implies that options B and C are incorrect.

- No. of atoms of Na = No. of moles of Na x Avogadro's number ($N_A$)

$= 1 \times N_A = 6.023 \times 10^{23}$ atoms

The fact that elemental sodium (Na) exists as individual atoms, not molecules, renders option A incorrect, making option D the correct answer.

**6. Correct option: (C)**

Given: 1. Mass of calcium carbonate ($CaCO_3$) = 50 g

To find: Number of oxygen atoms = ?

Calculation:

- Moles of $CaCO_3$ = $\frac{\text{Mass of } CaCO_3}{\text{Molecular weight of } CaCO_3}$

Moles of $CaCO_3$ = $\frac{50\text{ g}}{100\text{ g/mol}}$ = 0.5 mol

- Since there are 3 oxygen atoms in one molecule of calcium carbonate ($CaCO_3$), the moles of oxygen will be three times the moles of calcium carbonate.

Moles of O = 3 × Moles of $CaCO_3$ = 3 × 0.5 mol = 1.5 mol

- Number of oxygen atoms = Moles of O × Avogadro's number

$= 1.5 \text{ mol} \times 6.023 \times 10^{23}$

$= 9.035 \times 10^{23}$

**7. Correct option: (B)**

Given: 1. Abundance of $^{54}Fe$ = 5%

2. Abundance of $^{56}Fe$ = 90%

3. Abundance of $^{57}Fe$ = 5%

To find: Atomic mass of Fe = ?

Calculation:

- Using weighted average formula for atomic mass calculation,

Atomic mass = $\frac{(^{54}\text{Fe Mass} \times {}^{54}\text{Fe Abundance}) + (^{56}\text{Fe Mass} \times {}^{56}\text{Fe Abundance}) + (^{57}\text{Fe Mass} \times {}^{57}\text{Fe Abundance})}{100}$

Atomic mass = $\frac{(54 \times 5) + (56 \times 90) + (57 \times 5)}{100}$ = 55.95

**8. Correct option: (B)**

Given: 1. Standard Temperature and pressure (STP) condition

To find: Molar volume of gas = ?

Calculation:

At STP, the temperature is 0 degrees Celsius (273.15 Kelvin) and the pressure is 1 atmosphere (atm).

- Calculating the molar volume of gas using the ideal gas equation.

$PV = nRT$

$1 \times V = 1 \times 0.08206 \times 273.15$

Molar volume of gas = V = 22.414 L

**9. Correct option: (D)**

Given: 1. Volume of $N_2$ gas = 1 cc = 0.001 L

2. Normal Temperature and pressure (NTP) condition

To find: Number of nitrogen molecules = ?

Calculation:

At NTP, the temperature is 25°C and pressure is 1 atm.

- Ideal gas law

No. of moles = n = $\frac{PV}{RT}$

n = $\frac{1\ atm \times 0.001L}{0.08206\ {}^{L.atm}/_{mol.K} \times 293K}$ = 4.1 x $10^{-5}$

- Number of nitrogen molecules = n × Avogadro's number

$= 4.1 \times 10^{-5} \times (6.023 \times 10^{23})$

$= 2.67 \times 10^{19}$

**10. Correct option: (D)**

Given: 1. Mass of nitrogen = 7g

2. Mass of oxygen = 8g

3. Standard Temperature and pressure (STP) condition

To find: Volume of mixture = ?

Calculation:

- Moles of nitrogen = $\frac{\text{Mass of nitrogen}}{\text{Molecular weight of nitrogen}} = \frac{7}{28} = \frac{1}{4}$
- Moles of oxygen = $\frac{\text{Mass of oxygen}}{\text{Molecular weight of oxygen}} = \frac{8}{32} = \frac{1}{4}$
- Total moles of mixture = moles of $N_2$ + moles of $O_2$

$= 0.25 + 0.25 = 0.5$

- Ideal gas law,

$PV = nRT$

$1 \times V = 0.5 \times 0.08206 \times 273.15$

V ~ 11220 ml

**11. Correct option: (D)**

Given: 1. Mass of $NO_2$ = 23g

Calculation:

- Calculating molecules of $NO_2$

Moles of $NO_2$ = $\frac{\text{Mass of NO2}}{\text{Molecular weight of NO2}} = \frac{23}{46} = 0.5$

No. of molecules of $NO_2$ = Moles of $NO_2$ x Avogadro's number = 0.5 x $N_A$

A) 8g of oxygen

- Moles of $O_2 = \frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{8}{32} = 0.25$
- No. of molecules of $O_2$ = Moles of $O_2$ x Avogadro's number = 0.25 x $N_A$

B) 28 g of carbon

- Moles of $C = \frac{\text{Mass of C}}{\text{Molecular weight of C}} = \frac{28}{12} = 2.33$
- No. of molecules of C = Moles of C x Avogadro's number = 2.33 x $N_A$

C) 16g of $SO_2$

- Moles of $SO_2 = \frac{\text{Mass of SO2}}{\text{Molecular weight of SO2}} = \frac{16}{64} = 0.25$
- No. of molecules of $SO_2$ = Moles of $SO_2$ x Avogadro's number = 0.25 x $N_A$

D) 22 g of $CO_2$

- Moles of $CO_2 = \frac{\text{Mass of CO2}}{\text{Molecular weight of CO2}} = \frac{22}{44} = 0.5$
- No. of molecules of $CO_2$ = Moles of $CO_2$ x Avogadro's number = 0.5 x $N_A$

**12. Correct option: (D)**

Given: 1. Mass of $CO_2$ = 44g

To find: Moles of oxygen atom = ?

Calculation:

- Moles of $CO_2 = \frac{\text{Mass of CO2}}{\text{Molecular weight of CO2}} = \frac{44}{44} = 1$
- 1 mole of $CO_2$ contains 2 moles of oxygen atoms

**13. Correct option: (A)**

Given: 1. 20 g mass of $H_2$

2. Standard Temperature and pressure (STP) condition

To find: Volume of $H_2$ = ?

Calculation:

- Moles of $H_2 = \frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{20}{2} = 10$
- Ideal gas law,

$$V = \frac{nRT}{P}$$

$$V = \frac{10 \times 0.08206 \ {}^{L.atm}/_{mol.K} \times 273K}{1 \ atm} = 224 \text{ L}$$

**14. Correct option: (C)**

Given: 1. Mass of He = 2g

2. Mass of $N_2$ = 7g

3. Standard Temperature and pressure (STP) condition

To find: Total volume of mixture = ?

Calculation:

- Moles of He = $\frac{\text{Mass of He}}{\text{Molecular weight of He}} = \frac{2}{4} = 0.5$
- Moles of $N_2$ = $\frac{\text{Mass of N2}}{\text{Molecular weight of N2}} = \frac{7}{28} = 0.25$
- Total moles of mixture = moles of $N_2$ + moles of He

  = 0.25 + 0.5 = 0.75
- Ideal gas law,

$PV = nRT$

$1 \times V = 0.75 \times 0.08206 \times 273.15$

V = 16.8 L

**15. Correct option: (B)**

Balancing the chemical reaction, $2\ CO + 1\ O_2 \rightarrow 2CO_2$

According to the balanced equation, 2 molecules of CO react with 1 molecule of $O_2$ to form 2 molecules of $CO_2$

**16. Correct option: (C)**

Calculation:

A) 28 gms of CO

- Moles of CO = $\frac{\text{Mass of CO}}{\text{Molecular weight of CO}} = \frac{28}{28} = 1$
- No. of molecules of CO = Moles of CO x Avogadro's number = $N_A$

B) 46 gms of $C_2H_5OH$

- Moles of $C_2H_5OH$ = $\frac{\text{Mass of C2H5OH}}{\text{Molecular weight of C2H5OH}} = \frac{46}{46} = 1$
- No. of molecules of $C_2H_5OH$ = Moles of $C_2H_5OH$ x Avogadro's number = $N_A$

C) 36 gms of $H_2O$

- Moles of $H_2O$ = $\frac{\text{Mass of H2O}}{\text{Molecular weight of H2O}} = \frac{36}{18} = 2$
- No. of molecules of $H_2O$ = Moles of $H_2O$ x Avogadro's number = $2N_A$

D) 54 gms of $N_2O_5$

- Moles of $N_2O_5$= $\frac{\text{Mass of N2O5}}{\text{Molecular weight of N2O5}} = \frac{54}{108} = 0.5$
- No. of molecules of $N_2O_5$= Moles of $N_2O_5$ x Avogadro's number = 0.5 $N_A$

**17. Correct option: (A)**

Given: 1. 5 g-atom of Fe

To find: mass of Fe = ?

Calcultion:

- 5 g-atom of Fe represents 5 mol atom of Fe
- Moles of Fe = $\frac{\text{Mass of Fe}}{\text{Molecular weight of Fe}}$

  Mass of Fe = 5 mol x 56 g/mol = 280g

**18. Correct option: (C)**

Calculation:

A) 2g Hydrogen

- Moles of $H_2$ = $\frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{2}{2} = 1$
- No. of molecules of $H_2$ = Moles of $H_2$ x Avogadro's number = $N_A$

B) 16g Oxygen

- Moles of $O_2$ = $\frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{16}{32} = 0.5$
- No. of molecules of $O_2$ = Moles of $O_2$ x Avogadro's number = $0.5N_A$

C) 4g oxygen

- Moles of $O_2$ = $\frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{4}{32} = 0.125$
- No. of molecules of $O_2$ = Moles of $O_2$ x Avogadro's number = $0.125N_A$

D) 16g $CO_2$

- Moles of $CO_2$ = $\frac{\text{Mass of CO2}}{\text{Molecular weight of CO2}} = \frac{16}{44} = 0.36$
- No. of molecules of $CO_2$ = Moles of $CO_2$ x Avogadro's number = $0.36N_A$

**19. Correct option: (B)**

Given: 1. Vapour density of $N_2O_4$ = 46

To find: Molar mass of $N_2O_4$ = ?

Calculation:

- Vapour Density of $N_2O_4$ = $\frac{\text{Molar mass of N2O4}}{\text{Molar mass of } H_2}$

Molar mass of $N_2O_4$ = 46 x 2 = 92g

**20. Correct option: ()**

# LEVEL-II

**1. Correct option: (A)**

Given: No. of He atoms = $6.022 \times 10^{23}$

To find: Mass of He = ?

Calculation:

- No. of atoms of He = Moles of He x Avogadro's number

  $6.022 \times 10^{23}$ = Moles of He x $6.022 \times 10^{23}$

  Moles of He = 1

- Moles of He = $\frac{\text{Mass of He}}{\text{Molecular weight of He}}$

  Mass of He = 4g/mol x 1 mol = 4g

**2. Correct option: (B)**

Calculation:

A) One gram atom of iron

- 1 g-atom of Fe represents 1 mole of Fe
- Mass of Fe = No. of moles of Fe x M.W of Fe = 1 x 56 = 56g

B) 5 mole of $N_2$

- Mass of $N_2$ = No. of moles of $N_2$ x M.W of $N_2$ = 5 x 28 = 140g

C) $10^{24}$ carbon atoms

- No. of C-atoms = No. of moles of C x $N_A$

  $10^{24}$ = No. of moles of C x $6.022 \times 10^{23}$

No. of moles of C = 1.66

- Mass of C = moles of C x M.W of C = 1.66 x 12 = 19.93g

D) 44.8 lit of He at STP

- No. of moles = n = $\frac{PV}{RT}$

  $= \frac{1\ atm \times 44.8\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K} = 2$

- Mass of He = moles of He x M.W of He = 2 x 4 = 8g

**3. Correct option: (A)**

Given: 1. Number of atoms of gas = $6 \times 10^{24}$

2. Mass of gas = 200 g

3. Homo diatomic gas ($X_2$)

To find: Molar mass of gas = ?

Calculation:

- Gas is homodiatomic which means that it consists of two identical atoms in a molecule.
- If 1 molecule has 2 gas atoms, then there are $3 \times 10^{24}$ molecules in $6 \times 10^{24}$ atoms.
- No. of molecules of gas = Moles of gas x Avogadro's number

  Moles of gas = $\frac{3 \times 10^{24}}{6.023 \times 10^{23}} = 4.98$

- Moles of gas = $\frac{\text{Mass of gas}}{\text{Molecular weight of gas}}$

  Molecular weight of gas = $\frac{200}{4.98} \sim 40$

**4. Correct option: (D)**

A mole is defined as the chemical substance contains the same number of chemical units present in $^{1}/_{12}{}^{th}$ mass of C-12 isotope.

**5. Correct option: (B)**

Given: 1. 1g of hydrogen contains 6 x $10^{23}$ atoms

To find: Atoms in 4g of $H_2$ = ?

Calculation:

- Moles of $H_2 = \frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{4}{2} = 2$
- Atoms of $H_2$ = No. of moles of $H_2$ x Avogadro's number

  = 2 x 6.023 x $10^{23}$ = 12 x $10^{23}$ atoms

**6. Correct option: (A)**

Given: 1. Mass of $H_2$ = 100g

2. Mass of $O_2$ = 100g

Reaction: $2H_2 + O_2 \rightarrow 2H_2O$

To find: Mass of $H_2O$ formed = ?

Calculation:

- Moles of $H_2 = \frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{100}{2} = 50$
- Moles of $O_2 = \frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{100}{32} = 3.125$
- 1 mole of $O_2$ reacts with 2 moles of $H_2$ to form 2 moles of $H_2O$

  Since, $O_2$ is the limiting agent

  Moles of $H_2O$ formed = 2 x moles of $O_2$ = 2 x 3.125 = 6.25
- Moles of $H_2O = \frac{\text{Mass of H2O}}{\text{Molecular weight of H2O}}$

  Mass of $H_2O$ = 6.25 x 18 = 112.5g

**7. Correct option: (B)**

Given:1. Atomic weight of X = 32

2. Vapour density = 32

3. Volume before and after combustion is equal

Reaction: X + $O_2$ → Y

To find: Atoms of oxygen =?

Calculation:

- Vapour density of Y = $\frac{\text{Molar mass of Y}}{\text{Molar mass of } H_2}$

  Molar mass of Y = 32 x 2 = 64 g/gmol

**8. Correct option: (9600g or 9.6 kg)**

Given: 1. Mass of ethylene = 2.8kg

To find: Mass of oxygen required = ?

Calculation

- Moles of ethylene = $\frac{\text{Mass of ethylene}}{\text{Molecular weight of ethylene}} = \frac{2800}{28} = 100$
- 1 mole of $C_2H_4$ reacts with 3 moles of $O_2$ to form 2 moles of $CO_2$ and 2 moles of $H_2O$

  Therefore,

  We need 300 moles of $O_2$ to react with 100 moles of $C_2H_4$
- Moles of $O_2$ = $\frac{\text{Mass of O2}}{\text{Molecular weight of O2}}$

  Mass of $O_2$ = 300 x 32 = 9600g or 9.6 kg

**9. Correct option: (BCD)**

Calculation:

- Absolute mass of Na = Relative mass of Na x amu

  = 23 amu

  = 23 x 1.66 x $10^{-24}$ g (1 amu = 1.66 x $10^{-24}$ g)

  = 23 x 1.66 x $10^{-27}$ kg (1 kg = 1000 g)

**10. Correct option: (A)**

A) One mole of Helium gas

- No. of atoms of He = No. of moles of He x $N_A$ = $N_A$

B) 22.4 ml of $CO_2$ at STP

- No. of moles = n = $\frac{PV}{RT}$

  = $\frac{1\ atm \times 22.4 \times 10^{-3}\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K}$ = 9.993 x $10^{-4}$

- No. of atoms of $CO_2$ = No. of moles of $CO_2$ x $N_A$ = 9.993 x $10^{-4}$ $N_A$

C) 11.2 litre of Hydrogen gas

- Moles (n) = $\frac{\text{Volume of gas at STP}}{22.4\ \text{L}} = \frac{11.2\ \text{L}}{22.4\ \text{L}} = 0.5$
- No. of atoms of $H_2$ = No. of moles of $H_2$ x $N_A$ = 0.5 $N_A$

D) 3.2g of methane

- Moles of $CH_4$ = $\frac{\text{Mass of } CH_4}{\text{Molecular weight of } CH_4} = \frac{3.2}{16} = 0.2$
- No. of atoms of $CH_4$ = No. of moles of $CH_4$ x $N_A$ = 0.2 $N_A$

**11. Correct option: (ABCD)**

A) 1g of hydrogen atom

$$\text{Moles of H} = \frac{\text{Mass of H}}{\text{Molecular weight of H}} = \frac{1}{1} = 1$$

B) $6.023 \times 10^{23}$ molecules of oxygen

No. of molecules of oxygen = No. of moles of oxygen x $N_A$

$$\text{Moles of oxygen} = \frac{6.023 \times 10^{23}}{6.023 \times 10^{23}} = 1$$

C) 2g of Hydrogen molecules

$$\text{Moles of } H_2 = \frac{\text{Mass of } H_2}{\text{Molecular weight of } H_2} = \frac{2}{2} = 1$$

D) 32 g of oxygen molecules

$$\text{Moles of } O_2 = \frac{\text{Mass of } O_2}{\text{Molecular weight of } O_2} = \frac{32}{32} = 1$$

**12. Correct option: (AB)**

Given: 1. Mass of Ammonia = 34g

Reaction: $N_2 + 3H_2 \rightarrow 2NH_3$

Calculation:

- $\text{Moles of } NH_3 = \frac{\text{Mass of } NH_3}{\text{Molecular weight of } NH_3} = \frac{34}{17} = 2$
- Based on stoichiometric ratio,

  To form 2 mole $NH_3$, we need 1 mole of $N_2$ & 3 mole of $H_2$
- $\text{Moles of } N_2 = \frac{\text{Mass of } N_2}{\text{Molecular weight of } N_2}$

  Mass of $N_2$ = 1 x 28 = 28g
- $\text{Moles of } H_2 = \frac{\text{Mass of } H_2}{\text{Molecular weight of } H_2}$

  Mass of $H_2$ = 3 x 2 = 6g

**13. Correct option: (AC)**

Given: 1. Moles of $KClO_3$ = 2

Reaction: $2KClO_3 \rightarrow 2KCl + 3O_2$

Calculation:

- Based on stoichiometric ratio,

2 moles of $KClO_3$ on thermal heating gives 2 moles of KCl and 3 moles of $O_2$

**14. Correct option: (A)**

Given: 1. No. of $H_2$ gas = $3.011 \times 10^{23}$

2. Normal temperature and pressure (NTP) condition To find: Volume of $H_2$ gas = ?

Calculation:

- No. of $H_2$ gas atoms = No. of moles of $H_2$ gas x $N_A$

$$\text{Moles of } H_2 \text{ gas} = \frac{3.011 \times 10^{23}}{6.023 \times 10^{23}} = 0.5$$

- Ideal gas law, PV = nRT

$$V = \frac{0.5 \text{ mol} \times 0.08206 \, {}^{\text{L.atm}}/_{\text{mol.K}} \times 273\text{K}}{1 \text{ atm}} = 11.19 \text{ L}$$

**15. Correct option: (A)**

Given: Mass of 1 atom of an element 'X' = 6.64 x $10^{-23}$ g

To find: Molecular weight of $X_2$ = ?

Calculation:

- Atomic weight of 'X' = Mass of 1 atom of 'X' x Avogadro's number

  Atomic weight of 'X'

  = 6.64 x $10^{-23}$ g x 6.022 x $10^{23}$ atoms/mol

  Atomic weight of 'X' = 40 g/mol
- Molecular weight of $X_2$ = 2 x Atomic weight of 'X' Molecular weight of $X_2$

  = 2 x 40 g/mol

  Molecular weight of $X_2$ = 80 g/mol

**16. Correct option: (B)**

Given: 1. Mass of $O_2$ gas = 1kg at 273 °C and 2 atm

To find: Gram-atoms of oxygen = ?

Calculation:

- $\text{Moles of } O_2 = \frac{\text{Mass of } O_2}{\text{Molecular weight of } O_2} = \frac{1000}{32} = 31.25$

1 mole of $O_2$ = 1 g-molecule of $O_2$ ….(given in the passage)

& 1 g-molecules of $O_2$ has 2g-atom of oxygen

Therefore, 62.5 g-atom of oxygen will be present in 1 kg $O_2$.

**17. Correct option: (B)**

Given: 1. Mass of $O_2$ = 16g

2. Standard temperature and pressure (STP) condition

To find: Volume of $O_2$ = ?

Calculation:

- $\text{Moles of } O_2 = \frac{\text{Mass of } O_2}{\text{Molecular weight of } O_2} = \frac{16}{32} = 0.5$

- In the passage it is given that,

  1 mole of gas at STP has 22.4 L volume occupiedBased on the correlation

  0.5 mole of gas at STP should have 11.2 L volume occupied.

**18. Correct option: (C)**

Given: 1. Volume of $N_2$ gas = 67.2 L

2. Standard temperature and pressure (STP) condition

To find: Mass of $N_2$ gas = ?

Calculation:

- No. of moles of $N_2$ gas = n = $\frac{PV}{RT}$

$$= \frac{1\ atm \times 67.2\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K} = 2.99$$

- Mass of $N_2$ gas = Moles of $N_2$ x M.W of $N_2$ = 2.99 x 28 = 83.99g

**19. Correct option: (a-R, b-S, c-Q, d-P)**

a) Vapour Density of gas $= \frac{\text{Molar mass of gas}}{\text{Molar mass of } H_2}$

Molar mass of gas = 2 x Vapour Density of gas

b) Avogadro number = $6.023 \times 10^{23}$

c) 1 mole of gas occupies in volume occupies 22.4 L or 22400 $cm^3$ volume
1 g-atom of carbon is equal to 12g because it represents the atomic mass of carbon expressed in g

**20. Correct option: (a-R,S b-P c-T,S, d-Q)**

a) 32g of $SO_2$

- Moles of $SO_2 = \frac{\text{Mass of } SO_2}{\text{Molecular weight of } SO_2} = \frac{32}{64} = 0.5$
- 0.5 mole of $SO_2$ at STP has 11.2 L volume occupied.

b) 22.4 lit Hydrogen at STP

- No. of moles of hydrogen = n = $\frac{PV}{RT}$

$$= \frac{1\ atm \times 22.4\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K} = 1$$

- Mass of hydrogen = Moles of hydrogen x M.W of hydrogen = 1 x 2 = 2g

c) 0.5 mole of $CH_4$

- Mass of $CH_4$ = Moles of $CH_4$ x M.W of $CH_4$ = 0.5 x 16 = 8g
- 0.5 mole $CH_4$ of at STP has 11.2 L volume occupied.

d) 32g methane

- Moles of $CH_4 = \frac{\text{Mass of CH4}}{\text{Molecular weight of CH4}} = \frac{32}{16} = 2$
- 2 mole of $CH_4$ at STP has 44.8 L volume occupied.

# MOLARITY AND MOLALITY

**1. Correct option: (A)**

A 10% (w/v) solution means that 10 grams of solute is dissolved in 100 milliliters of the solution. Since 1 milliliter (ml) is equivalent to 1 cubic centimeter (cc), the correct option is A).

**2. Correct option: (D)**

A) Molar solution: A solution that contains one mole of solute dissolved in one liter of solvent.

B) Normal solution: A solution that contains one equivalent weight of a solute dissolved inone liter of solvent.

C) "Mole solution" and "molar solution" are synonymous terms, both denoting a solutionwhere the solute concentration is expressed in moles per liter

D) Standard solution: A standard solution is a solution with a precisely known concentrationof a solute.

**3. Correct option: (C)**

A) $\text{Volume percentage} = \frac{\text{Volume of solute}}{\text{volume of solution}} \text{ x } 100$

B) $\text{Weight percentage} = \frac{\text{mass of solute}}{\text{mass of solution}} \text{ x } 100$

C) When a solute is present in trace quantities, it is convenient to express the concentration in parts per million (ppm) or parts per billion (ppb).

D) $\text{Normality} = \frac{\text{Number of equivalents of solute}}{\text{Volume of solution in liters}}$

**4. Correct option: (B)**

The strength of a solution refers to the concentration or amount of solute present in a givenvolume of the solution, and it is commonly expressed in units of grams per liter (g/L)

**5. Correct option: (C)**

A 10% (w/w) solution means that 10 grams of solute is dissolved in 100 grams of thesolution.

**6. Correct option: (B)**

The strength of a solution refers to the concentration or amount of solute present in a givenvolume of the solution, and it is commonly expressed in units of grams per liter (g/L)

**7. Correct option: (A)**

$$\text{Strength of solution} = \frac{\text{Mass of solute in gms}}{\text{Volume of solution in litres}}$$

This formula also represents the concentration of solute in solution.

**8. Correct option: (C)**

Given: 1. Volume of solute = 25ml
2. Volume of solvent = 250 ml

To find: % by volume = ?

Calculation:

- $\text{Volume percentage} = \frac{\text{Volume of solute}}{\text{volume of solute+volume of solvent}} \text{ x } 100 = \frac{25}{25+250} \text{ x } 100 = 9.09$

**9. Correct option: (B)**

A) $gL^{-1}$: Unit of concentration of solution

B) $moleL^{-1}$: Unit of molarity

C) $Equivalent.L^{-1}$: Unit of normality

D) $mol.kg^{-1}$: Unit of molality

**10. Correct option: (A)**

Molarity provides a convenient and commonly utilized measure of concentration in various scientific disciplines.

**11. Correct option: (B)**

**12. Correct option: (B)**

Given: Dilution factor = x

Calculation:

$$\text{New Molarity} = \frac{\text{Old Molarity}}{\text{Dilution factor}} = \frac{\text{Old Molarity}}{x}$$

As evident from the equation, when a solution is diluted by a factor of x, its molarity decreases proportionally by the same factor x.

**13. Correct option: (B)**

Calculation:

- We define %w/v as

$$\%w/v = \frac{\text{mass of solute in g}}{\text{Volume of solution in ml}} \times 100 \qquad \text{.....(1)}$$

- Converting mass of solute to number of moles of solute in eq(1)

Mass of solute = moles of solute x M.W of solute

$$\%w/v = \frac{\text{moles of solute x M.W of solute}}{\text{Volume of solution in ml}} \times 100 \qquad \text{.....(2)}$$

- Converting solution volume from ml to litres in eq(2)

$$\%w/v = \frac{\text{moles of solute x M.W of solute}}{\text{Volume of solution in L} \times 10^3} \times 100 \qquad \text{.....(3)}$$

- We know that,

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

Substituting the molarity formula in eq(3)

$$\%w/v = \frac{\text{Molarity x M.W of solute x 100}}{10^3} \qquad \text{.....(4)}$$

- Rearranging equation (4)

$$\text{Molarity (M)} = \frac{\%w/v \times 10}{\text{M.W of solute}}$$

**14. Correct option: (A)**

Calculation:

- We define %w/W as

$$\%w/W = \frac{\text{mass of solute in g}}{\text{mass of solution in g}} \times 100 \quad \text{.....(1)}$$

- Converting mass of solute to number of moles of solute in eq(1)

Mass of solute = moles of solute x M.W of solute

$$\%w/W = \frac{\text{moles of solute x M.W of solute}}{\text{mass of solution in g}} \times 100 \quad \text{.....(2)}$$

- Converting solution mass to volume in eq(2)

Mass of solution (g) = density of solution (g/ml) x volume of solution (ml)

$$\%w/W = \frac{\text{moles of solute x M.W of solute}}{\rho\,(\frac{g}{ml}) \times \text{Volume of solution(ml)}} \times 100 \quad \text{.....(3)}$$

- Converting solution volume from ml to litres in eq(3)

$$\%w/W = \frac{\text{moles of solute x M.W of solute}}{\rho \text{ x Volume of solution in L x } 10^3} \times 100 \quad \text{.....(4)}$$

- We know that,

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

Substituting the molarity formula in eq(4)

$$\%w/W = \frac{\text{Molarity x M.W of solute x 100}}{\rho \text{ x } 10^3} \quad \text{.....(5)}$$

- Rearranging equation (4)

$$\text{Molarity (M)} = \frac{\%\frac{w}{W} \text{ x } \rho \text{ x } 10}{\text{M.W of solute}}$$

**15. Correct option: (A)**

Calculation:

- Finding the mass of water in 1 litre

The density of water = 1 kg/L. Therefore, mass of water in 1 litre is 1kg.

- $\text{Moles of } H_2O = \frac{\text{Mass of } H_2O}{\text{M.W of } H_2O} = \frac{1000}{18} = 55.55 \text{ mol}$

- $\text{Molarity of water} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}} = \frac{55.55 \text{ mol}}{1 \text{ L}} = 55.55\text{M}$

**16. Correct option: (D)**

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litre}}$$

Since 1 Litre (L) is equivalent to 1000 millilitre (ml), the correct option is D).

**17. Correct option: (B)**

Given: 1. Mass of $Na_2CO_3$ = 5.3g
2. Volume of solution = 1L
To find: Molarity of solution = ?
Calculation:

- $\text{Moles of } Na_2CO_3 = \frac{\text{Mass of Na2CO3}}{\text{Molecular weight of Na2CO3}} = \frac{5.3}{106} = 0.05$
- $\text{Molarity of solution} = \frac{\text{Number of moles of Na2CO3}}{\text{Volume of solution in litre}} = \frac{0.05 \text{ mol}}{1 \text{ L}} = 0.05\text{M}$

**18. Correct option: (B)**

Given: 1. Volume of solution = 2L
2. Molarity of solution = 0.5M
To find: Moles of NaOH = ?
Calculation:

$$\text{Molarity of solution} = \frac{\text{Number of moles of NaOH}}{\text{Volume of solution in litre}}$$

Number of moles of NaOH = 0.5M x 2L = 1

**19. Correct option: (A)**

Given: 1. Moles of solute = $10 \times 10^{-3}$
2. Molarity of solution = 0.08M
To find: Volume of solution = ?
Calculation:

- $\text{Molarity of solution} = \frac{\text{Moles of solute}}{\text{Volume of solution}}$

$$\text{Volume of solution} = \frac{10 \times 10^{-3}}{0.08} = 125 \text{ ml}$$

**20. Correct option: (**1.6 L**)**

Given: 1. $M_1$ = 1M

2. $M_2$ = 5M

3. M = 2M

4. V = 2 L

To find: $V_1$ and $V_2$

Calculation:

- Dilution Equation, $M_1V_1=M_2V_2$

$1M \times V_1 = 5M \times V_2$

$V_1 = 5 \times V_2$

- $MV = M_1V_1 + M_2V_2$

$4 = 1 \times 5\ V_2 + 5 \times V_2$

$V_2$ = 0.4 L

$V_1 = V - V_2 = 2 - 0.4 = 1.6$ L

## LEVEL-II

**1. Correct option: (C)**

Given: 1. %w/v = 10%

2. mole of glucose = 1

To find: volume = ?

Calculation:

- $\text{Moles of glucose} = \frac{\text{Mass of glucose}}{\text{Molecular weight of glucose}}$

  $\text{Mass of glucose} = 1 \times 180 = 180 \text{ g}$

- $\%w/v = \frac{\text{Mass of glucose}}{\text{Volume of solution}} \times 100$

  $10 = \frac{180}{\text{Volume of solution}} \times 100$

  $\text{Volume of solution} = 1800 \text{ ml}$

**2. Correct option: (D)**

Given: 1. $V_1 = 750$ ml

2. $M_1 = 0.5M$

3. $V_2 = 250$ ml

4. $M_2 = 2M$

To find: M =?

Calculation:

- $V = V_1 + V_2 = 750 + 250 = 1000$ ml
- Dilution formula, $MV = M_1V_1 + M_2V_2$

  $M \times 1000 = 0.5 \times 750 + 2 \times 250$

  $M = 0.875M$

**3. Correct option: (C)**

Given: 1. $V_1 = 250$ ml

2. $M_1 = 0.25M$

3. $V_2 = 500$ ml

To find: $M_2$ =?

Calculation:

- $M_1V_1 = M_2V_2$

  $0.25 \times 250 = M_2 \times 500$

  $M_2 = 0.125$ M

**4. Correct option: (D)**

Given: 1. $V_1 = 250$ ml

2. $M_1 = 6$ M

3. $V_2 = 650$ ml

4. $M_2 = 3$ M

5. M = 3 M

To find: $V_{water}$ = ?

Calculation:

- $V_{mix} = V_1 + V_2 = 650 + 250 = 900$ ml
- $M_{mix} V_{mix} = M_1V_1 + M_2V_2$

  $M_{mix} \times 900 = 6 \times 250 + 3 \times 650$

$M_{mix}$ = 3.83 M

- $M_f V_f = M_{mix} V_{mix}$

  3 x $V_f$ = 3450

  $V_f$ = 1150 ml

- $V_f = V_{mix} + V_{water}$

  $V_{water}$ = 250 ml

**5. Correct option: (B)**

Given: 1. Molarity of solution = 0.8 M

2. Moles of solute = 0.1

To find: Volume of solution = ?

Calculation:

- $\text{Molarity of solution} = \frac{\text{Moles of solute}}{\text{Volume of solution}}$

  $\text{Volume of solution} = \frac{0.1}{0.8} = 0.125$ L o 125ml

**6. Correct option: (B)**

Given: 1. Molarity of solution = 0.1M

2. Volume of solution = 400 ml
3. Molecular weight of $H_2SO_4$ = 98

To find: Mass of $H_2SO_4$ = ?

Calculation:

- $\text{Molarity of solution} = \frac{\text{Moles of } H_2SO_4}{\text{Volume of solution}}$

  $\text{Moles of } H_2SO_4 = 0.1 \text{ M} \times 0.4 \text{ L} = 0.04$

- $\text{Moles of } H_2SO_4 = \frac{\text{Mass of } H_2SO_4}{\text{Molecular weight of } H_2SO_4}$

  $\text{Mass of } H_2SO_4 = 0.04 \times 98 = 3.92\text{g}$

**7. Correct option: (A)**

Given: 1. Mass of urea = 6g

2. Volume of solution = 200 ml
3. Molecular weight of urea = 60

To find: Molarity of solution = ?

- $\text{Moles of urea} = \frac{\text{Mass of urea}}{\text{Molecular weight of urea}} = \frac{6}{60} = 0.1$

- $\text{Molarity of solution} = \frac{\text{Moles of urea}}{\text{Volume of solution}} = \frac{0.1}{0.2} = 0.5\text{M}$

Calculation:

**8. Correct option: (AD)**

Explanation:

A) A decimolar solution refers to a solution with a concentration of 0.1 M (0.1 moles per liter).

B) A molar solution is a solution with a concentration of 1 mole of solute per liter of solution, represented as 1 M.

C) A molal solution is a solution with a concentration of 1 mole of solute dissolved in 1kilogram of solvent, represented as 1 mol/kg or 1 m.

D) 0.1-gram molecule is equivalent to 0.1 mole

$$\text{Molarity of solution} = \frac{\text{Moles of urea}}{\text{Volume of solution}} = \frac{0.1}{1} = 0.1\text{M}$$

**9. Correct option: (ABCD)**

A) Weight percentage (w%)= $\frac{\text{mass of solute}}{\text{mass of solution}}$ x 100

B) Volume percentage (v%) = $\frac{\text{Volume of solute}}{\text{volume of solution}}$ x 100

C) %w/v = $\frac{\text{mass of solute in g}}{\text{Volume of solution in ml}}$ x 100

D) %v/W = $\frac{\text{Volume of solute in mL}}{\text{Weight of solution in g}}$ x 100

**10. Correct option: (BC)**

A) $\frac{v}{v+V} \times 100$ is the formula for volume percentage (v%)

D) $\frac{M}{m+M} \times 100$ is the formula for weight percentage (w%)

**11. Correct option: (BC)**

B) Doubling the weight of the solvent will effectively decrease the concentration of the solute in the solution, resulting in a halved molarity.

C) Doubling the volume of the solvent while keeping the amount of solute constant will dilute the solution, resulting in a lower concentration and a halved molarity.

**12. Correct option: (ABC)**

A) 5.85g of sodium chloride in 100ml solution

$$\text{Molarity of solution} = \frac{\text{Moles of NaCl}}{\text{Volume of solution}} = \frac{5.85}{58.44 \text{ x } 0.1} = 1\text{M}$$

B) 0.32g of methanol in 10ml of solution

$$\text{Molarity of solution} = \frac{\text{Moles of methanol}}{\text{Volume of solution}} = \frac{0.32}{32.04 \text{ x } 0.01} = 1\text{M}$$

C) 111g of calcium chloride in 1000 ml of solution

$$\text{Molarity of solution} = \frac{\text{Moles of calcium chloride}}{\text{Volume of solution}} = \frac{111}{111 \text{ x } 1} = 1\text{M}$$

D) 4gm of NaOH in 1L solution

$$\text{Molarity of solution} = \frac{\text{Moles of NaOH}}{\text{Volume of solution}} = \frac{4}{40 \text{ x } 1} = 0.1\text{M}$$

**13. Correct option: (A)**

Given: 1. Molarity of solution = 2 M
2. Moles of LiF = 6

To find: Volume of solution = ?

Calculation:

- $\text{Molarity of solution} = \frac{\text{Moles of LiF}}{\text{Volume of solution}}$

$$\text{Volume of solution} = \frac{6}{2} = 3\text{L}$$

**14. Correct option: (B)**

Given: 1. Molarity of solution = 3 M

2. Mass of NaCl = 526g

To find: Volume of solution = ?

Calculation:

- Moles of NaCl $= \frac{\text{Mass of NaCl}}{\text{Molecular weight of NaCl}} = \frac{526}{58} \sim 9$ mol
- Molarity of solution $= \frac{\text{Moles of NaCl}}{\text{Volume of solution}}$

Volume of solution $= \frac{9}{3} = 3$L

**15. Correct option: (B)**

Given: 1. Volume of solution = $5.00 \times 10^2$ $cm^3$ = 0.5 L

2. Molarity of solution = 5 M

To find: Moles of $CaCl_2$ = ?

Calculation:

- Molarity of solution $= \frac{\text{Moles of CaCl2}}{\text{Volume of solution}}$

Moles of $CaCl_2$ = 5 x 0.5 = 2.5 moles

**16. Correct option: (A-S, B-R, C-Q, D-P)**

A) Weight to weight percent $= \frac{\text{Weight of solute}}{\text{Weight of solution}}$ x 100

B) Weight to volume percent $= \frac{\text{Weight of solute}}{\text{Volume of solution}}$ x 100

C) Volume to volume percent $= \frac{\text{Volume of solute}}{\text{Volume of solution}}$ x 100

D) Volume to weight percent $= \frac{\text{Volume of solute}}{\text{Weight of solution}}$ x 100

**17. Correct option: (A-Q, B-P, C-S, D-R)**

A) M/2 = Semi molar. A semi-molar solution typically refers to a solution with aconcentration of 0.5 M (0.5 moles per liter).

B) M/10 = Deci molar. A decimolar solution refers to a solution with a concentration of 0.1 M (0.1 moles per liter).

C) M/100 = Centi molar. A centimolar (cM) solution refers to a solution with a concentration of 0.01 moles per liter or 10 millimoles per liter.

D) M/1000 = Milli molar. A millimolar (mM) solution refers to a solution with a concentration of 0.001 moles per liter or 1 millimole per liter.

**18. Correct option: (2%)**

Given: 1. Mass of solute = 2g

2. Mass of solvent = 98g

To find: Mass percent = ?

Calculation:

Mass % $= \frac{\text{Mass of solute}}{\text{Mass of solute+Mass of solvent}} = \frac{2}{2+98} = 2\%$

**19. Correct option: (1g/mL)**

Given: 1. %w/w= 20%

2. Molarity of solution = 5 M

To find: Density of solution = ?

Calculation:

$$\text{Molarity (M)} = \frac{\%\frac{w}{W} \times \rho \times 10}{\text{M.W of NaOH}}$$

$$5 = \frac{20 \times \rho \times 10}{40}$$

$\rho = 1$ g/ml

**20. Correct option: (2M)**

Given: 1. Mass of $CaCl_2 = 11.1$g

2. Volume of solution = 100ml

Calculation:

- Moles of $CaCl_2 = \frac{11.1}{111} = 0.1$

Reaction: $CaCl_2$ (aq) → $Ca^{2+}$ (aq) + $2Cl^-$

$$\text{Molarity of solution} = \frac{\text{Moles of CaCl2}}{\text{Volume of solution}} = \frac{0.2}{0.1} = 2\text{M}$$

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## NORMALITY AND MOLE FRACTION

**1. Correct option: (A)**

$$\text{A) Normality} = \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$$

$$\text{B) Molality} = \frac{\text{No.of.moles of solute}}{\text{Mass of the solvent in kg}}$$

$$\text{C) Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution (L)}}$$

$$\text{D) Mole fraction} = \frac{\text{Number of moles of the component}}{\text{Total number of moles of all components in the solution}}$$

**2. Correct option: (D)**

A normal solution is a solution that contains 1 gram equivalent weight of solute per liter ofsolution. Since 1 litre is equivalent to 1000ml , the correct option is D).

**3. Correct option: (B)**

A) mole/kg: Unit of molality

B) gram equivalent/lit: Unit of normality

C) mol/lit: Unit of molarity

**4. Correct option: (D) 0.001 N)**

A millinormal solution is a solution with a concentration of 0.001 gram equivalent weight of solute per liter of solution, often denoted as "mN"

**5. Correct option: (A)**

A) $\text{Normality} = \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

B) $\text{Formality} = \frac{\text{Number of formula weights of solute}}{\text{Volume of solution(L)}}$

C) $\text{Molality} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent (kg)}}$

D) $\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution(L)}}$

**6. Correct option: (A)**

A) gram. eq. perlitre: Unit of normality

B) mol per lit: Unit of molarity

C) mol per kg: Unit of molality

**7. Correct option: (D)**

The formulas for calculating the equivalent weight of an element differ depending on the specific reactive species involved

- For acids and bases:

$$\text{Equivalent weight of acid} = \frac{\text{Mol.wt.of theacid}}{\text{Basicity}}$$

$$\text{Equivalent weight of base} = \frac{\text{Mol.wt.of the base}}{\text{Acidity}}$$

- $\text{General Equivalent weight} = \frac{\text{At.wt}}{\text{Valency}}$

**8. Correct option: (C)**

$$\text{Equivalent weight of base} = \frac{\text{Mol.wt.of the base}}{\text{Acidity}}$$

Where, Acidity = Number of moles of hydroxide ions (OH-) donated per molecule

**9. Correct option: (D)**

For ions, the formula for equivalent weight transforms slightly. In this case,valency is signified by the total ionic charges

**10. Correct option: (A)**

**11. Correct option: ()**

**12. Correct option: (C)**

Phosphoric acid is a triprotic acid. The basicity of $H_3PO_4$ is 3.

$$\text{Equivalent weight of Phosphoric acid} = \frac{\text{Mol.wt.of the phosphoric acid}}{\text{Basicity}} = \frac{\text{Mol.wt.of theacid}}{3}$$

**13. Correct option: (B)**

Oxalic acid is a diprotic acid. The basicity of oxalic acid is 2.

$$\text{Equivalent weight of oxalic acid} = \frac{\text{Mol.wt.of the oxalic acid}}{\text{Basicity}} = \frac{\text{Mol.wt.of theacid}}{2}$$

**14. Correct option: (D)**

In the context of a salt, the formula for equivalent weight undergoes a slight modification. Here, the valency is represented as the total number of either positive or negative charges.

**15. Correct option: (A)**

- $\text{Molarity} = \dfrac{\text{No.of moles of solute}}{\text{Volume of solution(L)}}$

$$= \frac{\text{Mass of solute}}{\text{Molecular mass of solute}} \text{ x } \frac{1}{\text{Volume of solution(L)}}$$

Similarly,

- $\text{Normality} = \dfrac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

$$= \frac{\text{Mass of solute}}{\text{Equivalent mass of solute}} \text{ x } \frac{1}{\text{Volume of solution}}$$

- $\dfrac{\text{Molarity}}{\text{Normality}} =$

$$\frac{\text{Mass of solute}}{\text{Molecular mass of solute}} \text{ x } \frac{1}{\text{Volume of solution}} \Bigg/ \frac{\text{Mass of solute}}{\text{Equivalent mass of solute}} \text{ x } \frac{1}{\text{Volume of solution}}$$

$$\frac{\text{Molarity}}{\text{Normality}} = \frac{\text{Equivalent mass of solute}}{\text{Molecular mass of solute}}$$

$$\text{Normality} = \text{Molarity x } \frac{\text{Molar mass of solute}}{\text{Equivalent mass of solute}}$$

**16. Correct option: (D)**

1) Unit of normality is gram. eq. $\text{litre}^{-1}$

2) $\text{Normality} = \text{Molarity x } \dfrac{\text{Mol.wt}}{\text{Eq.wt}}$

For dibasic acid, valency factor = 2

$$\text{Eq. wt} = \frac{\text{Mol.wt}}{2}$$

Normality = Molarity x 2

3) $\text{Normality} \times \dfrac{\text{GEW}}{\text{GMW}} = \text{Molality.}$

GEW: Gram equivalent weight

GMW: Gram molecular weight

4) The solution of molality is always greater than molarity. This is because molarity is calculated as mol per unit L (volume of solution), whereas molality is calculated as moles per unit Kg (i.e. mass of solvent). So, in any case, the mass of the solvent is less than the volume of the solution, and thus the molality is greater

**17. Correct option: (A)**

Given: 1) Mass of KOH = 35 g

2) Volume of solution = 400 ml

To find: Normality of solution: ?

Calculation:

- Gram equivalent $= \frac{35}{56} = 0.625$ g/eq

- Normality $= \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

$$= \frac{0.625 \times 1000}{400} = 1.56 \text{ N}$$

**18. Correct option: (A)**

Given: 1) Mass of oxalic acid = 31.5 g
2) Volume of solution = 2.5L
3) Mol.wt. of oxalic acid = 126

To find: Normality of solution: ?

Calculation:

- Gram equivalent $= \frac{31.5}{63} = 0.5$
- Normality $= \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

$$= \frac{0.5}{2.5} = 0.2 \text{ N}$$

**19. Correct option: (A)**

Given: 1) Percent of K in empirical formula = 26.6%
2) Percent of Cr in empirical formula = 35.4%
3) Percent of O in empirical formula = 38.1%

To find: Empirical formula = ?

Calculation:

Let the total mass be 100g

| Atoms | Mass | Moles | Ratio |
|---|---|---|---|
| **K** | 26.6 | $^{26.6}/_{39.1} = 0.68$ | $^{0.68}/_{0.68} = 1$ |
| **Cr** | 35.4 | $^{35.4}/_{52} = 0.68$ | $^{0.68}/_{0.68} = 1$ |
| **O** | 38.1 | $^{38.1}/_{16} = 2.38$ | $^{2.38}/_{0.68} = 3.5$ |

Empirical formula = $K_1Cr_1O_{3.5}$

Multiplying to get a whole

numberEmpirical formula =

$K_2Cr_2O_7$

**20. Correct option: (B)**

Given: 1) Percent of C in empirical formula = 10.4%
2) Percent of H in empirical formula = 0.84%
3) Percent of Cl in empirical formula = 89.12%

To find: Empirical formula = ?

Calculation:

Let the total mass be 100g

| Atoms | Mass | Moles | Ratio |
|---|---|---|---|
| C | 10.4 | $^{10.4}/_{12} = 0.8$ | $^{0.8}/_{0.8} = 1$ |
| H | 0.84 | $^{0.84}/_{1} = 0.84$ | $^{0.84}/_{0.8} = 1.05$ |
| Cl | 89.12 | $^{89.12}/_{35.5} = 2.5$ | $^{2.5}/_{0.8} = 3.125$ |

Empirical formula = $C_1H_1Cl_3$

**21. Correct option: (B)**

Given: 1) Percent of Fe in empirical formula = 69.94%

2) Percent of O in empirical formula = 30.06%

To find: Empirical formula = ?

Calculation:

Let the total mass be 100g

| Atoms | Mass | Moles | Ratio |
|---|---|---|---|
| Fe | 69.94 | $^{69.94}/_{55.85} = 1.25$ | $^{1.25}/_{1.25} = 1$ |
| O | 30.06 | $^{30.06}/_{16} = 1.88$ | $^{1.88}/_{1.25} = 1.5$ |

Empirical formula = $Fe_1O_{1.5}$

Multiplying to get a whole number

Empirical formula = $Fe_2O_3$

**22. Correct option: (B)**

**23. Correct option: (C)**

Given: Empirical formula mass = 30

Calculation:

- $n = \frac{\text{Molar mass}}{\text{Empirical formula mass}} = \frac{180}{30} = 6$

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## LEVEL-II

**1. Correct option: (A)**

Given: 1. Molarity of solution = 0.1M

**1.** M.W of water = 18

To find: Mole fraction of solvent = ?

Calculation:

Relation b/w molality and mole fraction, $m = \frac{1-X_B}{X_B} \times \frac{1000}{M_B}$

$\frac{0.1\times18}{1000} X_B = 1 - X_B$

$1.8 \times 10^{-3} X_B = 1$

$X_B = 0.998$

**2. Correct option: (C)**

Given: 1. Mass of urea = 6g

2. Mass of water = 90g

To find: Mole fraction of urea = ?

Calculation:

- Moles of urea (A) $= \frac{6}{60} = \frac{1}{10}$
- Moles of water (B) $= \frac{90}{18} = 5$
- Mole fraction of urea $= \frac{n_A}{n_A + n_B} = \frac{0.1}{5.1} = \frac{1}{51}$

**3. Correct option: (A)**

Given: 1. 36% water and 64% acetaldehyde

To find: Mole fraction of acetaldehyde = ?

Calculation:

- Basis = 100g solution
- Moles of water (B)$= \frac{36}{18} = 2$
- Moles of acetaldehyde (A) $= \frac{64}{44} = 1.42$
- Mole fraction of acetaldehyde $= \frac{n_A}{n_A + n_B} = \frac{1.42}{3.42} = 0.42$

**4. Correct option: (C)**

Given: 1. molar mass = 78

2. n = 6

Calculation:

- $n = \frac{\text{Molar mass}}{\text{Empirical formula mass}}$

Empirical formula mass $= \frac{78}{6} = 13$

Checking options to find empirical formula of mass 13, we get CH

**5. Correct option: (B)**

Given: 1. No. of S atoms = $3.0115 \times 10^{23}$

2. Volume of solution = 500 ml

To find: Normality = ?

Calculation:

- No. of S atoms = No. of S moles x NA

  $3.0115 \times 10^{23}$ = S moles x $6.023 \times 10^{23}$

  No. of S moles = 0.5
- 1 mol $H_2SO_4$ has 1 mol S atoms

0.5 mol S atoms are present in 0.5 mol of $H_2SO_4$

- Molarity of $H_2SO_4$ solution $= \frac{\text{No.of moles of solute}}{\text{Volume of solution(L)}}$

$$= \frac{0.5}{0.5} = 1 \text{ mol/L}$$

- n-factor = 2

Normality = Molarity x n-factor = 2 x 1 = 2N

**6. Correct option: (C)**

Given: 1. Mass of NaOH = 0.4g

2. Volume of H2SO4 solution = 100 ml

3. Normality of H2SO4 solution = 0.1N

Calculation:

- NaOH

$$\text{No. of gram equivalent} = \frac{\text{mass}}{\text{GEW}} = \frac{0.4}{40} = 10^{-2}$$

- $H_2SO_4$

$$\text{No. of gram equivalent} = N \times V = 100 \times 10^{-3} \times 0.1 = 10^{-2}$$

- (No. of gram equivalent)NaOH = (No. of gram equivalent)$H_2SO_4$

Therefore, solution is neutral.

**7. Correct option: (D)**

Given: 1. $V_1$ = 250ml

2. mass of oxalic acid = 6.3g

3. M.W of oxalic acid = 1264.

4. $N_2$ = 0.1N

To find: Volume of water = ?

Calculation:

- $\text{Equivalent weight} = \frac{\text{M.W}}{\text{n}} = \frac{126}{2} = 63$
- $\text{No. of gram equivalent} = \frac{\text{Mass of solute}}{\text{equivalent weight}} = \frac{6.3}{63} = 0.1$
- $\text{Normality} = \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}} = \frac{0.1}{0.25} = 0.04N$
- $N_1V_1 = N_2V_2$

  $0.04 \times 250 = 0.1 \times V_2$

  $V_2 = 1000$ ml
- Volume of water = $V_2 - V_1 = 1000 - 250 = 750$ ml or 0.75L

**8. Correct option: (AB)**

Molarity and normality are temperature dependent because they involve volume of solutions.Volume is dependent on temperature.

Molality, mole fraction and weight percentage does not depend on temperature because they involve masses of solute and solvent.

**9. Correct option: (AD)**

A) The solution of molality is always greater than molarity. This is because molarity is calculated as mol per unit L (volume of solution), whereas molality is calculated as moles perunit Kg (i.e. mass of solvent). So, in any case, the mass of the solvent is less than the volume of the solution, and thus the molality is greater.

B) The way we calculate formality and molarity is exactly the same but formality is used whenever the solute does not have a definite molecule; i.e., like ionic solids.

For ionic solids, we have formula mass instead of molecular mass since there are no distinct molecules to talk about.

C) Mass fraction and mole fraction are different concepts and represent different ways of expressing the composition of a mixture. It is highly unlikely for them to have the same valuesimultaneously, unless there are specific circumstances or simplifications that lead to that specific scenario.

$$\text{D) Normality} = \text{Molarity x } \frac{\text{Molar mass of solute}}{\text{Equivalent mass of solute}}$$

$$= \text{Molarity x } \frac{98}{49}$$

$$\text{Normality} = \text{Molarity x } 2$$

**10. Correct option: (AD)**

A) 100 mL of 0.1 N HCl + 50 mL of 0.1 N NaOH: In this case, the amount of NaOH is less than the amount of HCl, so there is excess acid. The resulting solution will be acidic.

B) 75 mL of 0.1 N HCl + 75 mL of 0.2 N NaOH: Here, the amounts of HCl and NaOH are equal. However, the concentration of NoOH is higher compared to HCl, so the resulting solution will be basic in nature.

C) 50 mL of 0.1 M H2SO4 + 100 mL of 0.1 N Ca(OH)2: In this case, we have a strong acid (H2SO4) reacting with a strong base (Ca(OH)2) in a 1:2 ratio. The resulting solution will be basic because there is excess base.

D) 100 mL of 0.1 M H2SO4 + 50 mL of 0.1 N Ca(OH)2: Similar to the previous case, we have a strong acid (H2SO4) reacting with a strong base (Ca(OH)2) in a 2:1 ratio. The resulting solution will be acidic because there is excess acid.

Therefore, the solutions A) and D) contain an acidic nature.

**11. Correct option: (A)**

Given: 1. Volume of water = 250g

2. Molal concentration of solution = 0.1m

To find: Mass of $H_2SO_4$ = ?

Calculation:

- $\text{Molality} = \frac{\text{No.of } H_2SO_4 \text{ moles}}{\text{kg of water}}$

No. of $H_2SO_4$ moles = 0.25 x 0.1 = 0.025

- Mass of $H_2SO_4$ = 0.025 x 98 = 2.45g

**12. Correct option: (B)**

Given: 1. Density of $Na_2CO_3$ solution = 1.018 g/ml

2. % w/v of $Na_2CO_3$ solution = 8.653

To find: Molality of solution = ?
Calculation:

- $\text{Molarity (M)} = \frac{\%\text{strengthof Solutionx density of solution x10}}{\text{Molecular mass of the solute}}$

$$= \frac{8.653 \times 1.018 \times 10}{106} = 0.83\text{M}$$

- $\text{Molality} = \frac{1000 \times \text{M}}{1000\rho - \text{MM}_1}$

$$= \frac{830}{1018 - 87.98} = 0.876 \text{ mol/kg}$$

**13. Correct option: (C)**

Given: 1. Density of NaCl solution = 1.25 g $mL^{-1}$

2. Molarity of NaCl solution = 3M

To find: Molality of NaCl solution = ?

Calculation:

- $\text{Molality} = \frac{1000 \times \text{M}}{1000\rho - \text{MM}_1}$

$$= \frac{3000}{1250 - 175.5} = 2.79 \text{ mol/kg}$$

**14. Correct option: (B)**

Given: Mole fraction of ethanol (B)= 0.9
To find: Molality of solution = ?
Calculation:

- Mole fraction of water (A) = 1- 0.9 = 0.1
- Molecular weight of ethanol (B) = 46
- $\text{Molality} = \frac{X_A}{1 - X_A} \text{ x } \frac{1000}{M_B} = \frac{0.1}{0.9} \text{ x } \frac{1000}{46} = 2.415$

**15. Correct option: (D)**

Given: Normality of solution = 3N

To find: Molarity of solution = ?
Calculation:

- Phosphoric acid is a triprotic acid. The basicity of $H_3PO_4$ is 3.

$$\text{Equivalent weight of Phosphoric acid} = \frac{\text{Mol.wt.of the phosphoric acid}}{\text{Basicity}} = \frac{\text{Mol.wt.of theacid}}{3}$$

- $\text{Normality} = \text{Molarity x } \frac{\text{Molar mass of solute}}{\text{Equivalent mass of solute}}$

$$3 = \text{Molarity x } 3$$

$$\text{Molarity} = 1\text{M}$$

**16. Correct option: (a-T, b-P, c-Q, d-R)**

A) Hypophosphorous acid (H3PO2) is a monobasic acid, meaning it can donate one proton (H+) per molecule. So, the equivalent weight (EW) is the same as the molar weight (MW) for hypophosphorous acid.

B) Phosphorous acid (H3PO3) is a diprotic acid, meaning it can donate 2 protons (H+) per molecule.

C) Phosphoric acid (H3PO4) is a tribasic acid, meaning it can donate three protons (H+) per molecule. So, the equivalent weight of phosphoric acid is one third of its molar weight.

D) Pyrophosphoric acid (H4P2O7) is a tetrabasic acid, meaning it can donate four protons (H+) per molecule. So, the equivalent weight of pyrophosphoric acid is one fourth of its molar weight.

**17. Correct option: (a-Q, b-S, c-P, d-R)**

A) $\text{Molality} = \frac{\text{No.of molesof solute}}{\text{Wt.of solvent in kg}}$

B) $\text{Mole fraction} = \frac{\text{No.of molesof one component}}{\text{No.of moles of all components}}$

C) $x_A + x_B = 1$

D) $\text{Molarity} = \frac{\text{No.of molesof solute}}{\text{Vol.of solutioninlit}}$

**18. Correct option: (2)**

Calculation:

$$\text{Normality} = \text{Molarity} \times \frac{\text{Molar mass of H2SO4}}{\text{Equivalent mass of H2SO4}}$$

$$= \text{Molarity} \times \frac{98}{49}$$

$$= \text{Molarity} \times 2$$

**19. Correct option: (5.33)**

Given: 1. Mass of nitrogen = 7g

2. Mass of oxygen = 8g

To find: Mole percentage = ?

Calculation:

- Mole fraction of oxygen $= \frac{8}{8+7} = 0.533$
- Mole percentage of oxygen = Mole fraction of oxygen x 100%

$$= 5.33 \times 10$$

**20. Correct option: (2)**

Given: 1. % w/v of $H_2SO_4$ = 0.98

2. Molecular weight of $H_2SO_4$ = 98.08 g/mol

To Find: Normality of $H_2SO_4$ solution

Calculation:

- Convert % w/v to g/L 0.98 g/100 mL = 9.8 g/L
- $\text{Molarity} = \frac{mass}{Molecular\ weight \times Volume} = \frac{9.8}{98 \times 1} = 0.1$ M
- N = Molarity x n = 0.1 x 2 = 0.2 N or $2 \times 10^{-1}$ N

Where,

n = number of H+ ions that can be donated by each molecule of $H_2SO_4$, which is 2 for $H_2SO_4$

# 4. Gaseous State

## Solutions

### LEVEL-I

**1. Correct option: (D) gases**

Matter has 3 states

(i) Solid (ii) Liquid (iii) Gas

Molecules of solids have least amount of space between them.

Molecules of liquids have little amount of space between them.

But molecules of gases have the largest number of space between them. This allows them to move freely. As a result if external pressure is applied, the volume of gas decreases.

Hence gases have highest compressibility.

**2. Correct option: (C) They take shape of the container in which they are stored**

The space between two gas molecules is large. Hence they are highly compressible.

Due to this they have infinite shape and volume. They gradually take shape of the container in which they are stored.

Due to random motion of gas particles,they exert pressure on the wall.

Hence option C is correct.

**3. Correct option: (B) Kelvin**

Kelvin (K) is the S.I and CGS unit of temperature.

Whereas Centigrade(C) is the unit of temperature in Celsius scale and Fahrenheit (F) is the unit of temperature in Fahrenheit scale.

These two units are used widely.

$K = 273+C$

$F = (9/5)\times C + 32$

Pascal is the unit of pressure.

**4. Correct option: (B) critical temperature**

Critical temperature is the lowest possible temperature at which gases start to convert into liquid.

Absolute zero temperature is lowest possible temperature at which heat and energy is absent in a particle.

The value of absolute zero is 0 K or -273.15°C

Boyle's temperature is temperature at which real gases obey ideal gases equation.

Transition temperature is a temperature at which sudden change in physical properties occurs in a substance.

**5. Correct option: (D) a,b c,d**

Mass – Gases have have a definite mass and it can be measured in grams.

Volume – Gases do not have definite volume. But it is measurable when it occupies a container. The volume of the container is the volume of gas occupied. It can be expressed in cc or Litre.

Pressure – Force exerted by gas per unit area of a wall is called called pressure of gas. It can be measured in Pascal, atm.

Temperature – The scales which is used to measure temperature of gases are Celsius, Fehrenheit and Kelvin

**6. Correct option: (A) I, II, III, IV**

Matter has 3 states

(i) Solid (ii) Liquid (iii) Gas

So matter exists in solid, liquid and gas.

Molecules of gases have the largest number of space between them. This allows them to move freely. As a result if external pressure is applied, the volume of gas decreases.
Hence gases are highly compressible.
Due to highly compressible, gases do not have definite shape or volume.
The gaseous state of solid and liquid is called vapour.
Example – vapour of water (liquid), vapour of iodine (solid)

**7. Correct option: (B) Pressure**

Torr is a unit of pressure.
It is defined as 1/760 of an atmosphere.
Unit of mass – Kg(SI), g (CGS)
Unit of volume – Litre (CGS), $m^3$(SI)
Unit of density – $kg/m^3$ (SI), $g/cm^3$ (CGS)

**8. Correct option: (C) Measurable properties**

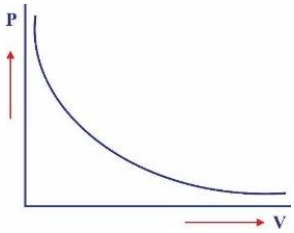
Gas laws give the relation between the different measurable properties (mass, pressure, volume, temperature) of gases.

**9. Correct option: (B)**

Boyle's law is a gas law that states that a gas's pressure and volume are inversely proportional.
Formula of Boyle's law
$P \propto 1/V$
Or $V \propto 1/P$
$P_1V_1 = P_2V_2$
Or $PV = K$ (K= proportionality constant)

**10. Correct option: (B) Hyperbola**

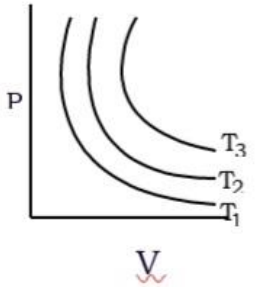
$PV = nRT$
$PV = C^2$ = hyperbola




**11. Correct option: (D) Isotherms**

A curve drawn at constant temperature is called an isotherm.
This shows the relationship between pressure (P) and volume (V).
The graph drawn at constant volume is known as Isochoric
The graph drawn at constant pressure is known as Isobars.

**12. Correct option: (B)**




According to ideal gas equation, $PV = nRT$, as n and R are constant.
$PV \propto T$
Hence $T_1 > T_2 > T_3$

**13. Correct option: (C) P/3**

According to Boyle's Law, volume of the gas is inversely proportional to the pressure at constant temperature.

$P_1V_1 = P_2V_2$

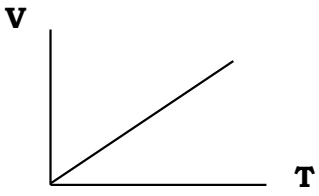
Now $V_2 = 3V_1$

$P_1V_1 = P_2 3V_1$

Hence pressure ($P_2$) will be = $P_1/3$

**14. Correct option: (B)**

B)

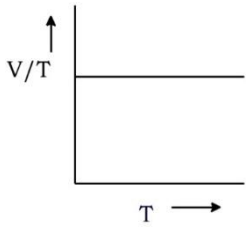



**Solution :**Charles' law states that at constant pressure, the volume occupied by a gas is directly proportional to the temperature.

That is V ∝ T at constant P

Hence option B is correct

**15. Correct option: (A)**




Charles' law states that at constant pressure, the volume occupied by a gas is directly proportional to the temperature.

That is V ∝ T at constant P

Or, V/T = Constant

So the graph will be parallel to x axis.

Hence option A is correct

**16. Correct option: (D) Isobars**

The graph drawn at constant pressure is known as Isobars.

A curve drawn at constant temperature is called an isotherm

The graph drawn at constant volume is known as Isochoric.

**17. Correct option: (A) V ∝ T**

Charles' law states that at constant pressure, the volume occupied by a gas is directly proportional to the temperature.

That is V ∝ T

**18. Correct option: (C) 450 K**

**19. Correct option: (C) Isochoric**

The graph drawn at constant volume is known as Isochoric

The graph drawn at constant pressure is known as Isobars.

The graph drawn at constant temperature is known as Isotherms.

**20. Correct option: (C) Equal volumes of gas contain equal number of moles under similar conditions of temperature and pressure.**

According to Avogadro's law, equal volumes of all gases at same temperature and pressure will have same moles. of molecules.

## LEVEL-II

**1. Correct option: (C) 273°C**

P = constant

$T_1$ = 0°C = 273 K

$T_2$ = ?

Let $V_1$ = V

Hence, $V_2$ = 2V

We know $V_1/T_1 = V_2/T_2$

$T_2 = (V_2T_1) / V_1$

$= (2V \times 273) / V$

= 546 K

= (546 – 273)°C

= 273°C

**2. Correct option: (B) 4 atm**

PV = nRT

P = nRT / V

P/(1/V) = slope

P = 60/15 = 4 atm

**3. Correct option: (A) 477°C**

From Charle's Law

$V_1/T_1 = V_2/T_2$

Let X be the volume of the vessel.

The volume of the air at the start ($V_1$) is x. After heating, 2/5 of the air fills a volume of x.

$(2/5)V_2=x$

⇨ $V_2=(5/2)x$

$=5x/2$

27 Celsius = 300 Kelvin

$x/300 = (5x/2)/T_2$

we get $T_2$ = 1500/2 = 750 K = 477°C

**4. Correct option: (C) 400 K**

PV = nRT

MP/RT = nM/V

MP/RT = d

P = dRT/M

When P, R and M is constant

Then $d_1= d_2T_2$

Or, $d_1$ = d

Or, $d_2$ = 0.75d

$T_1$ = 27°C = 300 K

$T_2 = d_1T_1/d_2$

$= d\times300/0.75d$

= 400

Hence, $T_2$ = 400 K

**5. Correct option: (B) $6 \times 10^5$ N/m−2**

P1 = $9 \times 10^5$ N/m², P2 = ?

V1 = 2.5 L, V2 = 1.5L

P2 = P1V1/V2

$= (9 \times 10^5 \times 2.5) / 1.5$

$= 15 \times 10^5\ N/m^2$

Hence additional pressure required

$(15 \times 10^5 - 9 \times 10^5) = 6 \times 10^5\ N/m^2$

**6. Correct option: (C) Decreased by 2 tunes**

Pressure is doubled means $P_2 = 2P_1$

$P_1V_1 = P_2V_2$

$V_2 = P_1V_1/P_2$

$V_2 = P_1V_1/2P_1$

$V_2 = V_1/2$

Hence volume is decreased to half.

**7. Correct option: (C) ½ atm**

Initial pressure $P_1 = 1atm$

Initial temperature $T_1 = 273°C = (273+273)K = 546\ K$

Final pressure $P_2 = ?$

Final temperature $T_2 = 0°C = 273K$

We know

$P_2/P_1 = T_2/T_1$

$P_2 = T_2 \times P_1/T_1$

$=( 273 \times 1)/546$

$= ½$ atm

**8. Correct option: (A)**

N = 10 moles

P = 0.821 atm

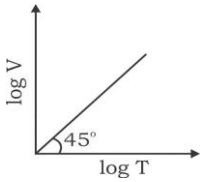
We know R = 0.0821

Now PV = nRT

$V = nRT/P$

$V = (10 \times 0.0821 \times T)/0.821$

$V = T$




Hence $(\log V/\log T) = 1 = \tan 45°$

Hence option A is correct.

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**9. Correct option: (A) Pa B) $Nm^{2-}$ C) Bar D) atm**

Pressure is force applied perpendicular to the surface of an object per unit area.

Pressure = P

Force = F

Area = a

P = F/a

Unit of force in SI = N

Unit of area in SI = $m^2$

So unit of pressure in SI = $Nm^{2-}$

Other units of pressure are

Pa, Bar, atm etc

**10. Correct option: (A)$V \propto T$, B) $(dV/dT)_p = k$**

According to Charles's law The volume of gas is directly proportional to its absolute temperature at constant pressure

$V \propto T$ at constant pressure

V = KT (K = constant)

V/T = K

Hence, $(dV/dT)_p = K$

**11. Correct options:**
**(A) Volume of a gas change as it pressure and temperature change**
**B) Pressure has little effect on solids and liquids when compared to gases.**
**D) Perfumes spread because of diffusion of gases.**
According to Charles's law The volume of gas is directly proportional to its absolute temperature at constant pressure.
According to Boyle's law the volume of a given amount of gas is inversely proportional to its pressure when temperature is constant.
Hence option A) is correct.
When pressure is applied on gas it becomes liquid and when pressure is applied on liquid it becomes solid.
Pressure has less effect on solid and liquid.
Hence option B) is correct.
The intermolecular attraction between gas molecules are weak. That is why they don't have definite shape or volume.
Hence option C) is incorrect.
The movement of molecules from a region of higher concentration to a region of lower concentration down the concentration gradient is known as diffusion.
As the particles of perfume are free to move, they move quickly through diffusion.
Hence option D) is correct.
**12. Correct option: (A) Litre B) Millilitre C) $dm^3$**
The amount of space occupied by any component is known as Volume.
Volume = length × width × height
Hence,
Unit of volume = unit of length × unit of width ×unit of height
= $m^3$(SI)
= $cm^3$ (CGS)
= $dm^3$
$Dm^3$ is also known as Litre.
So, Litre is also a unit of volume
1 Litre = 1000 $cm^3$ = 1000 millilitre
2 Milliliter is also a unit of volume
Hence option A), B and C) is correct.
Gram is a unit of mass in CGS.
**13. Correct option: (C) 273°C**
Let the volume of gas at 0°C = V ml
$V_1$ = V ml
$V_2$ = 2V ml
$T_1$ = (0+273) K = 273 K
$T_2$ = ?
From Charle's law,
$V_1/T_1 = V_2/T_2$
$T_2 = V_2T_1/V_1$
=( 2V × 273) / V
= 546 K
= (546 – 273)°C
= 273° C
Hence option C is correct.

**14. Correct option: (D) 21.38 mL**

$V_1$ = 20mL

$T_1$ = (15+273) K = 288 K

$T_2$ = ( 35 + 273) K = 308 K

$V_2$ = ?

From Charle's law

$V_1/T_1 = V_2/T_2$

$V_2 = V_1T_2/T_1$

$=(20\times308)/288$

= 21.38 mL

Hence option D is correct.

**15. Correct option: (B) double**

Let, $T_1 = T$

Then, $T_2 = 2T$

$V_1 = V$

$V_2$ = ?

Form Charle's law

$V_1/T_1 = V_2/T_2$

$V_2 = V_1T_2/T_1$

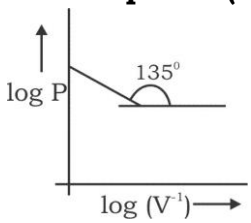
$V_2 = (V\times2T)/T$

2V

Hence the volume will be doubled.

Option B is correct

**16. Correct option: (C)**

C) 


From Boyle's law $P \propto 1/V$

PV = k (constant)

So it is hyper bola.

Taking log both sides

Log P = log K – log V

Log P = log K + log $(V^{-1})$

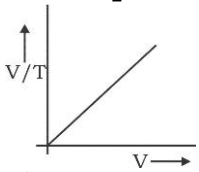
Comparing, y = mx + log K

For 45 ° the slope of the graph is +1

For 135 ° the slope of the graph is –1.

It doesn't obey Boyle's law.

**17. Correct option: (D)**

D) 

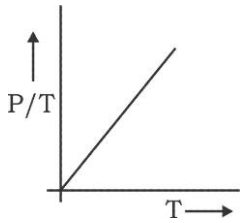

Charles' Law states that the volume of a fixed mass of a gas is directly proportional to the temperature at constant pressure.

$V \propto T$ at constant P

V = KT

i.e. y = mx
So the graph V vs T is straight line.
This law applies to ideal gases held at a constant pressure, where only the volume and temperature are allowed to change.
V/T = K
Log V = log K + log T
Comparing, y = mx + logK
Hence the slope for the graph of log V vs log T is tan 45° or +1.
V= KT
$VT = KT^2$
That is y = mx
Thus, VT vs $T^2$ will be straight line.
V = KT
V/T = K
Thus V/T vs T will be parallel to X axis.

**18. Correct option: (D)**




From gay-lusacc's law
P ∝ T
P = KT
Comparing y = mx
So it will be straight line.
Log P = log K + log T
Comparing y = mx + log K
M = tan 45° = 1
So it will be positive.
P = KT
P/T = K
So P/T vs P graph will be parallel to X axis
Hence option D is correct

**MATRIX MATCHING TYPE**

**19. Correct Matching is: a → S), b → Q), c → R), d → P)**

Charles' Law states that the volume of a fixed mass of a gas is directly proportional to the temperature at constant pressure.
V∝T at constant P
Gay-Lussac's Law states that the pressure of gas varies directly with the absolute temperature of the gas, when the volume is constant.
P ∝ T at constant V
According to Boyle's Law, volume of the gas is inversely proportional to the pressure at constant temperature.
P ∝ 1/V(at constant T)
Avogadro's law states that equal volumes of all gases, at the same temperature and pressure, have the same number of molecules
V ∝ n (at constant T &P)

**20. Correct Matching is: a) →Q), b) → P), c) → S), d) → R)**

According to Boyle's Law, volume of the gas is inversely proportional to the pressure at constant temperature.

$P \propto 1/V$(at constant T)

That is $P_1V_1 = P_2V_2$ =constant

Curve drawn at constant temperature is called an isotherm.

This shows the relationship between pressure (P) and volume (V).

The graph drawn at constant pressure is known as Isobars.

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## Ideal Gas Equation
## LEVEL-I

**1. Correct option: (B) 2 Cal**

From ideal gas equation,

PV = nRT

R = PV/nT

Here n = 1

So, PV/T = R

= 8.314 J $K^{-1}$ $mol^{-1}$

= 2 Cal $K^{-1}$ $mol^{-1}$

Hence option B is correct.

**2. Correct option: (A) 0.5**

From ideal gas equation,

PV = nRT

N = PV/RT

= (22.4 × 1) / (0.082 × 546)

= 0.5

**3. Correct option: (C) 1.75 atm**

2 g of hydrogen = 2/2 = 1 mole hydrogen

8 g of $O_2$ = 8/32 = 0.25 moles of $O_2$

22 g of $CO_2$ = 22/44 = 0.5 mole of $CO_2$

Total number of moles (n) = 1+0.25+0.5 = 1.75

V = 22.4 Litre, T = (0 + 273) = 273 K

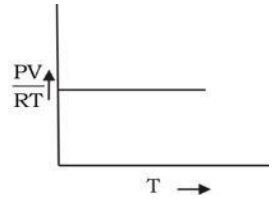
From ideal gas equation PV = nRT

P = nRT/V

= (1.75 × 0.082 × 273) /22.4

= 1.75 atm

**4. Correct option: (A)**




From ideal gas equation

PV = nRT

PV/RT = n, n is independent to temperature T

So the graph will be parallel to X axis.

Hence option A is correct.

**5. Correct option: (C) 44.8 lit**

We know that at 273 K temperature and 1 atm pressure gas occupies 22.4 Litre molar volume

Here t = (273+273) = 546 K

P = 1 atm

N = 1 mole

$V_1$ = 22.4 L

$T_1$ = 273K

$V_2$ = ?

$T_2$ = 546K

Volume is directly proportional to temperature at constant pressure.

$V_1/T_1 = V_2/T_2$

$V_2 = V_1T_2/T_1$

$=( 22.4 \times 546) / 273$

$= 44.8$

Hence option C is correct

**6. Correct option: (D) Ideal gas**

According to Charles's law The volume of gas is directly proportional to its absolute temperature at constant pressure.

According to Boyle's law the volume of a given amount of gas is inversely proportional to its pressure when temperature is constant.

Gay-Lussac's Law states that the pressure of gas varies directly with the absolute temperature of the gas, when the volume is constant.

The gas which obeys all the above mentioned laws at all temperature and pressure is called ideal gas.

**7. Correct option: (D) High temperature and low pressure**

Real gases show ideal gas behavior at high temperature and low pressure because at high pressures and low temperature, molecules of gases are very close to each other

Real gases show ideal behavior when intermolecular forces are practically negligible. This can only happen at very low pressure and very high temperature

**8. Correct option: (A)Low temperature and high pressure**

At Low temperature and high pressure, gas atoms or molecules get closer to each other and attracted by weak Vander Waals force. Hence a gas deviates the most from its ideal behavior.

**9. Correct option: (C)273°C**

1 mole oxygen occupies a volume of 22.4 L at 1 atm and 273 K

We know PV = nRT ( Ideal gas equation)

If P,V and R are constant then

N is inversely proportional to T

Hence $n_1T_1 = n_2T_2$

$n_1$ = 1 mol

$T_1$ = 273K

$n_2$ = 16/32 = 0.5 mol

$T_2 = n_1T_1/n_2$

$= 273/0.5$

$= 546$ K

$= (546 – 273)°$ C

$= 273 °$ C

**Hence option C is correct**

**10. Correct option: (C) 400k**

PV = nRT

MP/RT = nM/V

MP/RT = d

P = dRT/M

When P, R and M is constant

Then $d_1T_1 = d_2T_2$

Or, $d_1 = d$

Or, $d_2 = 0.75d$

$T_1 = 27°C = 300$ K

$T_2 = d_1T_1/d_2$

$= d \times 300/0.75\ d$

$= 400$

Hence, $T_2 = 400$ K

**11. Correct option: (A) NO**

1 mole gas occupies 22.4 L of volume at STP.

Thus 5.6 L of gas at STP contains 0.25 moles.

Number of moles = (weight / Molecular weight)

0.25 = 7.5/M

M = 30

Molecular weight of NO = (14+16) = 30

So the gas is NO

**12. Correct option: (C)149.24 lit**

Number of moles of hydrogen (n)= 10/2 = 5

P = 1.5 atm

T = (273+273) K = 546 K

From ideal gas equation,

PV = nRT

V = nRT/P

V = ( 5 × 0.082 × 546) / 1.5

V = 149.24 litre

Hence option C is correct

**13. Correct option: (B) 2:1**

Molar mass of $NH_3 = 14+3 = 17$

Molar mass of $PH_3 = 31+3 = 34$

The ratio of masses of,

$PH_3$ and $NH_3$

= 34: 17

= 2 : 1

**14. Correct option: (C) 1.5 atm**

Number of moles of Oxygen, $n_1 = 8/32 = 0.25$

Number of moles of nitrogen, $n_2 = 14/28 = 0.5$

Volume (V) = 11.2 L

Temperature = 273 K

So, $n = n_1 + n_2 = (0.25 + 0.5 ) = 0.75$

From ideal gas equation PV = nRT

P = nRT/V

= (0.75 × 0.082 × 273) / 11.2

= 1.5 atm

**15. Correct option: (C) 8.314 $J.K^{-1}.mole^{-1}$**

PV = nRT

SI unit of,

P = $N/m^2$

V = $m^3$

N = mole

Work = N.m = joule

R = PV/nT

= $(N \times m^3)/ (m^2 \times mole \times K)$

= $(N.m) / (mole \times K)$

= $J.mol^{-1}.K^{-1}$

At STP, for 1 mole of gas (n=1 mol),

The value of,

Temperature (T) = 273 K,

Pressure (P) = $1.01 \times 10^5$ $N/m^2$

Volume (V) = $22.4 \times 10^{-3}$ $m^3$

From PV = nRT

R = PV/nT

= $(1.01 \times 10^5 \times 22.4 \times 10^{-3})/273$

= 8.314 J.mol–1.K–1

**16. Correct option: (D) All of these**

From ideal gas equation

PV = nRT

R = PV/nT

Unit of

P = atm

V = litre

N = mole

T = K

So unit of R = $litre.atm.mol^{-1}.K^{-1}$

Again

R = $(N \times m^3)/ (m^2 \times mole \times K)$

= $(N.m) / (mole \times K)$

= $J.mol^{-1}.K^{-1}$

= $erg.mol^{-1}.K^{-1}$

= $Cal.mol^{-1}.K^{-1}$

**17. Correct option: (B) $dynes.cm^{-2}$ ; $cm^3$**

$erg.K^{-1}.mol^{-1}$, this CGS unit of R

So pressure and volume will be taken in CGS unit.

CGS unit of,

Pressure = $dyne.cm^{-2}$

Volume = $cm^3$

Hence option B is correct.

**18. Correct option: (B) Only I,II,III**

At STP, for 1 mole of gas (n=1 mol),

The value of,

Temperature (T) = 273 K,

Pressure (P) = $1.01 \times 10^5$ $N/m^2$

Volume (V) = $22.4 \times 10^{-3}$ $m^3$

From PV = nRT
R = PV/nT
= $(1.01 \times 10^5 \times 22.4 \times 10^{-3})/273$
= 8.314 $J.mol^{-1}.K^{-1}$
= $8.314 \times 10^7$ ergs $K^{-1}.mole^{-1}$
Again we know
R = 0.0821 $lit.atmK^{-1}.mol^{-1}$
= $0.0821 \times 10^3$ ml. $atm.K^{-1}.mol^{-1}$
= 82.1 ml. $atm.K^{-1}.mol^{-1}$
= $0.0821 \times 760$ lit.mm. $K^{-1}.mole^{-1}$
= 62.4 lit.mm. $K^{-1}.mole^{-1}$

Hence I), II) & III) are correct.

**19. Correct option: (A) dyne. Cm**

An erg is the amount of work done by a force of one dyne exerted for a distance of one centimetre.
So
Unit of erg = Unit of force in CGS × unit of distance in CGS
= dyne.cm

**20. Correct option: (D)273°C and 1 atm**

From ideal gas equation
PV = nRT
MP/RT = nM/V
MP/RT = d
M & R are constant
So the density will be minimum if P/T ratio is minimum.
At STP
P = 1 atm, T = 273 K
P/T = 1/273 = 0.0037
For 0 ° C and 2 atm
P/T = 2/273 = 0.0073
For 273° C and 2 atm
P/T = 2/546 = 0.0037
For 273° C and 1 atm
P/T = 1/546 = 0.0018

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## LEVEL-II

**1. Correct option: (B)1.25 g/1**

From ideal gas equation
PV = nRT
MP/RT = nM/V
MP/RT = d
M = 28
P = 1 atm
T = 273K
Hence, d = (28×1)/(273×0.082)
= 1.25 g/1

**2. Correct option: (B) 8g**

From ideal gas equation

PV = nRT

MP/RT = nM/V

MP/RT = d

M = 32,

P = 380 torr = 0.5 atm,

T = 273 K,

V = 11.2 L

D = (32×0.5)/(273×0.082)

= 0.714

Mass = d × V

= 0.714 × 11.2

= 8 g

**3. Correct option: (B) 89.6L**

From ideal gas equation PV = nRT

MP/RT = nM/V

MP/RT = d

M = 2

P = 1.5 atm

T = (276 + 276) K = 546 K

D = (2×1.5)/(0.082×546)

= 0.067

We know mass = d × V

Hence V = 6/0.067

= 89.6 L

**4. Correct option: (A) ¾**

Initial pressure $P_1$ = 1 atm

Initial temperature $T_1$ = 327°C = 600 K

Final pressure $P_2$ = 1 atm

Final temperature $T_2$ = 527°C = 800 K

$P_1V_1/T_1 = P_2V_2/T_2$

$V_2 = (1\times800\times V_1)/(600\times1) = 4/3\ V_1$

Volume of air expelled, $= V_2 - V_1$

$= 4/3\ V_1 - V_1$

$= V_1/3$

Hence fraction of air repelled

$= (V_1/3)/(4V_1/3)$

= ¼

Thus fraction of air remained in the flask

= (1 – ¼)

= ¾

**5. Correct option: (A) 10 %**

From Charle's law

$V_1/T_1 = V_2/T_2$

Let,

$V_1 = V$

$V_2$ =?

$T_1 = T$

$T_2 = 11T/10$
$V_2 = (V_1T_2)/T_1$
$= (V \times 11T) / 10T$
$= 11V/10$
Change in V,
$= (11V/10) – V$
$= V/10$
So percentage of change in V
$= (V/10) \times (100/V)$
$= 10\%$

**6. Correct option: (B) 2 atm/25°C**

From ideal gas equation
$PV = nRT$
$PM = dRT$
$P \propto d$
$P_1/d_1 = P_2/d_2$
$P_1 = 1$ atm
$d_1 = 1.458$ mg/L
$d_2 = 2.916$
$P_2 = P_1d_2/d_1$
$= 1 \times 2$
$= 2$ atm
Temperature will be same

**7. Correct option: (A) 40°C**

P = 0.5 atm
V = 51.4 L
T = ?
We know PV = nRT
T = PV/nR
$=(0.5 \times 51.5) /(0.082 \times 1)$
$= 313$ K
$T = (313 – 273)°$ C
$= 40°$ C

**8. Correct option: (D) 68.25 mL**

From ideal gas equation
PV = nRT
P = 4 atm
N = 0.5 mole
$R = 2$ Cal.mol$^{-1}$.K$^{-1}$
T = 273 K
V = ?
V = nRT/P
$= (0.5 \times 2 \times 273) / 4$
$= 68.25$ mL

ONE OR MORE THAN ONE OPTION CORRECT TYPE

**9. Correct option: (A) 0.0821L.atm.K$^{-1}$.mole$^{-1}$ B)1.99cal.K$^{-1}$.mole$^{-1}$**
D)8.314J.K$^{-1}$.mole$^{-1}$

At STP, for 1 mole of gas (n=1 mol),
The value of,

Temperature (T) = 273 K,
Pressure (P) = $1.01 \times 10^5$ N/m²
Volume (V) = $22.4 \times 10^{-3}$ m³
From PV = nRT
R = PV/nT
$= (1.01 \times 10^5 \times 22.4 \times 10^{-3})/273$
$= 8.314$ J.mol$^{-1}$.K$^{-1}$
$= 1.99$ cal. K$^{-1}$. Mole$^{-1}$
Again we know
R = 0.0821 lit.atmK$^{-1}$.mol$^{-1}$
$= 8.314$ m³. Pa. K$^{-1}$.mol$^{-1}$
$= 8.314 \times 10^3$ Pa. dm³.K$^{-1}$.mol$^{-1}$

**10. Correct option: (A) Volume increases if ideal nature is consider and 'P' & 'n' are kept constant, C)Vm = 22.4litres at 25°C assuming gas behaves ideally**

PV=nRT
If P and n constant.
V∝T
Volume increases with temperature if P and n kept constant and ideal nature is considered.

**11. Correct option: (B)Their pressure are same if 'Y' is of half the size of 'X'**
**C)Temperature ratio can be made 1:1 by doubling the moles in 'Y' (PV being maintained)**
**D)Temperature ratio can be made 1:1 by doubling the moles in 'X' (PV being maintained).**

From ideal gas equation PV = nRT
PY = nR2T/2V = nRT/V
PX = nRT/V
Option A is correct
PY = nR2T/(V/2) = 4nRT/V
PX = nRT/V
Hence option B is incorrect
2TY = PV/2nR
TY = PV /4nR
TX = PV/nR
Hence option C is incorrect
Again
TX =PV/2nR
2TY = PV/nR
TY = PV/2nR
Hence option D is correct

**12. Correct option: (A) PV = $N/N_A$ RT B) PV = $N_K$T**

PV = $N/N_A$ RT, $N/N_A$ = n
That is PV = nRT,hence it is correct.
PV = $N_K$T
Where $N_K$ = nR, as K = R/N, hence $N_K$ = nR
It is also correct.
PV = nRT
PV = W/M RT
Hence PV = ρ/m RT is incorrect

$PV = MK_BT/m$
$PV = nK_BT = nRT$
Hence it is also incorrect

**13. Correct option: (B) Velocity**

The simplest known example of an equation of state is the one relating the pressure P, the volume V, and the absolute temperature T of n mole of an ideal gas—that is, the ideal gas law PV = nRT, in which R is the universal gas constant.

**14. Correct option: (B) 4 atm**

From ideal gas equation
PV = nRT
P =(1/V)RT
Slope of P vs 1/V plot
M = 1/nRT
Again, V = 15 Litre
1/P = V/nRT
1/P = 15/60
P = 60/15
P = 4 atm

**15. Correct option: (B) 4 atm**

$P_1V_1 = P_2V_2$
$P_2 = P_1V_1 / V_2$
$= (2\times6)/4$
$= 3$ atm
Again $P_1/T_1 = P_2/T_2$
$P_2 = P_1T_2/T_1$
$= (3\times400)/300$
$= 4$ atm

**16. Correct option: (A) 27.5L**

We know
$P_1V_1/T_1 = P_2V_2/T_2$
$V_2 = P_1V_1T_2/T_1P_2$
$= (6 \times 10 \times 550)/(400\times 3)$
$= 27.5$

**17. Correct Matching is: a) → Q), b) → T), c)→P), d) → R)**

$R = PV/nT$
$= (1.01 \times10^5 \times 22.4 \times 10–3)/273$
$= 8.314$ J.mol$^{-1}$.K$^{-1}$ → SI unit of R
$= 8.314 \times 10^8$ ergs.k$^{-1}$.mol$^{-1}$ = CGS unit of R
V = 22.414 lit → gram molar volume at STP
Standard pressure at STP → 1 atm = 760 mm of Hg

**18. Correct Matching is: a) → Q), b) →P), c) →R), d) → T)**

Standard ambient temperature = 298.15 K
Gram molar volume at STP = 22.4 L
Volume at SATP (P = 1 bar) = 24.4 L
Volume at SATP (P = 1 atm) = 24.7 L

**19. Correct Answer is: 7g**

From ideal gas equation
PV = nRT
PV = (W/M)RT

(380/760)22.4 = (W/28) × 0.082 × 546
W = 7g

**20. Correct Answer is: 6.4 g**

From ideal gas equation
PV = nRT
PV = (W/M)RT
16 × 9 = (W × 300 × 0.082) / 16
W = x = 93.6 g
100-x = 100 – 93.6 = 6.4 g

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## Dalton's Law of Partial Pressures

## LEVEL-I

**1. Correct option: (C) 0.7 atm**

The ratio of molecular weight of $SO_2$ and $O_2$ is 2:1
Hence,Mole fraction of $SO_2$ = (w/64+w/32)/(w/64)
=1/3
$p_{SO2}$ = 1/3 ×$P_T$
=⅓ × 2.1
= 0.7 atm

**2. Correct option: (A) 1.9 atm**

According to Dalton's law,
$P = P_1 + P_2 + P_3$
Hence,
$9.8 = P_{CO2} + 3.7 + 4.2$
$P_{CO2}$ = 9.8 – 3.7 – 4.2
= 1.9 atm

**3. Correct option: (B) ⅔ of total pressure**

Number of moles of $H_2$ = 3/2 moles
Number of moles of $O_2$ = 24/32 = ¾ moles
Total number of moles = (3/2 + ¾) = 9/4
$p_{H2}$ = (3/2)/(9/4) × PT
= ⅔ PT

**4. Correct option: (C)3/7**

Number of moles of hydrogen = 4/2 = 2 molss
Number of moles of Nitrogen = 42/28 = 3/2 moles
Total number of moles = (2 + 3/2) = 7/2
Total pressure exerted by Nitrogen
= (3/2)/(7/2) × P
= 3/7
Hence option C is correct

**5. Correct option: (A) 500 torr**

From ideal gas equation and Dalton's law.
$P = (P_1V_1 + P_2V_2)/V$
= (600×1000 + 800×500)/2000
= 500 torr

**6. Correct option: (A) 8.4**

Number of moles of $H_2$ = ½ moles
Number of moles of He = 5/4 moles

Total number of moles = (½ + 5/4) = (7/4) = 1.75 moles
T = 20°C = 293 K
V = 5 L
From ideal gas equation
PV = nRT
P = nRT/V
= (1.75 × 0.0821× 293)/5
= 8.41 atm

**7. Correct option: (A) 0.2**

Number of moles of $H_2$ = 2/2 = 1 mole
Number of moles of $O_2$ = 8/32 = 0.25 moles
Total number of moles = (1+0.25) = 1.25 moles.
Hence Mole fraction of $O_2$ in the given mixture
= 0.25/1.25
= 0.2

**8. Correct option: (B)⅕**

A vessel contains methane and oxygen in the mass ratio 2:1
Hence Mole fraction of methane = 2/16 = ⅛
Mole fraction of Oxygen = 1/32
Total mole fraction = (1/8 + 1/32 ) = 5/32
$P_{oxygen}$ = (1/32)/(5/32)× $P_t$
= ⅕ × $P_t$

**9. Correct option: (B)44 atm**

Two identical vessels,
44 g of $CO_2$ =1 mol and exerts 2atm pressure.
Now, 44g of $H_2$ = 44/2 =22 mol and will exert =22×2atm pressure.
∴ Pressure of $H_2$ = 44 atm.

**10. Correct option: (D) 8/9 th of total pressure**

Let the weight ratio of $O_2$ and He is 1:1
Hence,
Mole fraction of He = ¼
Mole fraction of $O_2$ = 1/32
Total mole fraction = (¼ + 1/32) = (9/32)
The pressure exerted by helium
= (¼)/(9/32) × $P_t$
= 8/9 × $P_t$

**11. Correct option: (C)8/9**

Let the weight ratio of methane and hydrogen is 1:1
Hence,
Mole fraction of methane = 1/16
Mole fraction of hydrogen = ½
Total mole fraction = (½ + 1/16) = (9/16)
The pressure exerted by hydrogen
= (½ )/(9/16) × $P_t$
= 8/9 × $P_t$

**12. Correct option: (B) 319. 2mm**

Mole% =Vol%
Total mole = 100 mole
Moles of CO = 42 moles

Moles of $H_2O$ = 100 – 42 = 58 moles
$X_{CO}$ = 42/(42+58) = 42/100
$P_{CO}$ = (42/100)×760 = 319.2 mm

**13. Correct option: (D) ⅓**

Let the weight ratio of methane and oxygen is 1:1
Hence, Mole fraction of methane = 1/16
Mole fraction of Oxygen = 1/32
Total mole fraction = (1/16 + 1/32) = (3/32)
The pressure exerted by oxygen
= (1/32)/(3/32) × $P_t$
= 1/3 × $P_t$

**14. Correct option: (D) 15/16**

Let the weight ratio of ethane and hydrogen is 1:1
Hence,
Mole fraction of ethane = 1/30
Mole fraction of hydrogen = ½
Total mole fraction = (½ + 1/30) = (16/30)
The pressure exerted by hydrogen
= (1/2 )/(16/30) × $P_t$
= 15/16 × $P_t$

**15. Correct option: (A) 0.37**

At STP,
T=273 K
P=1 atm
From ideal gas equation
$M_{mix}$ = dRT/$P_{mix}$
$M_{mix}$ =( 1.7 × 0.082×273)/1 = 38.11 g
The molecular weight of the mixture ($M_{mix}$) can also be calculated by using the expression,
$M_{mix}$ = $X_{CO}$×$M_{CO}$ +(1−$X_{CO}$).$M_{CO2}$
38.11 = $X_{CO}$ × 28 + (1–$X_{CO}$) × 44
$X_{CO}$ = 0.37

**16. Correct option: (C) 1/9**

Let the weight ratio of methane and hydrogen is 1:1
Hence,
Mole fraction of methane = 1/16
Mole fraction of hydrogen = ½
Total mole fraction = (½ + 1/16) = (9/16)
The pressure exerted by Methane
= (1/16 )/(9/16) × $P_t$
= 1/9× $P_t$

**17. Correct option: (D) 5 atm**

Two gases A and B, having the mole ratio of 3:5 in a container.
Hence $n_A/n_B$ = ⅗
If $n_A$ = 3x
Then $n_B$ = 5x
Thus,
$p_B$ =( 5x/8x ) × 8 = 5 atm

**18. Correct option: (A)25.215atm**

Number of moles of $O_2$ = 4/32 = 0.125 moles

Number of moles of $H_2$ = 2/2 = 1 mole
Total number of moles = 1 + 0.125 = 1.125 moles
V = 1 L
T = 0° C = 273 K
Hence, PV = nRT
P = nRT/V
=(1.125×. 0.0821× 273)/1
= 25.215 atm

**19. Correct option: (A) 10 L**

Number of moles of $H_2$ =2/2 = 1 mole
Number of moles of He = 4/4 = 1 mole
Hence, total number of moles = 2 moles
Partial volume of $H_2$ = ½ × 20 = 10 L

**20. Correct option: (C) 6 atm**

According to Dalton's law
$P_t = P_1 + P_2$
$P_t$ = 4+2 = 6 atm

## LEVEL-II

**1. Correct option: (B) 250 mm**

Two parts having pressure of 250 mm Hg,
According to Dalton's law after removing the partition the total or final pressure will same as initial which is 250 mm Hg.

**2. Correct option: (B) 370**

Number of moles of Oxygen = 16/32 = 0.5 moles
Number of moles of Nitrogen = 28/28 = 1 moles
Number of moles of methane = 8/16 = 0.5 moles
Total number of moles = (1+0.5+0.5) = 2 moles
$p_{N2}$ = ½ × 740 = 370 mm

**3. Correct option: (C) P/2**

The total mole fraction of the system = 1
Then, $X_{N2}$ = ½
$X_{O2}$ = ½
$P_{N2}$ = ½ × P
$P_{O2}$ = ½ × P
If $N_2$ is removed then,
$P_T = P_{O2}$ = ½ P = P/2

**4. Correct option: (B) 6,4**

Raoult's Law states that the partial vapour pressure of a component in a mixture is equal to the vapour pressure of the pure component at that temperature multiplied by its mole fraction in the mixture.
In a mixture of liquids of A and B
$p_A = X_A \times P$
$p_B = X_B \times P$
Number of moles of Nitrogen = 56/28 = 2 moles
Number of moles of Oxygen = 96/32 = 3 moles
Total number of moles = 3 + 2 = 5 moles
$p_{O2}$ = ⅗ × 10 = 6 atm
$p_{N2}$ = ⅖ × 10 = 4 atm

**5. Correct option: (A) P/5**

Mole fraction of A, $X_A$ = ⅕

Mole fraction of B, $X_B$ = ⅘

$p_B = X_B \times P$

$= ⅕ \times P$

$= P/5$

**6. Correct option: (A) 1 g of $H_2$**

Number of moles of $H_2$ = ½ moles

Number of moles of $N_2$ = 7/28 = 0.25 moles

Number of moles of He = 4/4 = 1 mole

Number of moles of $O_2$ = 8/32 = 0.25 moles

As we know p = X × P

Thus the gas with highest number of moles will show higher partial pressure.

Hence option A is correct

**7. Correct option: (A) Remains same**

A vessel has two equal compartments A and B containing H2 and O2

Respectively, each at 1 atm pressure.Both are separated by a compartment. If the compartment is removed, the pressure will remain same in A and B.

In both the compartments, the volume, temperature and pressure are same. Hence, the number of moles of hydrogen and oxygen are equal.

When the number of moles and the volume are doubled at constant temperature, the pressure remains constant.

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**8. Correct option: (D) 3 atm**

Number of moles of $H_2$ = 2/2 = 1 moles

Number of moles of $O_2$ = 32/32 = 1 moles

Total number of moles = (1+1) = 2 moles

Mole fraction of $H_2$ = Mole fraction of O2 = ½

Partial pressure of $H_2$,

$p_{H2} = ½ \times 6 = 3$ atm

**9. Correct options: (A)$P_t = P_1 + P_2 + P_3 \ldots\ldots P_n$**

**B) $P_t = (n_1 + n_2 + n_3 \ldots\ldots n_n) RT/V$**

**C) $P_t = ( n_1RT/V) + (n_2RT/V) + (n_3RT/V) + \ldots\ldots\ldots+(n_nRT/V)$**

Dalton's law states that in a mixture of non-reacting gases, the total pressure exerted is equal to the sum of the partial pressures of the individual gases.

Hence,

$P_t = P_1 + P_2 + P_3 \ldots\ldots P_n$

Again form ideal gas equation we know

P = nRT/V

Then, $P_t = (n_1RT/V) + + (n_2RT/V) + (n_3RT/V) +$

$\ldots\ldots\ldots+(n_nRT/V)$

$= (n_1 + n_2 + n_3 \ldots\ldots n_n) RT/V$

**10. Correct option: (C) Mass of liquid water in X increases,D) Mass of liquid water in Y decreases.**

Y is higher pressure region and X is lower pressure region.

So water will flow from Y region to X region to maintain the balance.

Hence the final pressure will be same as X region which is 22 mm Hg.

And consequently, Mass of liquid water in X increases.

Mass of liquid water in Y decreases.

**11. Correct option: (C) 721.2 mm Hg**

Using principal of Dalton's Law Of Partial Pressure. According to which the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture.

P(dry gas) = P(moist gas) – Aqueous tension

P(dry gas) = 745 – 23.8

= 721.5 mm of Hg

**12. Correct option: (B) 650 torr**

From ideal gas equation PV=nRT

Hence, n = PV/RT

Number of moles of A, $n_A$ = (500×1000)/RT

Number of moles of B, $n_B$ = (1000×800)/RT

$N = n_A + n_B$

Final pressure P

= nRT/V

={(500×1000)/RT + (1000×800)/RT}RT/2000

= 650 torr

**13. Correct option: (B) 1466mm**

Number of moles of $N_2$,

$n_{N2}$ = (200×720)/RT

Number of moles of $O_2$, $n_{O2}$ = (400×750)/RT

$N = n_{N2} + n_{O2}$

Final pressure P

= nRT/V

={(200×720)/RT + (400×750)/RT}RT/ 300

= 1466 mm

**14. Correct option: (D) A,C**

Dalton's law of partial pressure is not applicable to a combination of CO and $Cl_2$,because they are reacting gases means they react with each other combines:

$CO+Cl_2 \rightarrow COCl_2$

Same for, B,C and B,D

CO and $F_2$ will not react together hence obeys Dalton's law.

**15. Correct option: (A) Mole fraction of the gas x Total pressure**

Partial pressure of a gas = mole fraction of the gas × Total pressure

**16. Correct option: (B) Amagat's law**

Amagat's law of partial volumes states that the total volume of a gas mixture is equal to the sum of the partial volumes each gas would occupy if it existed alone at the temperature and pressure of the mixture

**17. Correct Matching is: a) →R), b) →P), c) → Q), d) →S)**

a)25% $N_2$, 30%, $CO_2$

Remaining 45% is $O_2$

Partial pressure of O2 = 45/100 × 3 =1.35 atm

b) Numbers of moles of $O_2$ = 3.2/32 = 0.1

Number of moles of $H_2$ = 0.2/2 = 0.1

Hence, n = 0.1 + 0.1= 0.2

P= nRT/V = (0.2×0.0821×273)/1.12 =4.00 atm

c) Number of moles of Nitrogen = 1/28

Number of moles of Oxygen = 1/32

Hence, The partial pressure of nitrogen

= (1/28)/(15/224) P
= 8/15 P
d) Number of moles of hydrogen = 3/2
Number of moles of Oxygen = 24/32 =3/4
Hence,
The partial pressure of hydrogen
= (3/2)/(9/4) P
= 2/3 P

**18. Correct Matching is: a) → R), b) → P), c) → Q), d) → S)**

According to Amagat's law the total volume of a gas mixture is equal to the sum of the partial volumes each gas would occupy if it existed alone at the temperature and pressure of the mixture.
Hence $V = \sum V_i$
Dalton's law states that in a mixture of non-reacting gases, the total pressure exerted is equal to the sum of the partial pressures of the individual gases.
Hence, $P_t = P_1 + P_2 + P_3 \ldots\ldots P_n$
P(dry gas) = P(moist gas) – Aqueous tension
Aqueous tension = P(dry gas) – P(moist gas)
$P_t = (n_1 + n_2)RT/V$ and
$V = V_1 + V_2 + V_3 + \ldots + V_n$
Can be obtained from ideal gas equation

**19. Correct Answer is: 2 arm**

Number of moles of $H_2$ =2/2 = 1 mole
Number of moles of $O_2$ = 64/32 = 2 moles
Total number of moles = 3 moles
The partial pressure of $O_2$ = ⅔ ×3 = 2 atm

**20. Correct Answer is: 2 atm**

The mole fraction of the gas = 40/100 = ⅖
Hence, Partial pressure by the gas = ⅖ × 5 = 2 atm

**21. Correct option: 3**

Let the total mole fraction= X,
$CH_4$ and $O_2$ contains Y g.
Hence, Number of moles of $CH_4$ = Y/16
Number of moles of $O_2$ = Y/32
Total number of moles = 3Y/32
Here 1/x = (Y/32)/(3Y/32)
1/x = ⅓
So x = 3

## Graham's Law of Diffusion

## LEVEL-I

**1. Correct option: (C) $N_2$**

The rate of diffusion of a gas is inversely proportional to the square root of the molecular mass of the gas.
$R \propto 1/\sqrt{M}$
Molecular weight of $N_2$ = 28
Molecular weight of $O_2$ = 32
Molecular weight of $SO_2$ = 64,
Hence $N_2$ gas has highest rate of diffusion.

**2. Correct option: (C) $r_1/r_2 = \sqrt{(M_2/M_1)}$**

Graham's law states that the rate of diffusion or effusion of a gas is inversely proportional to the square root of its molar mass.

So, $r_1/r_2 = \sqrt{(M_2/M_1)}$ is correct mathematical equation for Graham's law

**3. Correct option: (D) 4**

$r_1/r_2 = \sqrt{(M_2/M_1)}$

Or, $V_1T_2/V_2T_1 = \sqrt{(M_2/M_1)}$

$T_2 = \sqrt{(M_2/M_1)} \times (V_2T_1/V_2)$

$T_2 = \sqrt{(32/2)} \times (120/2)$

$= 4 \times 60$

$= 4$ hours

**4. Correct option: (A) 9 : 10**

According to Graham's law

$r_1/r_2 = \sqrt{(M_2/M_1)}$

$r_1/r_2 = \sqrt{81/100} = 9/10$

$r_1 : r_2 = 9:10$

**5. Correct option: (A) 50ml / sec**

Molecular weight of Sulphur dioxide = 64

Molecular weight of methane = 16

Hence,

$r_1/r_2 = \sqrt{(16/64)}$

$r = \sqrt{(16/64)} \times 100$

$= 4/8 \times 100$

$= 50$ ml /sec

**6. Correct option: (C) 1 : 4**

Let B gas diffuses at a rate r

Then gas A diffuses at a rate 2r

Hence

$r/2r = \sqrt{(M_A/M_B)}$

$M_A/M_B = r^2/4r^2 = ¼$

Hence,

$M_A : M_B = 1:4$

**7. Correct option: (C) 100**

100 ml of $CO_2$ diffuses in some time.

$CO_2$ and $N_2O$ have same molecular mass = 44 g/mol

The rate of diffusion by same molecular masses is tsame. So, $N_2O$ also diffuses 100 ml in same time as $CO_2$ diffuses.

**8. Correct option: (A) 2**

$r_1/r_2 = \sqrt{(M_2/M_1)}$

Or, $V_1t_2/V_2T_1 = \sqrt{(M_2/M_1)}$

$T_2 = \sqrt{(M_2/M_1)} \times (V_2T_1/V_2)$

$= \sqrt{(64/16)} \times (120 \times 90/180)$

$= 2 \times 60$ seconds

$= 2$ minutes

**9. Correct option: (A) Mol.wt**

Graham's law states that at same pressure and temperature, rate of diffusion of a gas is inversely proportional to its molecular mass.

**10. Correct option: (A)1/25**

Let B diffuses at rate r

Then rate of diffusion of A is 5r
We know,
$5r/r = \sqrt{(d_B/d_A)}$
Hence, $d_B/d_A = 25/1$
So, $d_A/d_B = 1/25$

**11. Correct option: (B)$r_1/r_2 = (P_1/P_2)\sqrt{(M_2/M_1)}$**

The value of compression of two gases, the rate of diffusion through a fine hole is Given by
$r_1/r_2 = (P_1/P_2)\sqrt{(M_2/M_1)}$

**12. Correct option: (A) $CH_4$**

Let the gas is x, and rate of diffusion of $SO_2$ is r.
Molecular weight of $SO_2$ = 64
Molecular weight of x= $M_x$
Hence $r_x = 2r$
From Graham's law of diffusion,
$r/2r = \sqrt{(M_x/64)}$
$½ \times 8 = \sqrt{M_x}$
$M_x = 16$
Methane has molecular weight of 16.

**13. Correct option: (A) 64**

Let the gas is X, and rate of diffusion of methane is r.
Molecular weight of methane = 16
Molecular weight of X= $M_x$
Hence $r_x = r/2$
From Graham's law of diffusion,
$2r/2 = \sqrt{(M_x/16)}$
$2 \times 4 = \sqrt{M_x}$
$M_x = 8^2 = 64$

**14. Correct option: (C) 16 :1**

$r_1/r_2 = \sqrt{(M_2/M_1)}$
Again,
M= 2d
So,
$r_1/r_2 = \sqrt{(d_2/d_1)}$
$¼ = \sqrt{(d_2/d_1)}$
$d_2/d_1 = 1/16$
$d_1 : d_2 = 16 : 1$

**15. Correct option: (A)1: 2**

$r_1/r_2 = \sqrt{(M_2/M_1)}$
$r_1/r_2 = \sqrt{(1/4)}$
$r_1/r_2 = ½$
Hence,
$r_1 : r_2 = 1:2$

**16. Correct option: (C) Twice that of He**

Molecular weight of hydrogen = 2
Molecular weight of He = 4
$r_{H2}/r_{He} = \sqrt{(4/2)}$
$r_{H2} = 2 \times r_{He}$

**17. Correct option: (D) 8g**

$r_1/r_2 = \sqrt{(M_2/M_1)}$

Or, $V_1/V_2 = \sqrt{(M_2/M_1)}$

Or, $n_1/n_2 = \sqrt{(M_2/M_1)}$

Number of moles of $H_2 = 2/2 = 1$

If the weight of $O_2$ is w

Then number of moles of $O_2 = w/32$

Hence, $1/(w/32) = \sqrt{(32/2)}$

$32/w = 4$

$W = 32/4 = 8$ g

**18. Correct option: (D) d/3**

$PV = nRT$

$P = dRT/M$

$D = PM/RT$

Hence,

$D' = ⅔ (PM/2RT) = d/3$

**19. Correct option: (D) 4 :1**

The rate of diffusion of a gas is inversely proportional to the square root of the molecular mass of the gas.

$R \propto 1/\sqrt{M}$

Molecular weight of,

$NO_2 = 46$

$NH_3 = 17$

$SO_2 = 64$

$CO_2 = 44$

$CH_4 = 16$

Hence, the rate of diffusion in $CH_4$ is fastest and in $SO_2$ is slowest.

Hence, $r_1/r_2 = \sqrt{(64/16)}$

$r_1/r_2 = 4 : 1$

**20. Correct option: (A) 200ml**

$CO_2$ and $N_2O$ have same molecular mass = 44 g/mol

The rate of diffusion by same molecular masses is tsame. So, $CO_2$ also diffuses 200 ml in same time as $N_2O$diffuses.

## LEVEL-II

**1. Correct option: (C)1/24**

$r_1/r_2 = \sqrt{(M_2/M_1)} = ¼$

$M_2/M_1 = 1/16$

Let the masses are $w_1$ and $w_2$.

Mole fraction of first gas = $(w_1/M_1)/\{(w_1/M_1)+(w_2/M_2)\}$

Mole fraction of second gas=

$(w_2/M_2)/\{(w_1/M_1)+(w_2/M_2)\}$

Ratio

$W_1M_2/w_2M_1 = ⅔$

$W_1/w_2 = ⅔ \times 1/16 = 1/24$

**2. Correct option: (C) $H_2$**

Molecular weight of,

$N_2 = 28$

$O_2 = 32$

$H_2$ = 1
Ne = 20
So, rate of diffusion is inversely proportional to the Molecular weight.
So $H_2$ will be filled first.

**3. Correct option: (D) 1:2**

Molecular mass of ammonia = 17
Molecular mass of hydrogen sulphide =34
Hence, ratio of their masses under the same
Conditions of temperature and pressure = 1:2

**4. Correct option: (A) D**

A gas with atomic number 7 → $N_2$ → molar mass → 28
B gas with atomic number 8 → $O_2$ → molar mass 32
C gas with atomic number 9 → $F_2$ → molar mass 38
D gas with atomic number 10→ Ne → molar mass → 20
We know rate of diffusion is inversely proportional to molar mass.
Hence gas D will diffuse very fast

**5. Correct option: (A) 14.14 $dm^3$**

Rate of diffusion of $SO_2$ = 20/60= ⅓ $dm^3/s$
Let the volume of $O_2$ is V,
Hence its rate of diffusion = V/30 $dm^3/s$
So, $r_{O2}/r_{SO2} = \sqrt{(M_{SO2}/M_{O2})}$
$(V/30)/(⅓) = \sqrt{(64/32)}$
$V = 14.14\ dm^3$

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**6. Correct options: (A) Helium escapes at a rate $2\sqrt{2}$ times as fast as $O_2$ does.**
**B) Helium escapes at a rate 4 times as fast as $SO_2$ does.**
**C) Helium escapes at a rate 2.65 times as fast as CO does.**

Molecular mass of He = 4
Molecular mass of $O_2$ = 32
Molecular mass of $SO_2$ = 64
Molecular mass of CO = 28
$r_{He}/r_{O2} = \sqrt{(32/4)}$
$r_{He} = 2\sqrt{2}\ r_{O2}$
$r_{He}/r_{SO2} = \sqrt{(64/4)}$
$r_{He} = 4\ r_{So2}$
$r_{He}/r_{CO} = \sqrt{(28/4)}$
$r_{He} = 2.65\ r_{CO}$
Hence option A, B and C are correct.

**7. Correct options: A) Inversely proportional to square root of its density**
**B) Inversely proportional to square root of its molecular mass**
**C) Equal to Volume diffused to Time Taken**

According to Graham's law of diffusion
The rate of diffusion of a gas is inversely proportional to the square root of the molecular mass of the gas.
$R \propto 1/\sqrt{M}$
Again M = 2d
$R \propto 1/\sqrt{d}$
R = Volume diffused / time taken
Hence option A,B and C are correct.

**8. Correct options: A) Hydrogen diffuses 4 times faster than Oxygen.**
**B) Hydrogen diffuses 2.83 times faster than Methane.**
**C) Helium escapes at a rate 2 times as fast as Methane.**
**D) Helium escapes at a rate 4 times as much as Sulphur dioxide.**
Molar mass of hydrogen = 2
Molar mass of Oxygen = 32
Molar mass of Helium = 4
Molar mass of Methane = 16
Molar mass of Sulphur dioxide = 64
Hence, $r_{hydrogen}/r_{oxygen} = \sqrt{(32/2}$
$r_{hydrogen} = 4 \times r_{oxygen}$
$r_{hydrogen} / r_{methane} = \sqrt{(16/2} = \sqrt{8}$
$r_{hydrogen} = 2.83 \times r_{methane}$
$r_{helium} / r_{methane} = \sqrt{16/4}$
$r_{helium} = 2 \times r_{methane}$
$r_{helium} / r_{sulphurdioxide} = \sqrt{(64/4)}$
$r_{helium} = 4\times r_{sulphurdioxide}$
Hence all options are correct
**9. Correct option: (A) $CO_2$ B) $N_2O$**
Gases that have same molar mass will diffuse at same rate
Molar mass of $CO_2$ = 44
Molar mass of $N_2O$ = 44
Molar mass of $H_2$ = 2
Molar mass of $O_2$ = 32
Hence, $CO_2$ and $N_2O$ will diffuse at same rate
**10. Correct option: (C) 8**
$r_x/r_{He} = \sqrt{(M_x/M_{He})}$
$(1.414/1)^2 = M_x/4$
$M_x = (1.414)^2\times4 = 8$
**11. Correct option: (A)$\sqrt{11}$: 4**
Molar mass of $SO_2$ = 64
Molar mass of $CO_2$ = 44
$r_{SO2}/r_{CO2} = \sqrt{(44/64)} = \sqrt{11/4}$
$r_{SO2} : r_{CO2} = \sqrt{11} : 4$
**12. Correct option: (D) 1 : $\sqrt{2}$ : 2**
Molar mass of $SO_2$ = 64
Molar mass of $O_2$ = 32
Molar mass of $CH_4$ = 16
$r_{SO2} / r_{O2} = \sqrt{(32/64)}$
$r_{SO2} : r_{O2} = 1 : \sqrt{2}$
$r_{O2} / r_{CH4} = \sqrt{(16/32)}$
$r_{O2} : r_{CH4} = 1: \sqrt{2} = \sqrt{2} : 2$
$r_{SO2} : r_{O2} : r_{CH4} = 1 : \sqrt{2} : 2$
**13. Correct option: (A) $P_1/P_2 \times\sqrt{(M_2/M_1)}$**
According to Graham's law at a given temperature,
$R \propto P/\sqrt{M}$
Hence,
$r_1/r_2 = P_1/P_2 \times \sqrt{} (M_2/M_1)$

**14. Correct option: (C) 8 g**

$r_{O2}/r_{H2} = \sqrt{(2/32)}$

$r_{O2} = ¼\ r_{H2}$

$r_{O2} = 3/8\ r_{H2}$

4 g $H_2$ = 2 mole $H_2$

$r_{O2} = 2/t \times 1/4 = 8 / t$

**15. Correct option: (A) Rate of diffusion**

If the volume of two gases 'X' and 'Y' are same under identical conditions of temperature and pressure, their molar mass will differ and hence rate of diffusion will differ

**16. Correct Matching is: a) →P,b) → T, c) →R, d) →Q**

According to Graham's law, the rate of effusion or diffusion of a gas is inversely proportional to the square root of its molar mass.

Among the gases $H_2$ has lowest molar mass and hence it has the highest rate of diffusion

Molar mass of $UF_6$ when it is gas is 352 g/mol.

So it has the least rate of diffusion

Diffusion is a spontaneous process

Movement of gas molecules are unaffected by gravity.

As kinetic energy of the gases are greater than the attractive force between them.

**17. Correct Matching is: a) → R, b) → S, c) →A, d)→Q**

Pressure exerted by the individual

A gas is called partial pressure of the gas.

Aqueous tension = $P_{dry\ gas} + P_{moist\ gas}$

Pressure of water vapour is also the total pressure of the solution.

$P_i = X_i.P_{total} = P_{recorded} – P_{H2O}$

**18. Correct Answer is: $H_2$**

According to Graham's law, the rate of effusion or diffusion of a gas is inversely proportional to the square root of its molar mass.

Among the gases $H_2$ has lowest molar mass and hence it has the highest rate of diffusion

**19. Correct Answer is: 1:4**

$r \propto 1/\sqrt{M}$

Again M = 2d

$R \propto 1/\sqrt{d}$

Hence

$r_1/r_2 = \sqrt{(d_2/d_1)}$

$r_1 / r_2 = \sqrt{1/16} = ¼$

$r_1 : r_2 = 1:4$

**20. Correct Answer is: 2:1**

$r_{SO2}/r_{CH4} = \sqrt{(64/16)} = 2/1$

r = V/t = V/400 for both gasses hence,

The ratio will be remain same.

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## Kinetic Theory of Gas

## LEVEL-I

**1. Correct option: (C) PV = ⅔ $E_k$**

The pressure P can be expressed in terms of translational kinetic energy of the molecules,

P = ⅔ E, $E = 1/2dc^2$

Hence,

$P = ⅔ \times ½ \times mc^2/v$

$PV = ⅔\ E_k$

**2. Correct option: (B) 819 Cal**

E = 3/2 nRT

N = 2/2 = 1

T = 273 K

E = 3/2 ×( 1 × 273 × 2)

= 819 Cal

**3. Correct option: (D) 16**

$K_{hydrogen}$

= 3/2 nRT = (3/2) × (8/2) × RT = 6RT = $16K_{oxygen}$

$K_{oxygen}$

= 3/2 nRT = (3/2) × (8/32) RT = ⅜ RT

**4. Correct option: (C) 4800**

K = 3/2 nRT

= 3/2 ×(4 × 2 × 400) Cal

= 4800 Cal

**5. Correct option: (D) 4 g of $H_2$ at 32°C**

We know K = 3/2 nRT

$K_{H2}$ (at 27°C)

= 3/2 nRT = (3/2) × (4/2) × 2 × 300 = 1800 Cal

$k_{O2}$ (at 0°C)

= (3/2) × (32/32) × 2 × 273

= 819 Cal

$K_{O2}$ (at 4°C)

= (3/2) × (32/32) × 2 × 277

= 831 Cal

$K_{H2}$ (at 32° C)

= (3/2) (4/2) 2 × 305

= 1830 Cal

**6. Correct option: (C) – 273°C**

K = 3/2 nRT

3/2 × 2 × T = 0

T = 0 K = –273° C

**7. Correct option: (A) X**

Kinetic energy of 1 mole gas is

K = 3/2 RT

Hence it is molar mass independent.

So the average kinetic energy of $SO_2$ at 27°C is also X

**8. Correct option: (B) 1:16**

$K_{oxygen}$ = (3/2)× (8/32) nRT

= ⅜ RT

$K_{hydrogen}$ = (3/2) × ( 8/2)nRT

= 6RT

Hence,

$K_{oxygen}$ : $K_{hydrogen}$ = ⅜ : 6 = 3:48 = 1:16

**9. Correct option: (B) Of the absence of attractive forces**

In an ideal gas, the intramolecular forces of attraction are negligible. Hence, theoretically, it cannot be liquified.

**10. Correct option: (C) Pressure is due to collisions between molecules**

In kinetic theory of gases pressure is due to the molecules colliding with the walls of the container and not when molecules collide with each other.

**11. Correct option: (B) is a measure of the volume and pressure of the gas**
**D) is a measure of the average kinetic energy of the molecules**

The absolute temperature of a gas is the measure of volume and pressure of the gas. Because it is found that at absolute zero (−273°C) the volume and pressure of gas becomes zero.

Here also the kinetic energy Is zero..

**12. Correct option: (B) 6J**

K.E. at –73° C ( 200 K )
= 3/2 NR × 200
= 300 NR = 3
So, N = 1/100 R
K.E at 127° C (400K)
= 3/2 (1/100R) R × 400
= 6 J

**13. Correct option: (C) 1: 2**

Let,
Molar mass of $O_2$ = M
Then molar mass of $SO_2$ = 2M
We know for a weight of a gas,
N = w/M,
T = 298 K
So,
$K_{SO2}$ = (3/2)W/2M RT
$K_{O2}$ = (3/2) W/M RT
Thus, $K_{SO2}$ : $K_{O2}$ = ½ : 1 = 1:2

**14. Correct option: (C)1:1**

We know,
K.E. = 3/2 nRT
$K_{H2}$ = 3/2 $n_{H2}$RT
$K_{He}$ = 3/2 $n_{He}$RT
Hence the ratio of Kinetic energy of $H_2$ and He = $n_{H2}$ : $n^{He}$
As, $n_{H2}$ = $n_{He}$
The ratio is,
1:1

**15. Correct option: (D) All will have equal value**

K.E = 3/2 nRT
= 3/2 RT ( For one mole)
When the temperature is same, the average kinetic energy per mole will be same for hydrogen, oxygen and methane as the average kinetic energy per mole is independent of its molar mass.

**16. Correct option: (A) 280K**

K.E of nitrogen at 280K
= 3/2 nR 280 = 420 × R
So,
420 × nR = 3/2 nRT
T = 280 K

**17. Correct option: (B) K.E is dependent on pressure**

Pressure does not have any effect on the kinetic energy of the molecule. Gas pressure is due to the molecules colliding to the walls of the container. The kinetic energy of the molecule directly proportional to the temperature.

**18. Correct option: (B)27° C**

$K_{O2}$ at 27° C = 3/2 nR × 200 = 300 nR

So, 300 nR = 3/2 nRT

T = (300×2)/3 = 200 K = 27°C

**19. Correct option: (A) PV = 2/3 KT**

According to Boyle's law at constant temperature pressure & volume of gas are re lated as PV= Constant

According to kinetic theory, pressure of gas is

$P = 1/3\ rV^2$

$P = 1/3\ mNV^2$

$PV = 2/3\ KT$

**20. Correct option: (B) 44**

$K_{O2} = 3/2\ (2/2)\ RT = 3/2 \times 1 \times RT$ - - - - (i)

$K_{CO2} = 3/2\ N\ RT$ - - - - (ii) (N= number of moles)

Comparing equation (i) and (ii) we get, N = 1

1 mole $CO_2$ = 44 g $CO_2$

Hence, n = 44

## LEVEL - II

**1. Correct option: (A) At constant temperature, the KE of all gas molecules will be the same**

We know, K.E = 3/2 nRT

R an T are constant.

Then if n is same the kE of all has molecules will be same.

**2. Correct option: (A) N molecules of $O_2$**

K.E of $O_2$ at -123°C ( 150 K)

= 3/2 NR × 150

= 225 NR = x

K.E of $O_2$ at 27°C (300K)

= 3/2 N'R × 300

= 450 N'R = 2 × 225NR

Hence, N' = N

**3. Correct option: (B) Same as that of a hydrogen molecules**

Average kinetic energy doesn't depend on molar so. Hence the molar will be same for He and hydrogen both at 298 K.

**4. Correct option: (C) is doubled**

K.E at 127° C ( 400K)

K.E. (1) = 3/2 nR × 400 = 600 nR

K.E at 527° C (800K)

K.E (2) = 3/2 nR × 800 = 1200 nR = 2 × 600 nR = 2× K.E(1)

**5. Correct option: (A) Greatest in the vapour state**

In three states of matter, maximum kinetic energy is possessed by gas molecules.

So, in a closed flask contains in all its three states solid, liquid and vapour at 0°C in this the average K.E. of water molecules will be

Greater in the vapour state.

**6. Correct option: (D) 900 cal**

Number of moles of $O_2$ = 32/32 = 1

Hence,

Average KE

= 3/2 nRT

= 3/2 ×(1 × 2 × 300)

= 900 Cal

**7. Correct option: (C) The root mean square velocity is directly proportional to square root of the Kelvin temperature**

We know K.E = 3/2 nRT

So pressure, volume and molar mass doesn't have any effect on kinetic energy of gas.

Now $V_{rms}$ = √(3RT)/M

Hence $V_{rms}$ is directly proportional to the root of the Kelvin temperature.

**8. Correct option: (A) C= √(3PV/m),C) C = √(3P/d), B) C = √(3RT/m),D) $C^2$ = 3PV/M**

When n= 1

C= √(3PV/m)

We know C = √(3RT/M)

P = 1/3 m n/V $C^2$

P = 1/3 d $C^2$

C = √(3P/d)

Again d= M/V

So, $C^2$ = 3PV/M

**9. Correct option: (A) Gases do not have a definite shape and volume.**

**B) Volume of the gas is equal to volume of container.**

**C) Enclosed gas exerts uniform pressure on the walls of its container in all directions.**

Gases do not have any definite shape or Volume. They take the shape or volume of the container.

Hence the volume of the gas is equal to the volume of container.

Enclosed gas exerts a uniform pressure on the walls of container in all directions

The mass of gas can be determined by weighing the container, filled with gas and again weighing this container after removing the gas. The difference between the two weights gives the mass of the gas.

**10. Correct option: (C) 2493 J**

Number of moles of N2 = 14/28 = ½

Hence,

K.E a 127° C (400K)

= 3/2 ×(½ × 8.3 × 400)

= 300× 8.3 = 2493 J

**11. Correct option: (C) 533 K**

KE of 0.40 mole Ar at 400 K

= 3/2 ×(0.40 × R × 400)

= 240R

kE of 0.30 mole He at T

= 3/2 (0.30 × RT)

Now,

240R = 3/2 (0.30 × RT)

T=( 240 × 2) / (3×0.30) = 533 K

**12. Correct option: (C) Equal**

At same temperature for same moles of gases, the kinetic energy will be same.

**13. Correct option: (B) All are same**

Average kinetic energy doesn't depend on molar mass.

At same temperature for same moles of gases, the kinetic energy will be same.

So in all the flasks average kinetic energy is same.

**14. Correct option: (B) $6.17 \times 10_{-21}$J**

N = 1

R =$1.38 \times 10^{-23}$ $JK^{-1}$

T = 298 K

KE = 3/2 (nRT)

= 3/2 ( $1.38 \times 10^{-23} \times 298$)

= $6.17 \times 10^{-21}$ J

**15. Correct option: (A)$1.24 \times 10^2$J**

Number of moles of O2 (n) = 1/32

Hence, T = (273+47) K = 320 K

KE

= 3/2 ($1/32 \times 8.314 \times 320$)

= $1.24 \times 10^2$ J

**16. Correct Matching is: a) →S), b) → P), c) → Q, d) → R)**

K = 3/2 nRT

$3/2 \times 2 \times T = 0$

T = 0 K = –273° C

AT – 273° C, K.E = 0

$nE_k$ = K.E = 3/2 nRT

K = Boltzman constant = R/N

We know, P = 1/3 m n/V $C^2$

P = 1/3 d $C^2$

C = $\sqrt{(3P/d)}$

**17. Correct Matching is: a) →R), b) →P), c) → S), d) → Q)**

K.E of 8 g $H_2$ at 27°C (300K)

= 3/2 ($8/2 R \times 300$)

= $1800 R = 16 \times 112.5 R = 16 \times K_{O2}$

K.E of 8 g $O_2$ at 27°C (300K)

= 3/2 ($8/32 R \times 300$)

= 112.5 R

K.E of 2 moles $H_2$ at 400 K

= $3/2 \times 2R \times 400$

= 1200 R

= $1200 \times 2$ = 2400 Cal

The KE of O at 27°C ( 300K)

= 3/2 ($2 \times 300$)

= 900

Average K.E of 8 g methane at 27°C (300K)

= 3/2 ($8/16 \times 8.314 \times 300$)

= 1870.6 J

**18. Correct Answer is: 6 atm**

$X_B = 1–0.4 = 0.6$

$P_B = X_B \times P_T = 0.6 \times 10 = 6$ atm

**19. Correct Answer is: 8 : 1**

K.E of 8 g He at 27°C (300K)

= 3/2 (8/4 RT)

= 3/2 × 2 × 300 × R

= 3 × 300R

K.E of 8 g of Oxygen at 27°C (300K)

= 3/2 (8/32 RT)

= 3/2 (¼ × 300 R)

= ⅜ × 300R

Hence the ratio

= 3: ⅜

= 24 : 3

= 8:1

**20. Correct Answer is: 1:1**

Kinetic energy is independent of mass.

Hence the ratio is 1:1

**21. Correct Answer is: 3**

K.E of 2 moles of $N_2$ at 27°C (300K)

= 3/2 (2× 8.314 × 300)

= 7482.6

= $7.4826 \times 10^3$

Hence x = 3

# 5. Metals and Non-Metals

## KEY LEVEL-I

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | C | 2) | C | 3) | C | 4) | A | 5) | C | 6) | D | 7) | B |
| 8) | A | 9) | C | 10) | B | 11) | C | 12) | C | 13) | C | 14) | D |
| 15) | A | 16) | B | 17) | D | 18) | A | 19) | D | 20) | B | 21) | D |
| 22) | D | 23) | D | 24) | A | 25) | B | 26) | C | 27) | A | 28) | A |
| 29) | D | 30) | B | 31) | B | 32) | D | 33) | D | 34) | C | 35) | B |
| 36) | D | 37) | B | 38) | C | 39) | B | 40) | D | 41) | A | 42) | B |
| 43) | B | 44) | B | 45) | B | 46) | D | 47) | C | 48) | A | 49) | B |
| 50) | A | 51) | D | 52) | D | 53) | C | 54) | A | 55) | C | 56) | B |
| 57) | C | 58) | A | 59) | A | 60) | D | 61) | A | 62) | B | 63) | C |
| 64) | A | 65) | D | 66) | A | 67) | C | 68) | A | 69) | D | 70) | A |
| 71) | D | | | | | | | | | | | | |

## KEY LEVEL-II

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | B | 2) | B | 3) | B | 4) | C | 5) | C | 6) | A | 7) | D |
| 8) | B | 9) | C | 10) | C | 11) | A | 12) | C | 13) | A | 14) | A |
| 15) | A | 16) | ABD | 17) | ACD | 18) | ACD | 19) | A | 20) | BCD | 21) | ABC |
| 22) | AC | 23) | B | 24) | A | 25) | D | 26) | C | 27) | A | | |
| 28) | A→ q, B→ r, C→ p, D→ s | | | | | 29) | (A→ r); (B→ s); (C→ p); (D→ q) | | | | | | |
| 30) | (A→ p); (B→ q); (C→ r); (D→ s) | | | | | 31) | 2 | 32) | 1 | 33) | 4 | 34) | 3 |
| 35) | 2 | | | | | | | | | | | | |

# 6. Hydrogen and its Compounds

## Solutions

### LEVEL – I

**1. Correct option: (B, C and D)**

When sodium (Na) reacts with water, it forms sodium hydroxide (NaOH) and hydrogen gas ($H_2$) in the following reaction:

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

Iron react with Sulfuric acid produces hydrogen.

$Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$

When magnesium (Mg) reacts with steam ($H_2O$), it undergoes a chemical reaction to produce magnesium oxide (MgO) and hydrogen gas ($H_2$). The balanced chemical equation for this reaction is:

$2\ Mg + 2H_2O \rightarrow 2MgO + H_2$

**2. Correct option: (C)**

Hydrogen gas ($H_2$) can be prepared by combining hydrochloric acid (HCl) with a reactive metal, such as magnesium (Mg). The chemical reaction that occurs is as follows:

$2HCl + Mg \rightarrow MgCl_2 + H_2$

**3. Correct option: (C)**

To determine which of the given elements will not displace hydrogen from an acid, you need to consider the reactivity series of metals. In this series, metals are listed in order of their reactivity, with more reactive metals displacing hydrogen from acids while less reactive metals do not.

Potassium (K) > Sodium (Na) > Calcium (Ca) > Magnesium (Mg) > Aluminum (Al) > Zinc (Zn) > Iron (Fe) > Lead (Pb) > Hydrogen ($H_2$) > Copper (Cu) > Mercury (Hg) > Silver (Ag) > Gold (Au)

Based on this reactivity series, we can observe that mercury (Hg) is less reactive than hydrogen. So, mercury (Hg) will not displace hydrogen from an acid.

**4. Correct option: (A)**

Hydrogen gas ($H_2$) cannot be produced by the action of dilute sulfuric acid ($H_2SO_4$) on copper (Cu). This is because copper is less reactive than hydrogen in the reactivity series of metals and dilute sulfuric acid is not strong enough to displace hydrogen from metals that are less reactive than hydrogen.

**5. Correct option: (A, B, C)**

Commercial hydrogen is primarily obtained from various sources, including natural gas, oil, and coal. Among the options you provided: Coal gas, Oil gas, Marsh gas (also known as methane, $CH_4$)

**6. Correct option: (B)**

Hydrogen is a light gas, and it is an oxidizing agent because it can support combustion. When hydrogen gas is exposed to air (which contains oxygen), it can readily combust, producing water vapor ($H_2O$) as a result of the reaction:

$2H_2 + O_2 \rightarrow 2H_2O$

**7. Correct option: (B)**

Cavendish's work in the late 18th century involved the reaction of zinc (Zn) with sulfuric acid ($H_2SO_4$), which produce zinc sulfate ($ZnSO_4$) and hydrogen gas ($H_2$). This experiment allowed him to collect and study the properties of hydrogen, leading to his discovery of the gas's unique characteristics.

**8. Correct option: (A)**

Carbon dioxide ($CO_2$) is soluble in water, while hydrogen gas ($H_2$) is not very soluble in water. Therefore, when the mixture is passed through water, the $CO_2$ will dissolve into the water, leaving behind the hydrogen gas. This is a common method for separating these gases in various industrial processes, including the water-gas shift reaction.

**9. Correct option: (B)**

The Bosch process is also known as the water-gas shift reaction, where water gas (a mixture of carbon monoxide and hydrogen) is reacted with steam to produce hydrogen gas and carbon dioxide.

**10. Correct option: (C)**

In an electric arc, high-energy conditions can dissociate diatomic hydrogen molecules ($H_2$) into atomic hydrogen (H). This dissociation occurs because of the high temperatures and energy levels generated in the arc. Atomic hydrogen is highly reactive and is used in various industrial processes, such as welding and surface cleaning.

**11. Correct option: (A)**

$NaandC_2H_5OH$ Both Sodium and ethyl alcohol combine together gives nascent hydrogen.

$Na + C_2H_5OH \rightarrow C_2H_5ONa + [H]$

**12. Correct option: (A)**

Nascent hydrogen is formed when hydrogen atoms (H) are in a highly reactive state due to excess energy. These highly reactive hydrogen atoms can readily participate in various chemical reactions, making them important in several chemical processes and reactions.

**13. Correct option: (B)**

Para hydrogen is more stable than ortho hydrogen. Ortho hydrogen and Para hydrogen are two different spin isomers of molecular hydrogen ($H_2$), and their stability depends on the relative alignment of the spins of the two hydrogen nuclei within the molecule.

**14. Correct option: (D)**

In Para-hydrogen, the spins of the electrons are in opposite directions (anti parallel), and as a result, the spins of the protons are also in opposite directions (anti parallel).

**15. Correct option: (B)**

The nucleus of deuterium, also known as heavy hydrogen, consists of one proton and one neutron. It is represented as D, and it is an isotope of hydrogen. It also has one electron orbiting the nucleus, giving it a single positive charge.

**16. Correct option: (A)**

Tritium is an isotope of Hydrogen.

**17. Correct option: (A)**

The most reactive state of hydrogen is Atomic hydrogen.

**18. Correct option: (D)**

$H_2$ (Two protium atoms)

$HD$ (One protium and one deuterium)

$HT$ (One protium and one tritium)

$D_2$ (Two deuterium atoms)

$T_2$ (Two tritium atoms)

$HT$ (One tritium and one deuterium)

So, there are a total of 6 possible diatomic molecules involving these hydrogen isotopes.

**19. Correct option: (B)**

Hydrogen, like the halogens, has one electron in its outermost energy level and can achieve a stable electron configuration by gaining a single electron to fill its valence shell, similar to how halogens readily gain one electron to complete their valence shells and achieve a stable noble gas configuration.

**20. Correct option: (B)**

Ionization enthalpy is the energy required to remove an electron from an atom or ion. For hydrogen, this energy is relatively low, making it relatively easy to ionize and form a hydrogen cation ($H^+$).

## LEVEL – II

**1. Correct option: (C)**

Tritium (symbol T or 3H) is a radioactive isotope of hydrogen, and it contains
1 proton, 1 electron, 2 neutrons.

**2. Correct option: (C)**

Finely divided nickel is commonly used as a catalyst in the hydrogenation of oils. This process is used in the food industry to convert unsaturated fats and oils into saturated fats, which increases their stability and shelf life.

**3. Correct option: (C)**

In ortho-hydrogen, the spins of the two protons are parallel (aligned in the same direction), while in Para-hydrogen, the spins of the two protons are anti parallel (aligned in opposite directions). This difference in the spin states of the protons leads to distinct nuclear spin states and affects the physical properties of these two forms of hydrogen.

**4. Correct option: (B)**

Palladium (Pd) is known to absorb hydrogen gas, forming a solid solution or alloy with hydrogen. This property of palladium is used in various applications, including hydrogen storage and purification processes. The absorption of hydrogen by palladium is reversible, making it a valuable material for storing and releasing hydrogen gas.

**5. Correct option: (A)**

Hydrogen gas is not obtained when zinc reacts with cold water. Zinc reacts with dilute hydrochloric acid (HCl) and dilute sulfuric acid ($H_2SO_4$) to produce hydrogen gas, but it does not readily react with cold water to release hydrogen gas.

**6. Correct option: (A, B, C)**

Zn, Sn, Al are amphoteric metals react with boiling alkali, in this case NaOH to liberate $H_2$ gas.

$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$
$Sn + 2NaOH + H_2O \rightarrow Na_2SnO_3 + 2H_2$
$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$

**7. Correct option: (A, B, C)**

Hydrogen is a reducing agent and can reduce metal oxides to their metals.
Hydrogen will reduce heated cupric oxide (CuO) to produce copper (Cu).
Hydrogen will reduce heated ferric oxide ($Fe_2O_3$) to produce iron (Fe).
Hydrogen will reduce heated zinc oxide (ZnO) to produce zinc (Zn).

**8. Correct option: (C)**

When sodium (Na) reacts with ethyl alcohol ($C_2H_5OH$), it can produce hydrogen gas ($H_2$) along with sodium ethoxide:

$2Na + 2C_2H_5OH \rightarrow 2C_2H_5ONa + H_2$

**9. Correct option: (D)**

Hydrogen forms covalent compounds by sharing electrons with other nonmetals, and it does not form a large number of ionic compounds by losing an electron to form a cation. Instead, hydrogen usually forms hydride ions when it reacts with metals, which are considered as compounds with covalent character.

**10. Correct option: (A)**

The mixture of carbon monoxide (CO) and hydrogen ($H_2$) formed by the action of steam on hydrocarbons is known as "water gas" or "synthesis gas."

**11. Correct option: (B)**

Hydrogen typically does not act as a cation in ionic salts. In most ionic compounds, other metals or elements take on the role of cations, while hydrogen more commonly participates in covalent compounds.

**12. Correct option: (A, B, C)**

**It is pollution-free and thus is a green fuel**.

Hydrogen, when burned or used in fuel cells, produces water vapor as its only byproduct, making it a clean and environmentally friendly fuel.

**It releases greater energy per unit weight compared to gasoline and other fuels.**

Hydrogen has a high energy-to-weight ratio, making it an efficient fuel in terms of energy output for its weight.

**Hydrogen can be transported in the form of a solid as metal hydride.**

Hydrogen can indeed be stored and transported in the form of metal hydrides, which can be more convenient and safer than handling gaseous hydrogen in certain applications.

**13. Correct option: (B, C)**

**Its reaction with oxygen is highly exothermic**: The combustion of liquid hydrogen with oxygen results in a highly exothermic reaction, producing a large amount of thrust, which is crucial for rocket propulsion.

**Small space it occupies**: Liquid hydrogen has a high energy density, which means it can provide a significant amount of energy for propulsion while occupying relatively little space in the rocket's fuel tanks. This is essential for space travel, where minimizing weight and maximizing energy content are critical factors.

**14. Correct option: (B)**

Hydrogen is often considered one of the cleanest fuels in terms of environmental pollution because when it is burned or used in fuel cells, it produces water vapor ($H_2O$) as the only byproduct. This means that hydrogen combustion does not release harmful greenhouse gases or air pollutants that contribute to environmental pollution, making it an environmentally friendly fuel.

**15. Correct option: (D)**

Hydrogen is the lightest of the gases given. It has the lowest molecular mass among common elements and gases, with a molecular weight close to 2 g/mol.

**16. Correct option: (A)**

The abundance of hydrogen ($H_2$) in Earth's atmosphere is very small because Earth's gravity is relatively weak when it comes to holding onto extremely light molecules like hydrogen. Hydrogen gas has a very low molecular weight, and its lightness allows it to

escape Earth's gravitational pull more easily compared to heavier gases like nitrogen and oxygen.

**17. Correct option: (C)**

Pure hydrogen can be obtained by carrying out electrolysis of a solution of barium hydroxide ($Ba(OH)_2$). This process involves the use of an electrolytic cell where the electrolyte is a solution of $Ba(OH)_2$, and the electrolysis of water takes place to produce hydrogen gas ($H_2$) at the cathode. The reaction at the cathode is as follows:

$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$

**18. Correct option: (B)**

Iron (Fe) reacts with steam (water vapor, $H_2O$) to produce hydrogen gas ($H_2(g)$) and $Fe_3O_4$

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

**19. Correct option: (A)**

Sodium (Na) is the metal that reacts with water at room temperature to produce hydrogen gas ($H_2(g)$). This reaction is highly exothermic and can be quite vigorous, often resulting in the evolution of hydrogen gas and the formation of sodium hydroxide (NaOH) as a byproduct.

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

**20. Correct Matching is: a-S; b-P; c-Q; d-R**

a) Protium is the most common isotope of hydrogen, and its nucleus contains:
1 proton, 0 neutrons, 1 electron.
Correct match for the option is S.
b) Deuterium is an isotope of hydrogen, and it has one electron, one proton, and one neutron.
Correct match for the option is P.
c) Tritium has 1 proton, 1 electron, and 2 neutrons.
Correct match for the option is Q.
d) Di hydrogen each hydrogen atom still has one proton and one electron.
Correct match for the option is R.

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## WATER

## LEVEL-I

**1. Correct option: (D)**

Pure water, in its ideal form, has very low conductance of electricity. This is because pure water contains very few ions, and its electrical conductivity primarily depends on the presence of ions or charged particles in the solution. In the absence of impurities or dissolved ions, water does not conduct electricity to a significant degree.

**2. Correct option: (C)**

Rainwater is considered one of the purest forms of natural water. It starts as water vapor in the atmosphere, condenses to form droplets, and then falls to the ground as rain. During this process, many impurities and contaminants are left behind, resulting in relatively pure water.

**3. Correct option: (D)**

The shape of a water molecule is V-shape. It has a bent molecular geometry due to the arrangement of its two hydrogen atoms around the central oxygen atom, resulting in a

bond angle of approximately 104.5 degrees between the hydrogen-oxygen-hydrogen atoms in a water molecule.

**4. Correct option: (B)**

The HOH bond angle in a water molecule is 104.5 degrees (104°5').

**5. Correct option: (A)**

The dipole moment of a water molecule is approximately 1.84 Debye. This dipole moment arises due to the unequal sharing of electrons between the oxygen and hydrogen atoms in the molecule, resulting in a partial positive charge on the hydrogen atoms and a partial negative charge on the oxygen atom.

**6. Correct option: (A)**

The two hydrogen atoms in a water molecule are joined to an oxygen atom by covalent bonds. Covalent bonds involve the sharing of electrons between atoms, and in the case of water, each hydrogen atom shares an electron with the oxygen atom to form a covalent bond. This sharing of electrons allows the atoms to achieve a stable electron configuration.

**7. Correct option: (D)**

In liquid water, the aggregates of varying numbers of water molecules are held together primarily by hydrogen bonds. Hydrogen bonds are a type of intermolecular force that occurs between the positively charged hydrogen atoms of one water molecule and the negatively charged oxygen atoms of neighboring water molecules.

**8. Correct option: (D)**

A water molecule ($H_2O$) can form a maximum of four hydrogen bonds.

**9. Correct option: (B)**

At its melting point, ice is lighter than water because of the unique structure of ice crystals. In the solid state (ice), water molecules arrange themselves into a hexagonal lattice structure with open spaces. This arrangement creates a less dense structure compared to liquid water, where the water molecules are more closely packed.

**10. Correct option: (C)**

The process that will produce hard water is the saturation of water with calcium sulfate (CaSO4). When water is saturated with calcium sulfate, it can lead to the formation of calcium ions ($Ca^{+2}$)and sulfate ions ($SO_4^{-2}$) in the water, which are responsible for water hardness. These ions can react with soap, forming insoluble precipitates, and create scale deposits in pipes and appliances.

**11. Correct option: (C)**

**12. Correct option: (B)**

**13. Correct option: (D)**

When water is boiled for some time, it becomes free from temporary hardness but not from permanent hardness. Temporary hardness is caused by the presence of bicarbonate ions ( $HCO_3$ in the water, which can be removed by boiling because the heat causes the bicarbonate ions to decompose into carbonate ions ($HCO_3^{-2}$) and water ($H_2O$). The carbonate ions can then precipitate as calcium carbonate ($CaCO_3$) or magnesium carbonate ($MgCO_3$), which can be removed by filtration.

**14. Correct option: (B)**

When Zeolite, which is a hydrated sodium aluminum silicate, is treated with hard water, the sodium ions in the Zeolite are exchanged with calcium ions (Ca^2+ ions).

**15. Correct option: (D)**

The structure of $H_2O_2$ (hydrogen peroxide) is non-planar. The molecule has a bent or V-shaped structure with an angle of approximately 104.5 degrees between the two O-H bonds.

**16. Correct option: (A)**

$H_2O_2$ (Hydrogen peroxide) is commonly prepared in the laboratory by the action of dilute $H_2SO_4$ (sulfuric acid) on $BaO_2$ (barium peroxide).

**17. Correct option: (C)**

Moist hydrogen peroxide cannot be dried over concentrated $H_2SO_4$ (sulfuric acid) because it can be oxidized by $H_2SO_4$. Concentrated sulfuric acid is a strong oxidizing agent, and it can react with and oxidize hydrogen peroxide, potentially leading to the release of oxygen gas and other reactions.

**18. Correct option: (D)**

An aqueous solution of $H_2O_2$ (hydrogen peroxide) is weakly acidic. It has a pH slightly below 7, around 6.

**19. Correct option: (B)**

The reaction $2H_2O_2 \rightarrow 2H_2O + O_2$ shows that hydrogen peroxide ($H_2O_2$) acts as an oxidizing agent. In this reaction, it is decomposed to produce water ($H_2O$) and oxygen ($O_2$).

**20. Correct option: (C)**

This statement is explaining about the fact that 30 volumes of hydrogen peroxide ($H_2O_2$) solution will release 30 volumes of oxygen ($O_2$) gas when decomposed at standard temperature and pressure (STP). This property is commonly used in titration and analytical chemistry to determine the concentration of hydrogen peroxide solutions.

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## LEVEL-II

**1. Correct option: (B)**

The maximum density of water is reached at a temperature of approximately 277 K.

**2. Correct option: (A)**

Potash alum, also known as potassium alum ($KAl(SO_4)_2 \cdot 12H_2O$), is commonly used in water treatment to remove impurities and clarify water. It helps in flocculation and coagulation of impurities, making them easier to settle and filter out of the water.

**3. Correct option: (C)**

Zeolites are crystalline, hydrated aluminosilicate minerals with a three-dimensional structure that contains pores and channels. They have a high ion exchange capacity, making them useful in various applications, including water softening and catalysis. Sodium aluminum silicate Zeolites are commonly used in ion exchange processes.

**4. Correct option: (A)**

The true structure of hydrogen peroxide ($H_2O_2$) is more accurately represented as a "bent" or "V" shape. This molecule has two oxygen atoms bonded to each other, and each oxygen atom is also bonded to a hydrogen atom. The molecular formula is H-O-O-H.

**5. Correct option: (B)**

Volume strength of $H_2O_2$ =5.6× normality

Volume strength of $H_2O_2 = 5.6 \times 15$

Volume strength of $H_2O_2 = = 8.4L$

**6. Correct option: (A)**

Temporary hardness in water is primarily caused by the presence of bicarbonates of calcium (Ca) and magnesium (Mg).

**7. Correct option: (B)**

The dihedral angle of $H_2O_2$ in gaseous phase is 90.2° and in solid phase is 111.5°.

**8. Correct option: (C)**

Hydrogen peroxide ($H_2O_2$) is diamagnetic. Diamagnetic substances do not have unpaired electrons, means they have no net magnetic moment and are not attracted to a magnetic field.

**9. Correct option: (A)**

Hydrogen peroxide ($H_2O_2$) can readily donate oxygen atoms (in the form of hydroxyl radicals, OH) to other substances, oxidizing them in the process.

**10. Correct option: (A, B)**

Hydrogen peroxide ($H_2O_2$) is used for various purposes, but among the given options provided, it is primarily used as:

A bleaching agent, an oxidizing agent.

$H_2O_2$ Is commonly used for bleaching purposes (such as bleaching hair or fabrics) and as an oxidizing agent in various chemical reactions.

**11. Correct option: (C)**

Hydrogen peroxide can act as both an oxidizing agent and a reducing agent depending on the reaction conditions and the substances involved. It can oxidize certain substances by donating oxygen atoms, while in other reactions; it can be reduced by accepting electrons.

**12. Correct option: (D)**

Permanent hardness in water, which is primarily caused by the presence of calcium and magnesium ions in the form of sulfates and chlorides, can be removed or reduced by using the following methods:

**Treatment with washing soda** (sodium carbonate, $Na_2CO_3$): Washing soda reacts with calcium and magnesium ions in water to form insoluble calcium carbonate ($CaCO_3$) and magnesium carbonate ($MgCO_3$), which can be removed by precipitation. This method is often used to soften hard water.

**13. Correct option: (A, B, C and D)**

$H_2O_2$ Is used for bleaching delicate articles like wool, hair, feathers and ivory.

**14. Correct option: (D)**

**15. Correct option: (B)**

The volume strength of a solution is defined as the number of liters of oxygen gas that can be obtained from 1 liter of the solution at STP.

The balanced chemical equation for the decomposition of hydrogen peroxide is:

$2H_2O_2 \rightarrow 2H_2O + O_2$

From the above equation, we observe that 2 moles of hydrogen peroxide decompose to give 1 mole of oxygen gas. Therefore, for a 1M solution of hydrogen peroxide, the volume strength is: $11.2\ L \frac{O_2}{mol} \times \frac{1\ mol}{L} = 11.2\ vol$

**16. Correct option: (A)**

$$Volume of H_2O_2 = \left(\frac{Volume\ of\ O_2}{22400}\right) \times \left(\frac{Molar\ volume\ at\ STP}{Volume\ strength}\right)$$

Here, the volume of $O_2$ is given as 0.80 L and the molar volume at STP is 22.4 L/mol. The volume strength of 40 volumes $H_2O_2$ is 40 mL/100 mL.

Substituting these values in the formula, we get:

$$Volume\ of\ H_2O_2 = \left(\frac{0.80}{22400}\right) \times \left(\frac{22.4}{40}\right)$$

$$Volume\ of\ H_2O_2 = 0.0008\ L$$

We know that the volume strength of 40 volumes $H_2O_2$ is 40 mL/100 mL. Therefore, we can convert the volume of $H_2O_2$ from liters to milliliters as follows:

$$Volume\ of\ H_2O_2 = \frac{0.0008 \times 1000}{1000}$$

$$Volume\ of\ H_2O_2 = 0.8\ mL$$

**17. Correct option: (A, B)**

Temporary hardness of water is caused by the presence of dissolved bicarbonates of calcium and magnesium. It can be removed by adding a base such as slaked lime $Ca(OH)_2$ or sodium carbonate $Na_2CO_3$

**18. Correct option: (A)**

Calgon is a trade name for a complex salt, sodium hexametaphosphate $(NaPO_3)_6$, which is used for softening hard water.

**19. Correct option: (B)**

The reagent commonly used to determine the hardness of water titrimetrically is di sodium salt of EDTA.

**20. Correct option: a-R, b-P, c-S, d-Q**

a)Hard water is due to presence of Calcium ions in water.

Correct match for the option is R.

b) Temporary hardness is due to the presence of $Ca(HCO_3)_2$.

Correct match for the option is P.

c) Sodium ions in Zeolite get exchanged with $Mg^{+2}$ ions.

Correct match for the option is S.

d) Paramagnet hardness is due to presence of $MgSO_4$.

Correct match for the option is Q.

# 7. Pollution of Air and Water

## Solutions

### LEVEL-I

**1. Correct option: (C) 6 months**

A PUC or pollution under control certificate of a new car/bike is valid for 1 year and after that renewal is required every 6 months.

**2. Correct option: (A) 21%**

The atmosphere is majorly composed of 78% nitrogen and 21% oxygen and other gases.

**3. Correct option: (C) Carbon dioxide**

**4. Correct option: (C) Visakhapatnam**

(NTPC-Simhadri) Simhadri Super Thermal Power Plant is a coal-fired power plant located in the outskirts of Visakhapatnam city in Andhra Pradesh.

**5. Correct option: (C) Nuclear power**

Nuclear energy originates from the splitting of uranium atoms – a process called fission. This generates heat to produce steam, which is used by a turbine generator to generate electricity.

**6. Correct option: (D) 1984**

The Bhopal Gas Tragedy, a catastrophic industrial disaster, occurred in the year 1984 due to a gas leak at the Union Carbide pesticide plant.

**7. Correct option: (C) Chemicals**

Industrial waste often contains various harmful chemicals that can pose environmental and health risks.

**8. Correct option: (D) All**

Pollutants released by thermal power plants include Sulphur Dioxide (SO2), Carbon Monoxide (CO), Oxides of Nitrogen (NOx) and Ozone (O3). Suspended Particulate Matter (SPM), Lead and Non-Methane Hydrocarbons.

**9. Correct option: (A) PAN (Peroxyacyl nitrate)**

PAN is a secondary pollutant formed by the reaction of nitrogen oxides and volatile organic compounds in the presence of sunlight.

**10. Correct option: (B) Yamuna**

The Taj Mahal is situated on the banks of the river Yamuna in Agra. Taj Mahal is one of the Seven Wonders of the world. It is situated on the banks of river Yamuna.

**11. Correct option: (C) Mercury**

The "Minamata disease" is caused by the ingestion of methylmercury-contaminated seafood, leading to severe neurological damage.

**12. Correct option: (A) 31%**

**13. Correct option: (D) All the above**

Chlorofluorocarbons (CFCs) are found in various appliances like air conditioners (AC), refrigerators, also used historically in jet planes.

**14. Correct option: (B) July**

Van Mahotsav is an annual festival celebrated in India in the month of July. It marks the celebration of forests and trees. It was initiated by K.M. Munshi in 1950.

**15. Correct option: (D) CNG (Compressed Natural Gas)**

CNG is considered an eco-friendly fuel as it produces lower emissions compared to gasoline or diesel when burned.

**16. Correct option: (A) Nitrates and phosphates**

Due to the discharge of untreated domestic sewage and surface runoff water, having residues of phosphate, nitrate fertilizers, from agricultural fields etc., into neighbouring

water bodies, the water bodies become rich in nutrients especially phosphates and nitrates ions. As a result of nutrient enrichment, water bodies become highly productive or eutrophic and this phenomenon is called as eutrophication.

**17. Correct option: (A) CFCs**

Chlorofluorocarbons (CFCs) are responsible for depleting the ozone layer in the stratosphere.

**18. Correct option: (B) CO2 and SO2**

Acid rain is primarily caused by the presence of sulfur dioxide (SO2) and nitrogen oxides (NOx) in the atmosphere, leading to the formation of sulfuric acid and nitric acid.

**19. Correct option: (A) Electrostatic precipitator (ESP)**

Electrostatic precipitators (ESP) are effective devices used to reduce suspended particulate matter (SPM) from industrial emissions.

**20. Correct option: (C) Methyl Isocyanate**

Bhopal Disaster happened on 3 Dec,1984 where about 45 tons of the dangerous gas methyl isocyanate escaped from Union Carbide Limited pesticide plant.

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## LEVEL – II

**1. Correct option: (D) All of these**

Renewable sources of energy should be used instead of burning of fossil fuels.

**2. Correct option: (C) 21%**

The atmosphere is majorly composed of 78% nitrogen and 21% oxygen and other gases.

**3. Correct option: (B) Nitrogen**

Nitrogen is an inert gas and does not affect the climate in any way.

**4. Correct option: (A) Infrared radiations**

Absorption of IR by CO2 heats the lower atmosphere, making it a greenhouse gas.

**5. Correct option: (A) Replace**

Water cannot be 'replaced'. The human body is roughly 60% water, it is essesntial for our survival. Water is also considered as the universal solvent.

**6. Correct option: (C) DDT**

DDT is a man-made persistent organic pollutant. It gets readily adsorbed to soil and sediments.

**7. Correct option: (B) Distillation**

Distillation removes 99.5% of the impurities from water.

**8. Correct option: (C) Drip irrigation**

Drip irrigation can reduce water use by 30 to 70% compared to conventional sprinkler irrigation. It saves water by supplying water directly to the roots.

**9. Correct option: (D) Chlorination**

Chlorination is an effective method to disinfect water. When Chlorine is added to water, it forms hypochlorous acid and hypochlorite ions which are the main disinfecting compound in chlorinated water.

**10. Correct option: (B) Water Pollution**

Corrosion deformation studies found H2S to be the main compound responsible for corrosion of the white marbles of Taj Mahal. Recent studies have established river yamuna as the dominant producer of H2S.

**11. Correct option: (C) Sulphur dioxide**

Sulphur dioxide can irritate the lining of nose, throat, lungs, and subsequently cause permanent lung damage.

**12. Correct option: (B) CFCs**
Chlorofluorocarbon (CFCs) is an organic compound that does not occur naturally. It is a man-made gas and causes ozone depletion.
**13. Correct option: (C) The sun's UV rays**
The ozone layer in the strastosphere absorbs UV radiation which is known to cause cancer.
**14. Correct option: (B) CO**
Carbon Monoxide (CO) is produced due to incomplete combustion of fuels and is toxic since it reacts with haemoglobin.
**15. Correct option: (A) Swimming in the pond**
Swimming in pond doesn't discharge as much pollutants as washing clothes or washing vehicles would release.
**16. Correct option: (B) Leakage of poisonous gas**
Bhopal Gas Tragedy happened on 3 Dec,1984 where about 45 tons of the dangerous gas methyl isocyanate escaped from Union Carbide Limited pesticide plant.
**17. Correct option: (C) Water Pollution**
Water pollution can be caused by sewage and wastewater, oil spills, radioactive substances, etc. It makes the water unfit for consumption and also threatens aquatic life.
**18. Correct option: (C) Reduced oxygen carrying capacity of RBCs**
Carbon monoxide competes with oxygen to bind to Hemoglobin. CO binds permanently with Hb forming carboxyhemoglobin. This causes reduced oxygen carrying capacity of RBCs.
**19. Correct option: (D) Large amount of salt**
Sea water is considered non-potable because of high concentration of salt. Drinking sea water will lead to death by dehydration.
**20. Correct option: (B) Reduce Ganga water pollution**
Ganga action Plan was started in Jan 14, 1986 by the then prime minister Rajiv Gandhi. It is a centrally sponsored scheme with the aim of pollution abatement from river ganga.
**21. Correct option: (C) Carbon monoxide**
Acid rain is caused by a chemical reaction between the oxides of sulfur and nitrogen and rainwater.
**22. Correct option: (D) All**
Chlorofluorocarbon (CFCs) is an organic compound that does not occur naturally. It is a man-made gas and causes ozone depletion.
**23. Correct option: (B) Acid Rain**
The slow corrosion of marble of a monument by acid rain is known as marble cancer.
**24. Correct option: (A) Spring Water**
High-quality spring water is rich in magnesium, potassium, calcium, sodium, and other trace minerals.
**25. Correct option: (C) Reduced oxygen carrying capacity of RBCs**
Carbon monoxide competes with oxygen to bind to Hemoglobin. CO binds permanently with Hb forming carboxyhemoglobin. This causes reduced oxygen carrying capacity of RBCs.
**26. Correct option: (C) By burning coal in thermal power stations**
Lead is released into the air when coal, oil, or waste is burned. Smelters, mining operations, waste incinerators, battery recycling also contribute to lead pollution.
**27. Correct option: (A) Increasing forest reserves**
Forests act as carbon sinks and are extremely efficient in removing pollutants from the atmosphere.
**28. Correct option: (A) Increasing forest reserves**
Increasing forest reserves help in combating air pollution since forests serve as carbon sinks.

**29. Correct option: (D) iii, i, ii**

A) Condensed wider vapor in air - fog

B) Asthma- caused by smog

C) Smog- Smoke + fog + fumes

**30. Correct option: (C) iii, i, iv, ii**

A) Greenhouse gas- Carbon dioxide

B) Infrared Radiations- Absorbed by $CO_2$

C) Oxides which affect vegetation- Nitrogen oxides

D) Sulphur dioxide- Polluting gas